Directed Cyclic Networks
Shortened Proof Compendium

4 October 2026

This edition replaces repeated standard arguments with citations and retains the model-specific coefficients, parameter constructions, spectral gaps and realizability proofs. P01–P05 and P07–P11 contain the completed arguments supplied in the source project. Earlier independent audits apply to that source edition, not automatically to this edited text. The supplied P08 global-interior extension also awaited independent audit.

P06 and P12–P18 were outlines in the source. They are retained as concise, explicitly conditional frameworks with their outstanding calculations identified. In particular P11 establishes a spectral Hopf candidate, not a completed nonlinear Hopf theorem.

All ten cited works were checked against primary publisher or author records. Full-text access and theorem-location limits are recorded in `CITATION_VERIFICATION.txt`. Research articles, reference articles, a preprint and one standard textbook are distinguished there. No unverified article-specific theorem number is used.

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1 Canonical model contract

This section specifies the equations and parameter domain used throughout.

1.1 Main demand-modulated model

For $i \in \mathbb{Z}_n$, define the vector field F^{DM} by

$$\dot{x}_i = ax_i - kx_i^2 + \frac{cx_{i-1}}{1+x_i} - \frac{gx_i}{1+x_{i+q}} - hx_ix_{i-q}. \tag{$\mathrm{DM}_{n,q}$}$$

The variables are dimensionless normalized activities. The preferred term is *demand-modulated model*. The positive input is called demand-modulated predecessor input; no standard Holling type-II classification is adopted. The intended initial conditions are $x(0) \in \mathbb{R}^n_{\geq 0}$. For local statements and boundary bifurcations, use the specified open extension domain $\mathcal{U} = (-1,\infty)^n$. This domain choice asserts no invariance theorem.

1.2 Primary matched bilinear comparator

Define F^{MB} by

$$\dot{y}_i = ay_i - ky_i^2 + cy_iy_{i-1} - gy_iy_{i+q} - hy_iy_{i-q}, \qquad i \in \mathbb{Z}_n. \tag{$\mathrm{MB}_{n,q}$}$$

This is the topology- and parameter-count-matched comparator in the main program. Its intended initial data are nonnegative; the polynomial vector field is specified on $\mathbb{R}^n$. No existence, invariance, boundedness or stability conclusion is included in this contract. Matching parameter names does not settle a quantitative calibration between the two laws.

1.3 Parameter and indexing contract

$$n \in \mathbb{Z}, \quad n \geq 5, \qquad q \in \{1,\dots,n-1\}, \qquad \boldsymbol{\pi} = (a,k,c,g,h) \in (0,\infty)^5.$$

All indices are modulo n, with representatives $0,\dots,n-1$. Arbitrary q is allowed. The interpretation as one connected step-q cycle requires $\gcd(n,q) = 1$; this condition is not imposed on the general analytical program. In particular, even n is not excluded when $q = 2$. Coincident offsets do not remove terms from either defining equation. The intended exact wave-number-selection program focuses on $q = 2$. The five-node Wu-Xing-inspired specialization is $n = 5, q = 2$.

1.4 Archived secondary six-term comparator

The following is a transcription of the blueprint's proposed arbitrary-n,q extension, not a verified attribution to the original Lin–Han publication:

$$\dot{y}_i = ay_i - by_i^2 + cy_iy_{i-1} - dy_iy_{i-q} - g_{\mathrm{c}}y_iy_{i+1} - hy_iy_{i+q}. \tag{$\mathrm{LB}_{n,q}$ archive}$$

Its archived five-node equation is obtained by specifying $n = 5, q = 2$ in this transcription. The six coefficient names are archival and are not a parameter identification with DM or MB. The original source, coefficient domain, index orientation, and attribution remain to be checked before analysis or use as a literature application.

1.5 Scope boundary

The main program covers homogeneous DM, primary comparator MB, their synchronous equilibria and spectra, the planned bifurcations, and the planned small-heterogeneity response theory. All requested P01–P18 tasks remain in the project; final main-text versus supplement placement is open. The heterogeneous coefficient assignment is not frozen by this contract. The six-term model is archived for a possible secondary literature application, never a replacement for MB. Source-saturating replacements, local motif theory, pathology extensions, adaptive set points, interventions, noise, delays, data calibration, and prior Lyapunov-spectrum reanalysis are postponed to other work.

2 Notation and assumptions

2.1 Standing assumptions

A1. $n \in \mathbb{Z}$, $n \geq 5$, and $q \in \{1, \ldots, n-1\}$.

A2. Homogeneous parameters $\boldsymbol{\pi} = (a, k, c, g, h) \in (0, \infty)^5$.

A3. Dimensionless real activities; initial states in the nonnegative orthant when forward biological-admissibility statements are sought. For DM the local extension domain is $\mathcal{U} = (-1, \infty)^n$.

A4, conditional. Impose $\gcd(n, q) = 1$ only for the connected step-q cycle interpretation; specify $q = 2$ or $(n, q) = (5, 2)$ when required by a module.

A5, conditional. Positivity of an equilibrium, invertibility, hyperbolicity, stability, Hopf nondegeneracy, and perturbation regularity are additional hypotheses to be stated and checked in the appropriate future proof. None is a standing established result.

2.2 Indices and cyclic shifts

Define $\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$, using representatives $0, \ldots, n-1$. For $z \in \mathbb{C}^n$, define the predecessor and successor shift matrices by

$$(Pz)_j = z_{j-1}, \qquad (P^{-1}z)_j = z_{j+1}, \qquad P_{j\ell} = \begin{cases} 1, & \ell \equiv j-1 \pmod{n}, \\ 0, & \text{otherwise.} \end{cases}$$

Thus the symbols P^q and P^{-q} denote repeated predecessor and successor shifts, respectively. The group-action convention is $\rho(r)z = P^r z$. For transcription of the original five-node labels, the original $x_1, \ldots, x_5$ are relabeled as $x_0, \ldots, x_4$ in the same cyclic order. This is a label translation only; no historical or physiological statement is certified.

2.3 Fourier vectors, transform and pairing

Use i for the imaginary unit and j for a node index. Define

$$\theta_m = \frac{2\pi m}{n}, \qquad v_j^{(m)} = \frac{1}{\sqrt{n}} \exp(\mathrm{i}\theta_m j), \qquad \mathsf{Q}_{jm} = v_j^{(m)}, \quad 0 \leq j, m < n.$$

The canonical transform convention is

$$\widehat{z}_m = \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} z_j \exp(-\mathrm{i}\theta_m j), \qquad z_j = \frac{1}{\sqrt{n}} \sum_{m=0}^{n-1} \widehat{z}_m \exp(\mathrm{i}\theta_m j).$$

The normalization is the unitary DFT convention. The blueprint's displayed unscaled Fourier vectors are not copied as normalized eigenvectors. Use A^* for conjugate transpose, A^{T} for transpose, and

$$\langle u, v\rangle = u^* v = \sum_j \overline{u_j} v_j, \qquad \|z\|_2 = (z^* z)^{1/2}.$$

The pairing is conjugate-linear in its first argument. Matrix 2-norms are induced Euclidean norms. The adjoint eigenvector equation, its phase/scale, Taylor tensors and first-Lyapunov-coefficient convention must be reconciled in P13 before any blueprint normal-form formula is imported.

2.4 State sets and equilibrium notation

$$\mathbb{R}^n_{\geq 0} = \{x \in \mathbb{R}^n : x_j \geq 0 \text{ for all } j\}, \qquad \mathbb{R}^n_{>0} = \{x \in \mathbb{R}^n : x_j > 0 \text{ for all } j\},$$

$$\mathbf{1} = (1, \ldots, 1)^{\mathsf{T}}, \qquad \mathcal{S} = \{s\mathbf{1} : s \in \mathbb{R}\}.$$

The phrase positive orthant means the strict set $\mathbb{R}^n_{>0}$; nonnegative orthant means $\mathbb{R}^n_{\geq 0}$. A synchronous state is a member of $\mathcal{S}$; a positive synchronous equilibrium, if established, is denoted $x^*\mathbf{1}$ with $x^* > 0$. The admissible synchronous domain for DM is $\mathcal{S} \cap \mathcal{U}$. No invariance or equilibrium classification is asserted by these definitions.

2.5 Spectrum, hyperbolicity and mode terminology

For a square matrix J, define

$$\operatorname{sa}(J) = \max\{\operatorname{Re}\lambda : \lambda \in \operatorname{spec}(J)\}.$$

An equilibrium of a differentiable vector field is hyperbolic when its Jacobian has no eigenvalue with zero real part. A matrix is Hurwitz if all its eigenvalues have negative real part, and normal if $JJ^* = J^*J$. These definitions assert neither stability nor normality for a model. Mode $m = 0$ is called the uniform or synchronous mode; modes $m \neq 0$ are called nonuniform or symmetry-breaking modes. The mode-pair notation is $\{m, (-m) \bmod n\}$, usually written $\{m, n-m\}$. A pair representative is chosen in $0 \leq m \leq \lfloor n/2 \rfloor$. Keep the uniform mode and, for even n, $m = n/2$ as singleton labels. Any eigenvalue-conjugacy or simplicity claim belongs to P04 or P11. When a spectrum has been derived, the selected-mode set will mean $\arg\max_m \operatorname{Re}\lambda_m$, including every tie. A first unstable mode additionally requires a specified equilibrium branch and oriented parameter path.

2.6 Reserved scalar notation

For continuity with the archive, reserve $A = k + h$ and $\mu = a + c - g$ as abbreviations, without asserting an equilibrium equation or bifurcation. Use ν for a future scalar continuation parameter and ε for a heterogeneity amplitude. Reserve $\omega > 0$ for a chosen positive Hopf frequency. Use $\operatorname{sa}(J)$ for spectral abscissa and $\delta = -\operatorname{sa}(J) > 0$ only under an explicitly stated Hurwitz hypothesis. This avoids reusing α as both a stability margin and a parameter perturbation. The spectral coefficient called T_0 in the blueprint will be named τ if needed, keeping T_{scale} available for a physical time scale. Heterogeneity profiles will be denoted collectively by $\boldsymbol{\zeta}$ if adopted; no coefficient-assignment convention is implied.

3 P01: Well-posedness and bounded absorbing sets

3.1 Domain, hypotheses, and set terminology

Use exactly the demand-modulated vector field in $(\mathrm{DM}_{n,q})$, denoted here by $F = (F_i)_{i\in\mathbb{Z}_n}$. The canonical assumptions imply the weaker hypotheses

$$n \geq 1, \quad q \in \mathbb{Z} \pmod{n}, \quad a \in \mathbb{R}, \quad k > 0, \quad c, g, h \geq 0. \tag{P01-W}$$

We prove the main results under (P01-W), without changing the canonical contract for later modules. Coincident offsets retain every term of the defining equation. Write

$$\mathcal{U} = (-1, \infty)^n, \qquad K_r = [0, r]^n \ (r \geq 0), \qquad X(t) = \max_{i\in\mathbb{Z}_n} x_i(t), \qquad s_+ = \max\{s, 0\}.$$

For a continuous real function v, define

$$D^+ v(t) = \limsup_{\tau\downarrow 0} \frac{v(t+\tau) - v(t)}{\tau}.$$

Distances use the maximum norm; for $x \in \mathbb{R}^n_{\geq 0}$,

$$\mathrm{dist}_\infty(x, K_r) = (\max_i x_i - r)_+. \tag{1}$$

A set is *positively invariant* if solutions starting there remain there for all forward times for which they exist. It *absorbs bounded subsets of* $\mathbb{R}^n_{\geq 0}$ if every bounded $E \subset \mathbb{R}^n_{\geq 0}$ has a finite T_E such that all solutions starting in E lie in the set for every $t \geq T_E$. Attraction means convergence of the distance to zero; uniform attraction of bounded sets means convergence uniform over each such E. We will also establish stronger absorption bounds with a single time valid for the entire orthant.

3.2 Finite maxima, local solutions, and coordinate boundaries

Lemma 3.1 (Finite maxima, including ties)**.** *For finitely many continuously differentiable functions y_j, set $v(t) = \max_j y_j(t)$ and $I(t) = \{j : y_j(t) = v(t)\}$. Then v is locally Lipschitz, its right derivative exists at every time with a right neighborhood in the interval, and*

$$D^+ v(t) = \max_{j\in I(t)} \dot{y}_j(t). \tag{2}$$

Proof. Bounded derivatives on compact time intervals make the finite maximum Lipschitz. At a fixed t, every nonmaximizing index has a positive gap from $v(t)$. Finiteness and continuity ensure that none of these indices maximizes at $t + \tau$ for sufficiently small $\tau > 0$. Uniformly over $I(t)$, $y_j(t+\tau) = v(t) + \tau\dot{y}_j(t) + o(\tau)$. Taking the maximum and dividing by τ proves the assertion. □

Proposition 3.2 (Local well-posedness and nonnegative invariance)**.** *For any finite real coefficients, F is C^∞ on $\mathcal{U}$. Every $x^0 \in \mathcal{U}$ has a unique maximal classical forward solution $x : [0, T_{\max}) \to \mathcal{U}$, with continuous dependence on initial data. If $c \geq 0$ and $x^0 \in \mathbb{R}^n_{\geq 0}$, then $x(t) \in \mathbb{R}^n_{\geq 0}$ for $t < T_{\max}$, regardless of the signs of the other coefficients.*

Proof. The denominators $1 + x_i$ and $1 + x_{i+q}$ are strictly positive on $\mathcal{U}$. Each component of F is a sum of polynomials and rational functions with nonvanishing denominators, hence is smooth and locally Lipschitz. Standard local ODE existence, uniqueness, and maximal continuation theory on an open domain applies.

On each coordinate face,

$$x_i = 0, \quad x \in \mathbb{R}^n_{\geq 0} \quad \Longrightarrow \quad F_i(x) = cx_{i-1} \geq 0. \tag{3}$$

To treat tangent faces rigorously, fix $T < T_{\max}$ and let $x^+(t)$ be the coordinatewise positive part of $x(t)$. The curves x, x^+ and their connecting line segments lie in a compact rectangle in $\mathcal{U}$. Choose a Lipschitz constant L for F on this rectangle, in the maximum norm, and put

$$V(t) = \max\{0, -x_0(t), \ldots, -x_{n-1}(t)\} = \left\|x(t) - x^+(t)\right\|_\infty .$$

If $V > 0$ and $-x_i = V$, then $x_i^+ = 0$, so $F_i(x^+) \geq 0$ and $-F_i(x) \leq |F_i(x) - F_i(x^+)| \leq LV$. If $V = 0$, any active coordinate has $x_i = 0$ and $-F_i(x) \leq 0$; the constant zero is also active. Lemma 3.1 gives $D^+V \leq LV$. Since V is Lipschitz, this inequality holds for its ordinary derivative almost everywhere. Integration of $(e^{-Lt}V)' \leq 0$, with $V(0) = 0$, gives $V = 0$ on $[0, T]$. The choice of T was arbitrary. □

Proposition 3.3 (Persistence and propagation of positivity)**.** *For $c \geq 0$ and nonnegative initial data, every initially positive coordinate remains strictly positive on $[0, T_{\max})$. In particular, strictly positive initial data stay strictly positive. If $c > 0$ and $x^0 \neq 0$, all coordinates are strictly positive at every $0 < t < T_{\max}$. If $c = 0$, initially zero coordinates remain zero. The zero initial state remains identically zero.*

Proof. Along a nonnegative solution write

$$\dot{x}_i = b_i(t)x_i + d_i(t), \qquad b_i = a - kx_i - \frac{g}{1 + x_{i+q}} - hx_{i-q}, \qquad d_i = \frac{cx_{i-1}}{1 + x_i} \geq 0.$$

These coefficients are continuous on every compact time interval in the maximal interval. The integrating-factor formula gives

$$x_i(t) = x_i^0 e^{\int_0^t b_i(s)\, ds} + \int_0^t e^{\int_s^t b_i(r)\, dr} d_i(s)\, ds. \tag{4}$$

This proves persistence of an initially positive coordinate. If $c > 0$, a predecessor positive at every $s > 0$ supplies a strictly positive integrand to its successor at every $s > 0$, making that successor positive at every $t > 0$. At most $n - 1$ iterations along the step-one cycle reach all nodes, independently of q. For $c = 0$ the integral vanishes, proving preservation of zero coordinates. Finally $F(0) = 0$ and uniqueness proves the zero-data assertion. □

3.3 Maximum comparison and the exact comparison radius

Under (P01-W), define

$$b(s) = a - ks + \frac{c}{1 + s}, \qquad H(s) = sb(s) \quad (s \geq 0),$$

and set

$$X_{\max} = R := \max\left\{0, \frac{a - k + \sqrt{(a + k)^2 + 4kc}}{2k}\right\}. \tag{5}$$

This radius is defined by a scalar upper comparison; it is not asserted to be the smallest invariant cube of the full vector field.

Lemma 3.4 (Dini inequality and scalar root)**.** *Every nonnegative solution under* (P01-W) *satisfies*

$$D^+X = \max_{i:x_i=X} F_i(x) \le H(X) - \frac{gX}{1+X} \le H(X). \tag{6}$$

If $a+c>0$, then $R>0$ is the unique positive zero of b, and

$$kR^2 + (k-a)R - (a+c) = 0. \tag{7}$$

If $a+c\le 0$, then $R=0$ and $H(s)\le -ks^2$ for $s\ge 0$. For $R>0$ and $s\ge R$,

$$H(s) = s\left[-k(s-R) + c\left(\frac{1}{1+s} - \frac{1}{1+R}\right)\right] \le -ks(s-R). \tag{8}$$

Proof. Lemma 3.1 includes every tied maximizing coordinate. At $x_i = X$, nonnegative invariance gives $0 \le x_{i-1}, x_{i+q}, x_{i-q} \le X$. Consequently

$$\frac{cx_{i-1}}{1+X} \le \frac{cX}{1+X}, \qquad -\frac{gX}{1+x_{i+q}} \le -\frac{gX}{1+X}, \qquad -hXx_{i-q} \le 0.$$

The middle inequality uses the negative prefactor and $g \ge 0$. Adding proves (6); no independence of neighbor coordinates is required.

Now $b'(s) = -k - c/(1+s)^2 < 0$, $b(0) = a+c$, and $b(s) \to -\infty$. If $a+c>0$, multiply $b(R)=0$ by $1+R$ to obtain (7). Its constant term is negative and its discriminant is $(a+k)^2 + 4kc$, so its positive root is (5). If $a+c\le 0$, then $k-a>0$ and

$$(a+k)^2 + 4kc - (k-a)^2 = 4k(a+c) \le 0.$$

Thus the raw root in (5) is nonpositive, $R=0$, and $b(s) \le a+c-ks \le -ks$. Subtracting $b(R)=0$ when $R>0$ gives (8); its inequality follows from $c\ge 0$ and $s\ge R$. □

Theorem 3.5 (Global continuation and invariant attracting cubes)**.** *Under* (P01-W)*, every $x^0 \in \mathbb{R}^n_{\ge 0}$ has a unique classical solution for all $t\ge 0$. The scalar solution*

$$\dot{u} = H(u), \qquad u(0) = X(0) \tag{9}$$

is global and nonnegative, and

$$0 \le X(t) \le u(t) \le \max\{X(0), R\}. \tag{10}$$

Every K_r with $r\ge R$ is positively invariant. Furthermore,

$$\operatorname{dist}_\infty(x(t), K_R) \le e^{-kRt}(X(0)-R)_+ \qquad (R>0), \tag{11}$$

$$X(t) \le \frac{X(0)}{1+kX(0)t} \qquad (R=0). \tag{12}$$

There are also initial-state-independent bounds, for $t>0$:

$$X(t) \le \frac{R}{1-e^{-kRt}} \qquad (R>0), \tag{13}$$

$$X(t) \le \frac{1}{kt} \qquad (R=0). \tag{14}$$

Thus K_R attracts all nonnegative solutions uniformly on bounded initial sets, and indeed uniformly over the entire orthant as $t\to\infty$. Every larger cube K_r, $r>R$, absorbs the entire orthant by the common time

$$T_r = \begin{cases} \dfrac{1}{kR}\log\dfrac{r}{r-R}, & R>0, \\ \dfrac{1}{kr}, & R=0. \end{cases} \tag{15}$$

Proof. The scalar field H is smooth on $(-1,\infty)$, $H(0)=0$, and $H(s)\le 0$ for $s\ge R$. Its solution starting nonnegative stays in $[0,M]$, where $M=\max\{u(0),R\}$: uniqueness prevents crossing zero, and the upper endpoint has nonpositive field. The latter barrier also follows by the positive-part argument below applied to $(u-M)_+$, using $H(M)\le 0$. Compact containment gives global scalar continuation.

We justify the vector comparison without assuming differentiability of X everywhere. On a compact subinterval of the vector maximal interval, let L be a Lipschitz constant for H on an interval containing X and u. The Lipschitz function

$$W=(X-u)_+=\max\{0,x_i-u : i\in\mathbb{Z}_n\}$$

satisfies $D^+W\le H(X)-H(u)\le LW$ where $W>0$. At $W=0$, any active coordinate has $x_i=u=X$, unless the constant zero is the only active function; Lemma 3.1 and (6) give $D^+W\le 0$. Integrating the almost-everywhere inequality, as in Proposition 3.2, yields $W=0$. This proves (10) on the maximal interval.

If $T_{\max}<\infty$, the vector trajectory stays in K_M, a compact subset of $\mathcal{U}$ at distance at least one from every singular hyperplane. The field is bounded on a compact neighborhood in $\mathcal{U}$, hence $\dot{x}$ is bounded. Thus $x(t)$ has a limit in K_M as $t\uparrow T_{\max}$. Local existence at that limit and uniqueness extend the solution past $T_{\max}$, a contradiction. This proves global continuation and absence of finite-time blow-up. If $X(0)\le r$ and $r\ge R$, (10) proves invariance.

For $R>0$, (8) gives $\dot{u}\le -ku(u-R)\le -kR(u-R)$ where $u\ge R$. It follows that $(u(t)-R)_+\le e^{-kRt}(u(0)-R)_+$, including solutions initially at or below R, and hence (11). If $u(0)>R$, set $w=1-R/u$ while $u>R$. Then

$$\dot{w}=\frac{R\dot{u}}{u^2}\le -kRw,\qquad 0<w(0)<1.$$

Solving this inequality gives (13) for u and therefore for X. If u reaches R, invariance supplies the same bound afterwards; if $u(0)\le R$, it is immediate. For $R=0$, integrating $(1/u)'\ge k$ where $u>0$, using $\dot{u}\le -ku^2$, proves (12) and (14); zero initial data follow by uniqueness.

Attraction follows from these estimates and (1). For $R>0$ the distance is at most $R/(e^{kRt}-1)$ independently of initial data; for $R=0$ use $1/(kt)$. Substitution of T_r into (13) or (14) gives $X(T_r)\le r$, and invariance supplies all subsequent times. □

3.4 Finite-time absorption at the comparison radius

Theorem 3.6 (Exact absorption criterion for K_R)**.** *Assume* (P01-W). *If $R=0$, no initially nonzero nonnegative solution enters $K_R=\{0\}$ at finite time. This cube is invariant and globally attracting but not absorbing.*

If $R>0$, the following are equivalent:

(i) K_R absorbs every nonnegative solution in finite time.

(ii) K_R absorbs bounded subsets of $\mathbb{R}^n_{\ge 0}$.

(iii) One finite absorption time is valid for the entire orthant.

(iv) The coefficients and offset satisfy

$$g>0 \quad or \quad [\,h>0 \text{ and } (c>0 \text{ or } q\equiv 0 \pmod n)\,]. \tag{16}$$

Canonically $R > 0$, $g > 0$, and the entire orthant enters K_R by

$$T_* = \frac{\log 2}{kR} + \frac{1+R}{g}. \tag{17}$$

If $g > 0$ and $X(0) \le M$, an alternative entry-time bound is

$$T_M = \frac{(M-R)_+(1+R)}{gR}. \tag{18}$$

Proof. When $R = 0$, Proposition 3.3 ensures that some coordinate of any initially nonzero solution stays positive at each finite time.

Now suppose $R > 0$ and set $t_0 = (\log 2)/(kR)$. By (13), every trajectory is in the invariant cube K_{2R} by t_0. If $g > 0$, then whenever $X \ge R$,

$$D^+X \le -\frac{gX}{1+X} \le -\frac{gR}{1+R} =: -\delta < 0, \tag{19}$$

using $H(X) \le 0$. Absolute continuity permits integration before the first entry into $X \le R$. If $X(0) \le M$, a longer time outside than $(M-R)_+/\delta$ would force $X \le R$, a contradiction. Invariance gives all later times. Applying this after t_0 with $M = 2R$ proves (17).

If $g = 0$, $h > 0$, and $q \equiv 0 \pmod n$, a maximizing coordinate has $x_{i-q} = X$. Thus $D^+X \le H(X) - hX^2 \le -hR^2$ when $X \ge R$, giving common absorption by $t_0 + 1/(hR)$.

For the remaining sufficient case $g = 0$, $h > 0$, $c > 0$, we give a coordinatewise quantitative argument. Put $M = 2R$ and fix any $\tau > 0$. On K_M,

$$\dot{x}_i \ge -\beta x_i + \eta x_{i-1}, \qquad \beta = (-a)_+ + (k+h)M + g, \quad \eta = \frac{c}{1+M} > 0. \tag{20}$$

Indeed $a \ge -(-a)_+$, $kx_i \le kM$, $g/(1+x_{i+q}) \le g$, $hx_{i-q} \le hM$, and $c/(1+x_i) \ge c/(1+M)$. On $[s, s+\tau]$ in this cube, the transformed variables $z_i(r) = e^{\beta(r-s)}x_i(r)$ satisfy

$$z_i(r) \ge x_i(s) + \eta \int_s^r z_{i-1}(v)\, dv.$$

Repeated substitution, keeping the remaining nested integral nonnegative, yields

$$x_i(s+\tau) \ge e^{-\beta\tau} \sum_{j=0}^{n-1} \frac{(\eta\tau)^j}{j!} x_{i-j}(s). \tag{21}$$

A j-fold integral of a constant over the time simplex equals $\tau^j/j!$, which gives the displayed coefficients.

For a trajectory that has not entered K_R by time $t \ge t_0 + \tau$, we have $X(t-\tau) > R$. The indices $i - j$, $0 \le j \le n-1$, cover the cycle once. Thus (21) gives, for every i,

$$x_i(t) \ge d := Re^{-\beta\tau} \min_{0 \le j \le n-1} \frac{(\eta\tau)^j}{j!} > 0. \tag{22}$$

At such times, every maximizing coordinate satisfies $F_i(x) \le H(X) - hXx_{i-q} \le -hRd$. Starting from $X \le 2R$ at $t_0 + \tau$, integration proves entry by $t_0 + \tau + 1/(hd)$, or earlier. We have proved (iv)⇒(iii).

The implications (iii)⇒(ii)⇒(i) are immediate. If (iv) fails and $g = h = 0$, choose all initial coordinates equal to $s_0 > R$. Direct substitution verifies the solution $x_i(t) = s(t)$ with $\dot{s} = H(s)$. It

stays above R at each finite time: a first meeting with the constant scalar solution R would violate uniqueness, also backwards near that meeting. Hence it never enters K_R. The other failure case is $g = 0$, $h > 0$, $c = 0$, $q \not\equiv 0 \pmod{n}$. Here $R = a/k > 0$. Choose one initial coordinate $x_j(0) > R$ and all others zero. The zero coordinates stay zero and $j - q \neq j$, so $\dot{x}_j = ax_j - kx_j^2$. It likewise stays strictly above a/k at every finite time. These cases exhaust failure of (iv) and contradict (i), completing the equivalence. □

3.5 Canonical conclusions and weaker-assumption limits

Corollary 3.7 (Canonical foundation)**.** *For canonical $n \geq 5$, $1 \leq q \leq n-1$ and $a, k, c, g, h > 0$, every nonnegative initial state has a unique global nonnegative solution. Every initially positive coordinate stays positive, and every nonzero initial state becomes strictly positive in every coordinate at each positive time. The cube $[0, X_{\max}]^n$, with*

$$X_{\max} = \frac{a - k + \sqrt{(a+k)^2 + 4kc}}{2k} > 0,$$

is positively invariant, globally attracting, and finite-time absorbing for the entire orthant with the common bound (17)*. All larger cubes are also invariant and absorbing. No condition on* $\gcd(n, q)$ *is required.*

Proof. The canonical assumptions imply (P01-W), $a + c > 0$, $c > 0$ and $g > 0$. Apply the preceding results. □

Remark 3.8 (Archived radius and the meaning of stability). The blueprint's candidate radius

$$B = \frac{a + \sqrt{a^2 + 4kc}}{2k}$$

is the larger real root of $as - ks^2 + c = 0$, and is nonnegative. If $c > 0$, then $B > 0$ and $H(B) = -c/(1 + B) < 0$, so $B > R$. Its cube is therefore also invariant and absorbing by Theorem 3.5, even when $g = h = 0$. The smaller radius R retains $cX/(1+X)$ instead of replacing it by c. If $c = 0$, then $B = R$ and the exact-radius criterion applies. Neither radius is claimed optimal after all g, h, q information is used.

Invariance concerns remaining inside; absorption concerns finite-time entry; attraction alone permits approaching from outside forever. The distance estimates also show Lyapunov stability of K_R as a set, relative to $\mathbb{R}^n_{\geq 0}$, because distance never exceeds its initial value. Under the convention that asymptotic stability of a set means Lyapunov stability plus attraction, K_R is globally asymptotically stable as a set. This is not convergence to a particular point, nor a claim that the cube is a minimal global attractor or is invariant in both time directions.

Proposition 3.9 (Global existence when $k = 0$ is allowed)**.** *Global nonnegative existence and the positivity conclusions still hold for $k \geq 0$, $a \in \mathbb{R}$, $c, g, h \geq 0$, any $n \geq 1$, and any residue q. For $k = 0$, no bounded absorbing cube is guaranteed over this parameter class.*

Proof. Local existence and nonnegative invariance did not require $k > 0$. Discarding nonpositive losses at a maximizing coordinate gives

$$D^+ X \leq aX + \frac{cX}{1+X} \leq (a_+ + c)X.$$

The Lipschitz/Dini argument yields $X(t) \leq X(0)e^{(a_+ + c)t}$. If the maximal time were finite, this would confine the trajectory to a compact subset of $\mathcal{U}$, and the earlier continuation argument would extend it. Positivity follows from Proposition 3.3. For the last assertion take $a > 0$, $k = c = g = h = 0$: an initially positive coordinate equals $x_i(0)e^{at}$ and is unbounded. □

4 P02: Synchrony and synchronous equilibria

The standing assumptions are the canonical A1–A3. In particular, $n \geq 5$, $1 \leq q \leq n-1$, $a, k, c, g, h > 0$, and the open extension domain is $\mathcal{U} = (-1, \infty)^n$. The nonnegative model state space is $\mathbb{R}^n_{\geq 0}$. Only P01's local uniqueness and, for nonnegative initial data, global continuation are needed here. The calculations below start from $(\mathrm{DM}_{n,q})$. No assertion about nonsynchronous equilibria, equilibrium stability, or bifurcation type is made.

4.1 Invariant synchronous subspace

Proposition 4.1 (Invariance of the synchronous subspace)**.** *The set*

$$\mathcal{S} \cap \mathcal{U} = \{s\mathbf{1} : s \in (-1, \infty)\}$$

is invariant for the canonical vector field for the entire maximal solution interval in $\mathcal{U}$. Its scalar restriction is

$$\dot{s} = f(s) := as - ks^2 + \frac{cs}{1+s} - \frac{gs}{1+s} - hs^2, \qquad s \in (-1, \infty). \tag{23}$$

This conclusion needs no coprimality or distinct-offset assumption. Indeed, the substitution and local invariance remain valid for any integer $n \geq 1$, any residue q modulo n, and any finite real coefficients. The canonical restrictions are retained for the equilibrium classification below.

Proof. Put $x = s\mathbf{1}$ with $s > -1$. For every $i \in \mathbb{Z}_n$, each of $x_i, x_{i-1}, x_{i+q}, x_{i-q}$ equals s. Direct substitution gives

$$F_i(s\mathbf{1}) = as - ks^2 + \frac{cs}{1+s} - \frac{gs}{1+s} - hs^2 = f(s).$$

Every component is identical, so $F(s\mathbf{1}) = f(s)\mathbf{1}$. Coincident offsets retain all the displayed terms; no distinctness of neighbors is used. Nor is the size or connectedness of a step-q cycle used.

The scalar field is smooth on $(-1, \infty)$. A scalar solution $s(t)$ lifts to the vector solution $s(t)\mathbf{1}$. Uniqueness from Proposition 3.2 makes this lift agree with the vector solution having initial value $s(0)\mathbf{1}$ wherever both exist. If the scalar maximal interval ended earlier while the vector solution continued in $\mathcal{U}$, the latter would supply a limit $s_*\mathbf{1} \in \mathcal{U}$ at that time. Local scalar continuation at $s_* > -1$ would extend the lift, a contradiction. Conversely, a longer scalar interval would extend the vector solution. Thus the maximal intervals coincide and synchrony persists throughout. The same local uniqueness argument applies on either side of an initial time. The scalar domain inherited from $\mathcal{U}$ is exactly $(-1, \infty)$. □

4.2 Scalar synchronous dynamics

Use the reserved abbreviations and one local auxiliary abbreviation

$$A = k + h > 0, \qquad \mu = a + c - g, \qquad b = a - A. \tag{24}$$

Here b is used only within P02; it is not the archived comparator's coefficient. In particular $b > -A$ because $a > 0$. The exact restriction is

$$f(s) = s\left(a - As + \frac{c-g}{1+s}\right) = \frac{s}{1+s}\big(\mu + bs - As^2\big), \qquad s > -1. \tag{25}$$

For $s(0) \geq 0$, P01 and Proposition 4.1 give a global nonnegative scalar solution. For $-1 < s(0) < 0$, only the maximal interval within this open domain is asserted. Even when $c = g$ cancels the two rational terms along synchrony, this does not enlarge the canonical domain or admit $s = -1$.

4.3 The origin and the nonzero equilibrium equation

Proposition 4.2 (Equilibrium equation with domain restrictions)**.** *The origin* $0\mathbf{1}$ *is an equilibrium for every canonical parameter tuple. Every other synchronous equilibrium in* $\mathcal{U}$ *is exactly* $s\mathbf{1}$*, where* $s \neq 0$*,* $s > -1$*, and*

$$\begin{aligned} 0 = Q(s) &:= As^2 - bs - \mu \\ &= (k+h)s^2 + (k+h-a)s - (a+c-g). \end{aligned} \tag{26}$$

Define the discriminant

$$\begin{aligned} D &:= b^2 + 4A\mu \\ &= (a-A)^2 + 4A(a+c-g) = (a+A)^2 + 4A(c-g). \end{aligned} \tag{27}$$

The complete list of real polynomial roots, when $D \geq 0$*, is*

$$s_\pm = \frac{b \pm \sqrt{D}}{2A} = \frac{a-(k+h) \pm \sqrt{(a+k+h)^2 + 4(k+h)(c-g)}}{2(k+h)}. \tag{28}$$

They are distinct for $D > 0$ *and coincide for* $D = 0$*. When* $D < 0$ *there are no real roots of* Q*.*

Proof. Directly in (23), all five terms vanish at $s = 0$. Thus the origin is always an equilibrium, before any division is performed. Now take $s \neq 0$ with $s > -1$. Dividing (25) by s and multiplying by the strictly positive number $1 + s$ gives

$$(a - As)(1+s) + c - g = \mu + (a-A)s - As^2 = 0.$$

Both operations are reversible on this stated set, giving (26). Completing the square yields

$$Q(s) = A\left(s - \frac{b}{2A}\right)^2 - \frac{D}{4A}. \tag{29}$$

Expansion of $b^2 + 4A\mu$ proves both discriminant forms and the quadratic formula gives (28). Finally,

$$Q(-1) = A + b - \mu = g - c. \tag{30}$$

Thus $s = -1$ is a polynomial root precisely when $g = c$, but it is never a point of the canonical domain. Multiplication by $1 + s$ does not authorize retaining that root. A root $s = 0$ of Q, when present, is the already counted origin rather than another equilibrium. □

4.4 Complete classification in the canonical parameter regime

In this subsection, a *mathematical root* means a real root of Q, including a zero or an inadmissible root. A *DM-domain root* additionally satisfies $s > -1$. A *nonnegative model equilibrium* has $s \geq 0$, and a *positive model equilibrium* has $s > 0$. The ever-present origin is added once to the admissible nonzero roots. All counts below count distinct values, not algebraic multiplicity.

Theorem 4.3 (Complete synchronous-equilibrium classification)**.** *Under the canonical assumptions, the synchronous equilibrium set is*

$$\{0\mathbf{1}\} \ \cup \ \{s\mathbf{1} : Q(s) = 0,\ s > -1,\ s \neq 0\}. \tag{31}$$

The following eleven cases are exhaustive. In the table, r *counts real nonzero polynomial roots,* d *counts those with* $s > -1$*,* N_0 *counts all nonnegative synchronous equilibria (including the origin), and* N_+ *counts the strictly positive ones. The total number of synchronous equilibria in* $\mathcal{U}$ *is* $1 + d$*.*

Case	*Parameter conditions*	*Real roots of Q and location*	r	d	N_0	N_+
I	$\mu > 0,\ g < c$	$s_- < -1 < 0 < s_+$	*2*	*1*	*2*	*1*
II	$\mu > 0,\ g = c$	$s_- = -1,\ s_+ = a/A > 0$	*2*	*1*	*2*	*1*
III	$\mu > 0,\ g > c$	$-1 < s_- < 0 < s_+$	*2*	*2*	*2*	*1*
IV	$\mu = 0,\ b > 0$	$s_- = 0,\ s_+ = b/A > 0$	*1*	*1*	*2*	*1*
V	$\mu = 0,\ b = 0$	$s_- = s_+ = 0$ *(double)*	*0*	*0*	*1*	*0*
VI	$\mu = 0,\ b < 0$	$-1 < s_- = b/A < 0 = s_+$	*1*	*1*	*1*	*0*
VII	$\mu < 0,\ D < 0$	*No real roots; any sign of b*	*0*	*0*	*1*	*0*
VIII	$\mu < 0,\ D > 0,\ b > 0$	$0 < s_- < s_+$	*2*	*2*	*3*	*2*
IX	$\mu < 0,\ D = 0,\ b > 0$	$s_- = s_+ = b/(2A) > 0$	*1*	*1*	*2*	*1*
X	$\mu < 0,\ D > 0,\ b < 0$	$-1 < s_- < s_+ < 0$	*2*	*2*	*1*	*0*
XI	$\mu < 0,\ D = 0,\ b < 0$	$-\frac{1}{2} < s_- = s_+ = b/(2A) < 0$	*1*	*1*	*1*	*0*

When $\mu < 0$ and $b = 0$, necessarily $D = 4A\mu < 0$, so case VII applies. Each appearance of $s_\pm$ refers to (28)*. The only repeated polynomial roots are in cases V, IX and XI.*

As zeros of the original smooth scalar field f on $(-1, \infty)$, all nonzero equilibria are simple except the double roots in IX and XI. The origin is simple if $\mu \neq 0$, double if $\mu = 0$, $b \neq 0$, and triple if $\mu = b = 0$. The algebraic continuation of the nonzero equilibrium curve meets the origin exactly at $\mu = 0$: through s_- in case IV, through s_+ in case VI, and through both coincident root labels in case V. No negative root represents a state in $\mathbb{R}^n_{\geq 0}$.

Proof. Proposition 4.2 proves completeness of (31). The Vieta relations are

$$s_- + s_+ = \frac{b}{A}, \qquad s_- s_+ = -\frac{\mu}{A}. \tag{32}$$

We first establish the domain test for every real pair of roots. Because $a > 0$ and $A > 0$,

$$s_+ \geq \frac{b}{2A} = \frac{a}{2A} - \frac{1}{2} > -\frac{1}{2} > -1.$$

For the smaller root,

$$s_- > -1 \iff \sqrt{D} < a + A \iff D < (a + A)^2 \iff g > c. \tag{33}$$

The middle equivalence uses $a + A > 0$ and $D \geq 0$. Equality gives $s_- = -1$ exactly when $g = c$; the reverse inequality gives $s_- < -1$ exactly when $g < c$. These are polynomial-root statements on the real-discriminant set, not assertions that such roots are admissible.

If $\mu > 0$, then $D > b^2$, so $s_- < 0 < s_+$ and both roots are nonzero and simple. The domain test gives I–III. When $g = c$, $D = (a + A)^2$ and the stated roots $-1, a/A$ follow directly.

If $\mu = 0$, then $Q(s) = s(As - b)$. If $b \neq 0$, its distinct roots are 0 and b/A. Since $b/A = a/A - 1 > -1$, the latter is always domain-admissible and has the sign of b. Their ordering gives IV and VI. If $b = 0$, $Q(s) = As^2$ gives V. The origin is counted only once in the equilibrium set in all three cases.

If $\mu < 0$, real roots exist precisely when $D \geq 0$. For $D < 0$ we obtain VII. For $D \geq 0$, the product in (32) is strictly positive and $b \neq 0$ (otherwise $D < 0$). Thus both roots are nonzero with the sign of their sum, namely the sign of b. They are distinct for $D > 0$ and double for $D = 0$. Also $g > a + c > c$, so both are admissible by (33). This gives VIII–XI, including the sharper negative repeated-root bound from $b > -A$. These arguments yield every count in the table.

On $s > -1$, the identity $f(s) = -sQ(s)/(1 + s)$ shows that the nonzero zeros have the same multiplicity as their polynomial roots. At the origin, use (25) without dividing: the numerator is

$\mu s + bs^2 - As^3$ and $1 + s$ does not vanish. This gives the stated orders one, two and three. Finally $Q(0) = -\mu$, so the nonzero equation's algebraic continuation contains $s = 0$ exactly when $\mu = 0$. The formulas at that locus give its root labels, and the local branch description below confirms the approach by nonzero roots. The counting and multiplicity assertions are purely algebraic and carry no stability conclusion. □

4.5 Positive synchronous equilibria: necessary and sufficient conditions

Theorem 4.4 (Exact positive-equilibrium alternatives)**.** *In the canonical parameter regime the following four alternatives partition parameter space:*

(i) There is no positive synchronous equilibrium if and only if

$$[\,\mu = 0,\ b \le 0\,] \quad \text{or} \quad [\,\mu < 0,\ (b \le 0 \text{ or } D < 0)\,]. \tag{34}$$

(ii) There is exactly one positive synchronous equilibrium and it is simple if and only if $\mu > 0$, or $\mu = 0$ and $b > 0$. It is $s_+\mathbf{1}$; in the second case $s_+ = b/A$.

(iii) There are exactly two distinct positive synchronous equilibria if and only if $\mu < 0$, $b > 0$, and $D > 0$. They are $s_-\mathbf{1}, s_+\mathbf{1}$, both simple.

(iv) There is exactly one positive synchronous equilibrium and it is repeated if and only if $\mu < 0$, $b > 0$, and $D = 0$. It is a double root, $s_-\mathbf{1} = s_+\mathbf{1} = (b/(2A))\mathbf{1}$.

Consequently, exactly one distinct positive equilibrium (without a simplicity qualification) occurs if and only if (ii) or (iv) holds. There are never more than two positive synchronous equilibria.

Proof. The cases of Theorem 4.3 prove every implication, including the necessity and all equalities. One can also reconstruct the positive-root count directly from $H(s) = \mu + bs - As^2$ for $s > 0$. If $b \le 0$, then $H'(s) = b - 2As < 0$ on this interval: it has one positive zero exactly when $\mu > 0$. If $b > 0$, its maximum on $s \ge 0$ is at $s = b/(2A) > 0$ and has value $D/(4A)$. For $\mu < 0$, that maximum is positive, zero, or negative precisely in alternatives (iii), (iv), or (i). For $\mu = 0$, the zero at the endpoint is excluded from the positive count and the other root is $b/A > 0$. For $\mu > 0$, the opposite-sign Vieta product leaves one positive root. This gives an independent sign-and-vertex justification of the list. □

4.6 Algebraic branch loci and degeneracies

Proposition 4.5 (Branch-intersection and repeated-root loci)**.** *For fixed positive a, k, c, h, define*

$$g_f := a + c + \frac{(a-A)^2}{4A} = c + \frac{(a+A)^2}{4A}, \qquad s_f := \frac{a-A}{2A}. \tag{35}$$

The following statements concern only the scalar equilibrium equation.

(i) The branch-intersection locus *is $\mu = 0$, or $g = a + c$. It is the set where the algebraic continuation of the nonzero equilibrium equation meets the origin branch.*

(ii) The repeated-root locus *of Q is $D = 0$, or $g = g_f$, and its root is s_f. Indeed,*

$$D = 4A(g_f - g), \qquad g_f - (a + c) = \frac{(a-A)^2}{4A} \ge 0. \tag{36}$$

The root is positive for $a > A$, negative for $0 < a < A$, and zero for $a = A$. It always lies in $(-1/2, \infty)$ and hence in the DM domain. It is a nonzero double equilibrium only when $a \ne A$.

(iii) The two parameter loci meet exactly at the algebraic degeneracy $a = A$, $g = a + c$. *There* $Q(s) = As^2$ *and*

$$f(s) = -\frac{As^3}{1+s}. \tag{37}$$

The origin is the sole synchronous equilibrium in the domain at that parameter tuple; it is a triple zero of f *and a double root of* Q*, not an additional nonzero equilibrium.*

Proof. The nonzero equation can be written, without approximation, as

$$\mu = As^2 - bs, \qquad g = a + c + bs - As^2 = g_f - A(s - s_f)^2. \tag{38}$$

This is its algebraic continuation, to be filtered by $s > -1$, $g > 0$, and $s \neq 0$ when counting nonzero model equilibria. At $s = 0$ it gives exactly $g = a + c$. Completing the square or using (29) gives $D = 0$, $g = g_f$, and $s = s_f$ as the unique repeated-root condition. The difference of the two parameter values in (36) vanishes exactly at $a = A$. Since $a > 0$, $s_f = a/(2A) - 1/2 > -1/2$. At the common locus, substitution into (25) proves (37) and all the remaining assertions. □

For additional local algebra at the origin, fix a, A, c and vary g (equivalently μ). If $b \neq 0$, the root that tends to zero is

$$s_0(\mu) = \frac{b - \operatorname{sgn}(b)\sqrt{b^2 + 4A\mu}}{2A} = -\frac{\mu}{b} + \frac{A\mu^2}{b^3} + O(\mu^3). \tag{39}$$

This follows by expansion of the explicit square root near $\mu = 0$; $g = a + c - \mu$ remains positive in a sufficiently small neighborhood. Thus nonzero domain-admissible roots approach the origin from the neighboring parameter regions. At the intersection itself,

$$f(s)\big|_{\mu=0} = \frac{s^2(b - As)}{1+s} = bs^2 - as^3 + O(s^4). \tag{40}$$

The coefficient of the usual quadratic scalar term is $b = a - A$. It vanishes when $a = A$; this is precisely the degeneracy above.

More explicitly, for $a = A$ the nonzero polynomial equation is $As^2 = \mu$. For $\mu > 0$ its roots are

$$s_\pm = \pm\sqrt{\mu/A}. \tag{41}$$

The negative one is admissible exactly when $0 < \mu < A$ (equivalently $c < g < a + c$); it is -1 when $\mu = A$, and below -1 when $\mu > A$. At $\mu = 0$ both labels give the origin; for $\mu < 0$ no real nonzero roots exist. These are algebraic branch descriptions only. Any bifurcation theorem requires the separate work assigned to P09 and P10; none is asserted here.

4.7 Canonical parameter corollary

Corollary 4.6 (Positive branches in the original canonical parameters)**.** *Let* $a, k, c, g, h > 0$, $A = k + h$*, and let* g_f *be as in* (35)*. If* $a \leq k + h$*, there is one simple positive synchronous equilibrium exactly for* $0 < g < a + c$*, and none for* $g \geq a + c$*. If* $a > k + h$*, the exact alternatives are*

Parameter range	*Positive synchronous equilibria*
$0 < g \leq a + c$	*One simple equilibrium,* $s_+\mathbf{1}$
$a + c < g < g_f$	*Two distinct simple equilibria,* $s_-\mathbf{1}, s_+\mathbf{1}$
$g = g_f$	*One double equilibrium,* $s_f\mathbf{1}$
$g > g_f$	*None*

In particular, $a+c<g$ does not by itself preclude positive synchrony: two distinct positive equilibria exist if and only if

$$a > k+h, \qquad a+c < g < a+c+\frac{(a-k-h)^2}{4(k+h)}. \tag{42}$$

The upper equality gives one repeated positive equilibrium when $a > k+h$; the lower equality gives one simple positive equilibrium and the origin. The origin exists in every row, independently of the displayed positive count. There is no dependence on n, q, $\gcd(n,q)$, or distinctness of the offsets.

Proof. The sign of b is the sign of $a-(k+h)$; the sign of μ is the sign of $a+c-g$; and $D = 4A(g_f-g)$ with $A>0$. Substitute these identities into Theorem 4.4. □

4.8 Weaker assumptions and limits of the classification

The synchronous substitution and the scalar identity require only homogeneity, finite real coefficients, and nonvanishing denominators. They therefore apply under P01's weaker assumptions as well. For $A>0$, the polynomial formula, discriminant, Vieta relations, and the positive-root alternatives of Theorem 4.4 remain valid for real a, c, g: their positive-root proofs use only $A>0$. This observation is not a generalization of the canonical negative-root table. In particular, the domain test (33) uses $a+A>0$, and the bound $s_f > -1/2$ uses $a>0$. P01 allows real a and does not supply these inequalities. The complete domain-filtered classification in this module is intentionally restricted to the canonical positive parameter regime. No classification for $A=0$ or $A<0$ is claimed.

4.9 Checks and possible failure points

Root substitution and discriminant reconstruction. For $D \geq 0$, put $d=\sqrt{D}$ and $r = (b+\varepsilon d)/(2A)$, $\varepsilon \in \{-1,1\}$. Direct expansion gives

$$\mu + br - Ar^2 = \mu + \frac{2b(b+\varepsilon d)-(b+\varepsilon d)^2}{4A} = \mu + \frac{b^2-d^2}{4A} = 0.$$

For $r \neq -1$, substitution in the original scalar expression gives

$$ar - kr^2 + \frac{cr}{1+r} - \frac{gr}{1+r} - hr^2 = \frac{r}{1+r}(\mu + br - Ar^2) = 0.$$

Only $r > -1$ is retained as a DM equilibrium. If $r=-1$, the two original rational terms are undefined even when $c=g$, so no substitution or canceled extension admits that point. At $\mu=0$, the specialized root $r=b/A$ also gives $br-Ar^2=0$. At $D=0$, $r=s_f=b/(2A)$ gives $\mu+br-Ar^2 = \mu+b^2/(4A)=0$. At $g=c$, the retained root a/A gives $ar-Ar^2=0$. At $b=0$, the square-root branches give $\mu-Ar^2=0$. Finally, direct substitution at $r=0$ gives zero for every parameter tuple. These checks include every exact root formula displayed above, with s_0 in (39) being one of $s_\pm$; its truncated expansion is not used as a root.

Independently expanding the polynomial discriminant gives

$$(A-a)^2 - 4A[-(a+c-g)] = a^2+2aA+A^2+4A(c-g) = (a+A)^2+4A(c-g).$$

Also the sum of the two explicit roots is b/A, and their product is $(b^2-D)/(4A^2) = -\mu/A$, confirming both Vieta relations.

Exact representative cells and equality boundaries. The following rational tuples realize every row of the classification. Set $k = h = 1/2$, hence $A = 1$, and $c = 1$ throughout. All original coefficients are strictly positive. Roots are obtained by the formulas already proved; the counts refer to the preceding table.

Case	a	g	μ	D	(r, d, N_0, N_+)
I	2	1/2	5/2	11	$(2, 1, 2, 1)$
II	2	1	2	9	$(2, 1, 2, 1)$
III	2	2	1	5	$(2, 2, 2, 1)$
IV	2	3	0	1	$(1, 1, 2, 1)$
V	1	2	0	0	$(0, 0, 1, 0)$
VI	1/2	3/2	0	1/4	$(1, 1, 1, 0)$
VII	1	3	-1	-4	$(0, 0, 1, 0)$
VIII	2	25/8	$-1/8$	1/2	$(2, 2, 3, 2)$
IX	2	13/4	$-1/4$	0	$(1, 1, 2, 1)$
X	1/2	49/32	$-1/32$	1/8	$(2, 2, 1, 0)$
XI	1/2	25/16	$-1/16$	0	$(1, 1, 1, 0)$

Cases II, IV–VI, IX and XI test all relevant equality surfaces, including the simultaneous equality $a = A$, $g = a + c$. For $b < 0, 0, > 0$ inside the $\mu > 0$ cases, one may respectively take $a = 1/2, 1, 2$, and $g = 1/2, 1, 1 + a/2$ for I, II, III. For all three signs of b in VII, take the same three a values and $g = g_f + 1$. These choices also check that omitted sign subcases do not alter the counts. Internal exact-arithmetic checks supplement these displayed examples; they are not numerical dependencies or independent verification of the module.

For reproducibility of that finite check, take $V = \{1/4, 1/2, 1, 2, 3, 4\}$ and all $(a, A, c, g) \in V^4$. For each $(a, A, c) \in V^3$, also take $g \in \{c, a + c, g_f\} + \{-\eta, 0, \eta\}$, where $\eta = \min\{a, A, c, 1\}/64$. The union contains 2,868 distinct positive tuples. Set $k = A/3$, $h = 2A/3$ for this grid; use the displayed $k = h = 1/2$ for the eleven table examples. Exact arithmetic in $u + v\sqrt{D}$, with rational u, v, D, checks root residuals, Vieta relations, domain comparisons, and every positive-count equivalence and its converse. Including the eleven table examples, 4,438 original scalar residuals vanish exactly and 217 occurrences of $s = -1$ are explicitly excluded. All eleven cells and all equality surfaces are represented. These finite checks found no counterexample; they supplement rather than establish the exhaustive proof.

Boundary continuity. The ordered formulas $s_\pm$ are continuous on the parameter set $D \geq 0$, since $A > 0$. At $D \downarrow 0$ they both tend to s_f; for $D < 0$ their absence as real roots is recorded, not bridged by a real continuation. At $\mu \to 0$ with fixed $b > 0$, $(s_-, s_+) \to (0, b/A)$; with fixed $b < 0$, $(s_-, s_+) \to (b/A, 0)$. With $b = 0$ and $\mu \downarrow 0$, both labels tend to zero as in (41). At $g \to c$, $(s_-, s_+) \to (-1, a/A)$; the first root is excluded from the DM domain at the boundary. At $b \to 0$ all limits within $D \geq 0$ are again given by the same continuous formulas. The sign-specific table is therefore compatible with every boundary, although admissible root counts can change there.

Logical completeness and attempted counterexamples. The origin was retained before division, and every other equilibrium is equivalent to a nonzero root of Q in $s > -1$. This proves that none was lost or introduced. In particular, the excluded root -1 at $g = c$ and the zero root at $\mu = 0$ cannot be counted as additional nonzero equilibria. Positivity, discriminant sign, and root order partition all parameter tuples as in the theorem; each implication in the necessary-and-sufficient statements is checked in every cell. The independent positive-half-line argument supplies a second check. Case VIII is an explicit canonical counterexample to the rejected claim that $a + c \le g$ rules out positive synchrony. Case IV is a counterexample even at equality. Case X shows that two admissible nonzero roots need not be positive; case XI shows that a repeated domain root need not be positive. Case V prevents counting a repeated zero as a positive equilibrium, and case II prevents admitting a canceled denominator root. All these distinctions are essential.

5 P03: Origin and homogeneous linearization

Use the canonical demand-modulated field ($\mathrm{DM}_{n,q}$), with $n \ge 5$, $1 \le q \le n-1$, $a, k, c, g, h > 0$, and $\mathcal{U} = (-1, \infty)^n$. All indices and equalities of indices are modulo n. Write $\mu = a+c-g$ and $A = k+h$ as in the project notation. P01 supplies smoothness on $\mathcal{U}$; P02 supplies the synchronous restriction and, only where explicitly invoked below, its equilibrium relation. No formula is imported from the archived blueprint. The stability equivalences below concern the *linearized system*. Nonlinear behavior at a zero spectral abscissa and bifurcation type are reserved for P09. The positive-state Fourier spectrum and the homogeneous normality theorem are reserved for P04.

5.1 General Jacobian

Let $\delta^{(n)}_{j\ell}$ be one if $j \equiv \ell \pmod n$ and zero otherwise. These are modular Kronecker symbols, not mutually exclusive case labels.

Lemma 5.1 (Jacobian entries)**.** *For every $x \in \mathcal{U}$ and $i, j \in \mathbb{Z}_n$, $J(x) = DF(x)$ has entries*

$$\begin{aligned} J_{ij}(x) = &\left(a - 2kx_i - \frac{cx_{i-1}}{(1+x_i)^2} - \frac{g}{1+x_{i+q}} - hx_{i-q} \right) \delta^{(n)}_{ji} \\ &+ \frac{c}{1+x_i}\delta^{(n)}_{j,i-1} + \frac{gx_i}{(1+x_{i+q})^2}\delta^{(n)}_{j,i+q} - hx_i\delta^{(n)}_{j,i-q}. \end{aligned} \tag{43}$$

All contributions are added when indices coincide; all other entries are zero. The three off-diagonal indices differ from i, but need not differ from each other.

Proof. Differentiate the five terms of ($\mathrm{DM}_{n,q}$) by the product and chain rules. In particular the denominator derivative of $-gx_i/(1+x_{i+q})$ is $+gx_i/(1+x_{i+q})^2$. When indices coincide, differentiation adds every contribution, as encoded by the modular Kronecker symbols. □

For a second formulation, put

$$D_i = a - 2kx_i - \frac{cx_{i-1}}{(1+x_i)^2} - \frac{g}{1+x_{i+q}} - hx_{i-q}, \quad C_i = \frac{c}{1+x_i}, \quad G_i = \frac{gx_i}{(1+x_{i+q})^2}, \quad H_i = hx_i.$$

The directional derivative is

$$(DF(x)z)_i = D_iz_i + C_iz_{i-1} + G_iz_{i+q} - H_iz_{i-q}. \tag{44}$$

Since $(P^r z)_i = z_{i-r}$ for every integer r, this gives

$$J(x) = \operatorname{diag}(D_i) + \operatorname{diag}(C_i)P + \operatorname{diag}(G_i)P^{-q} - \operatorname{diag}(H_i)P^q. \tag{45}$$

The diagonal factors are on the left because their coefficients belong to receiving row i. This general matrix need not be circulant.

5.2 Linearization at the origin

Direct substitution gives $F(0) = 0$. In (43), $D_i(0) = a - g$, $C_i(0) = c$, and $G_i(0) = H_i(0) = 0$. Both derivatives of the bilinear h term contain a state coordinate and vanish. The denominator-coordinate (target) derivative of the input contains x_{i-1} and vanishes; that of the rational loss contains x_i and vanishes as well. The numerator derivative of the loss survives as $-g$ on the diagonal. Thus, including every collision case,

$$\boxed{J(0) = (a - g)I + cP.} \tag{46}$$

Proposition 5.2 (Origin spectrum)**.** *With the project's DFT convention $\theta_m = 2\pi m/n$ and $v_j^{(m)} = n^{-1/2} e^{\mathrm{i}\theta_m j}$, the complete spectrum is*

$$\boxed{\lambda_m(0) = a - g + ce^{-\mathrm{i}\theta_m}, \qquad 0 \le m < n.} \tag{47}$$

Every eigenvalue is algebraically and geometrically simple over $\mathbb{C}$. Its real and imaginary parts are

$$\operatorname{Re}\lambda_m(0) = \mu - c(1 - \cos\theta_m), \qquad \operatorname{Im}\lambda_m(0) = -c\sin\theta_m. \tag{48}$$

The unique mode attaining the largest real part is $m = 0$, and

$$\boxed{\operatorname{sa}(J(0)) = \lambda_0(0) = \mu = a + c - g.} \tag{49}$$

Proof. Apply Gray's circulant diagonalization theorem [1, Theorem 3.1] to (46). With our positive-exponent columns, $Pv^{(m)} = e^{-\mathrm{i}\theta_m} v^{(m)}$; Gray's opposite exponent is reconciled by relabelling $m \mapsto -m$. Distinct roots of unity and $c > 0$ give simple eigenvalues. In particular

$$\det(\lambda I - J(0)) = (\lambda - (a - g))^n - c^n. \tag{50}$$

Taking real and imaginary parts gives the stated formulas. Since $1 - \cos\theta_m > 0$ for every $m \neq 0$, only $m = 0$ attains μ. □

5.3 Origin stability and the nonhyperbolic boundary

Proposition 5.3 (Exact linear stability and hyperbolicity conditions)**.** *For $\dot{z} = J(0)z$, zero is asymptotically (indeed exponentially) stable if and only if $\mu < 0$, and unstable if and only if $\mu > 0$. At $\mu = 0$ it is Lyapunov stable but not asymptotically stable. The origin of the nonlinear field is nonhyperbolic if and only if*

$$g = a + c\cos\frac{2\pi m}{n} \quad \text{for some } m \in \{0, \dots, n-1\}, \tag{51}$$

with $g > 0$ as always. Nonhyperbolicity is therefore not restricted to the stability boundary $\mu = 0$: every locus in (51) with $m \neq 0$ has $\mu > 0$.

Proof. In the orthonormal basis of Proposition 5.2, each coefficient solves $\dot{\widehat{z}}_m = \lambda_m(0)\widehat{z}_m$. For $\mu < 0$, all coefficients decay exponentially, with $\|z(t)\|_2 \le e^{\mu t}\|z(0)\|_2$ for $t \ge 0$. For $\mu > 0$, the real synchronous solution $z(t) = \varepsilon e^{\mu t}\mathbf{1}$ proves instability for arbitrarily small $\varepsilon > 0$. For $\mu = 0$, every nonsynchronous coefficient decays and the synchronous coefficient is constant. Therefore $\|z(t)\|_2 \le \|z(0)\|_2$, while nonzero constant synchronous solutions prevent attraction. This also proves both converses without a nonlinear stability theorem. The definition of hyperbolicity and (48) give (51). For $m \neq 0$ its value of μ is $c(1-\cos\theta_m) > 0$. □

The spectral-abscissa sign and the linear stability threshold depend on neither n, q, nor $\gcd(n,q)$. Individual eigenvalues and the *full set* of nonhyperbolic loci depend on n, but not on q or a coprimality condition. For example, $n = 6$, $a = 2$, $c = 1$, $g = 5/2$ gives $\mu = 1/2 > 0$ and purely imaginary eigenvalues in modes $1, 5$; mode zero is already unstable. This is not a Hopf claim. More generally, using $0 \le m \le \lfloor n/2 \rfloor$ as pair representatives, each interior representative in (51) contributes exactly the two simple nonzero imaginary eigenvalues in modes $m, n-m$. Strict decrease of the cosine on $[0,\pi]$ excludes any other simultaneous zero-real-part mode. The endpoint cases are treated below.

Proposition 5.4 (Simple zero at the stability boundary)**.** *At $g = a + c$ the only zero eigenvalue is $\lambda_0(0) = 0$. Its algebraic and geometric multiplicities are both one, with kernel* $\operatorname{span}\{\mathbf{1}\}$*. Every nonzero mode has strictly negative real part; none can simultaneously have zero real part.*

Proof. At $g = a + c$, (47) becomes $c(e^{-\mathrm{i}\theta_m} - 1)$. Only $m = 0$ is zero, and every other real part is negative. Simplicity follows from Proposition 5.2; $Pz = z$ is equivalent to $z \in \operatorname{span}\{\mathbf{1}\}$. □

Remark 5.5 (A zero eigenvalue away from the stability boundary). Over *all* canonical parameter values, a nonzero-mode zero is possible. Its imaginary part requires $\sin\theta_m = 0$; the only nonzero possibility is even n and $m = n/2$. The real part then vanishes exactly at $g = a - c$, which is canonical precisely when $a > c$. That zero is also simple, but $\mu = 2c > 0$, so the origin is already linearly unstable. For example, $a = 2$, $c = g = 1$ works in every even canonical dimension. Thus $c > 0$ excludes nonzero-mode zeros at $g = a + c$, not at every parameter value. For odd n there is no nonzero-mode zero.

These are spectral and linear-system conclusions. "Nonhyperbolic" and "linearly unstable" are not disjoint categories. The Hurwitz criterion is not an assertion that nonlinear asymptotic stability holds if and only if $\mu < 0$. Linearization alone does not decide nonlinear behavior at $\mu = 0$, either in $\mathcal{U}$ or relative to $\mathbb{R}^n_{\ge 0}$. The simple zero does not establish a transcritical bifurcation or a nonlinear nondegeneracy condition.

5.4 Jacobian at a synchronous state

Let $x = x^*\mathbf{1}$ be any synchronous state with $x^* > -1$ and set $s = 1 + x^* > 0$. Here s denotes the positive denominator, not the scalar state variable denoted by s in P02.

Proposition 5.6 (Homogeneous Jacobian)**.** *At every such state, whether or not it is an equilibrium, define*

$$d = a - 2kx^* - \frac{cx^*}{s^2} - \frac{g}{s} - hx^*, \tag{52}$$

$$C = \frac{c}{s}, \qquad G = \frac{gx^*}{s^2}, \qquad H = hx^*. \tag{53}$$

Then

$$\boxed{J(x^*\mathbf{1}) = dI + CP + GP^{-q} - HP^q.} \tag{54}$$

Its entries are $d\delta^{(n)}_{ji} + C\delta^{(n)}_{j,i-1} + G\delta^{(n)}_{j,i+q} - H\delta^{(n)}_{j,i-q}$*, with sums retained at coincident indices.*

Proof. Substitute $x_i = x^*$ in (43). The index $i+q$ corresponds to P^{-q} and $i-q$ to P^q; coincident terms still add. □

Always $C > 0$. If $x^* > 0$, then $G, H > 0$, and the separate neighbor contributions have signs $+C, +G, -H$. If $x^* = 0$, then $G = H = 0$; if $-1 < x^* < 0$, then $G, H < 0$. No fixed sign for d at an arbitrary synchronous state is assumed. Signs at colliding offsets must be assessed after addition.

Optional equilibrium simplification using P02. Only now suppose $x^* \neq 0$ and $x^*\mathbf{1}$ is an equilibrium. P02's (25), divided by this nonzero state, gives

$$a - (k+h)x^* + \frac{c-g}{1+x^*} = 0. \tag{55}$$

Substitution of $a = (k+h)x^* - (c-g)/s$ in the unsimplified diagonal coefficient yields

$$d = -kx^* - \frac{c}{s} - \frac{cx^*}{s^2} = -kx^* - \frac{c(1+2x^*)}{s^2}. \tag{56}$$

This applies in particular at a positive synchronous equilibrium. It is not available at an arbitrary state or at the origin merely because the origin is an equilibrium. No stability conclusion is drawn.

5.5 Circulant representation and offset collisions

Since $P^n = I$, (54) is the ordinary polynomial $dI + CP + GP^{n-q} - HP^q$. Each power of P has entries depending only on the cyclic difference $i - j$; their linear combination has rows obtained by cyclic shifts of the preceding row. This proves that the homogeneous Jacobian is circulant without requiring four distinct cyclic diagonals. The standard consequence is Fourier diagonalizability; the actual homogeneous spectrum and normality theorem are left to P04.

The diagonal cannot collide with a neighbor since $1 \not\equiv 0$ and $q \not\equiv 0$ modulo n. Neighbor collisions are exhausted by

$$\begin{aligned} i - 1 \equiv i - q &\iff q = 1, \\ i - 1 \equiv i + q &\iff q = n - 1, \\ i + q \equiv i - q &\iff 2q \equiv 0 \pmod{n}. \end{aligned}$$

The last case means even n and $q = n/2$. For $n \geq 5$, these three cases are mutually exclusive; no triple collision occurs. A nontrivial $\gcd(n, q)$ creates no further coincident offsets. The resulting matrices are

$$q = 1: \quad J = dI + (C - H)P + GP^{-1}, \tag{57}$$

$$q = n - 1: \quad J = dI + (C + G)P - HP^{-1}, \tag{58}$$

$$n \text{ even},\ q = n/2: \quad J = dI + CP + (G - H)P^{n/2}. \tag{59}$$

For other canonical q the three neighbor offsets are distinct. Even then a coefficient may vanish; the representation lists contributions, not a claim that every entry is nonzero.

At nonsynchronous states the same rule gives, for $q = 1$, the entry $c/(1+x_i) - hx_i$ at $j = i - 1$; for $q = n - 1$, the entry $c/(1+x_i) + gx_i/(1+x_{i-1})^2$ at $j = i - 1$; and for even n, $q = n/2$, the entry $gx_i/(1+x_{i+q})^2 - hx_i$ at $j = i + q = i - q$. The actual repeated coordinate is identified before combining contributions. At homogeneous states $C - H$ or $G - H$ can vanish through cancellation. The summed polynomial remains valid.

Differentiating P02's synchronous field also gives the consistency identity

$$f'(r) = a - 2(k+h)r + \frac{c-g}{(1+r)^2} = d + C + G - H \quad (x^* = r). \tag{60}$$

6 P04: Arbitrary-(n, q) Fourier spectrum and normality

The standing assumptions are $n \geq 5$, $1 \leq q < n$ and $a, k, c, g, h > 0$. Use the canonical domain $\mathcal{U} = (-1, \infty)^n$, indices in $\mathbb{Z}_n$, predecessor shift $(Pz)_j = z_{j-1}$ and the unitarily normalized, positive-exponent Fourier vectors from Section 2. No coprimality condition is imposed. This is supporting, known Fourier machinery specialized to the verified homogeneous Jacobian, not a novelty claim.

For any synchronous state $x^*\mathbf{1} \in \mathcal{U}$, set $s = 1 + x^* > 0$. Here s is the denominator, whereas the scalar state variable below is r. Import Proposition 5.6, without re-deriving its Jacobian:

$$J := DF(x^*\mathbf{1}) = dI + CP + GP^{-q} - HP^q, \tag{61}$$

$$d = a - 2kx^* - \frac{cx^*}{s^2} - \frac{g}{s} - hx^*, \qquad C = \frac{c}{s}, \quad G = \frac{gx^*}{s^2}, \quad H = hx^*. \tag{62}$$

All coefficients are real; $C > 0$, while G, H have the sign of x^* (and vanish at zero). The spectral statements through the synchronous-mode identity apply even when the state is not an equilibrium. At a nonstationary state they describe its Jacobian, not an autonomous linearization about an equilibrium.

6.1 Fourier diagonalization

Theorem 6.1 (Complete homogeneous Fourier spectrum)**.** *Let* $\theta_m = 2\pi m/n$, $v_j^{(m)} = n^{-1/2} e^{\mathrm{i}\theta_m j}$, *and* $Q := \mathsf{Q} = [v^{(0)} \ \cdots \ v^{(n-1)}]$. *Then* $Q^*Q = QQ^* = I$, *and the full spectrum, listed with algebraic multiplicity, is*

$$\boxed{\lambda_m = d + Ce^{-\mathrm{i}\theta_m} + Ge^{\mathrm{i}q\theta_m} - He^{-\mathrm{i}q\theta_m}, \qquad m = 0, \ldots, n-1.} \tag{63}$$

In particular,

$$Jv^{(m)} = \lambda_m v^{(m)}, \qquad Q^*JQ = \Lambda := \mathrm{diag}(\lambda_0, \ldots, \lambda_{n-1}), \qquad J = Q\Lambda Q^*. \tag{64}$$

These statements do not require distinct shift offsets, distinct numerical eigenvalues, nonzero combined coefficients, or $\gcd(n, q) = 1$.

Proof. Equation (61) is a polynomial in the cyclic shift, so Gray's theorem [1, Theorem 3.1] gives a unitary Fourier diagonalization, also when offsets or numerical eigenvalues coincide. The project's convention fixes the phases as

$$Pv^{(m)} = e^{-\mathrm{i}\theta_m} v^{(m)}, \quad P^{-q}v^{(m)} = e^{\mathrm{i}q\theta_m} v^{(m)}, \quad P^q v^{(m)} = e^{-\mathrm{i}q\theta_m} v^{(m)}. \tag{65}$$

Substitution gives (63) and (64), hence

$$\det(\zeta I - J) = \prod_{m=0}^{n-1} (\zeta - \lambda_m). \tag{66}$$

We reuse this diagonalization throughout. □

6.2 Real and imaginary dispersion relations

Writing $\lambda_m = r_m + \mathrm{i}\omega_m$, Euler's identities give

$$\boxed{r_m = d + C\cos\theta_m + (G-H)\cos(q\theta_m),} \tag{67}$$

$$\boxed{\omega_m = -C\sin\theta_m + (G+H)\sin(q\theta_m).} \tag{68}$$

Indeed, the imaginary contributions of the three shift terms are $-C\sin\theta_m$, $+G\sin(q\theta_m)$ and $+H\sin(q\theta_m)$: the last plus sign is the product of $-H$ and the negative imaginary part of $e^{-\mathrm{i}q\theta_m}$. Their real contributions are $C\cos\theta_m$, $G\cos(q\theta_m)$ and $-H\cos(q\theta_m)$. Here ω_m is a signed imaginary part, possibly zero; it is not the reserved positive frequency ω of a future Hopf theorem.

6.3 Conjugacy and multiplicities

For a label m, write $\bar{m} = (-m) \bmod n$ in $\{0, \dots, n-1\}$; in particular $\bar{0} = 0$, not an additional label n. The phases at $\bar{m}$ are conjugates of those at m, so

$$v^{(\bar{m})} = \overline{v^{(m)}}, \qquad \lambda_{\bar{m}} = \overline{\lambda_m}, \qquad r_{\bar{m}} = r_m, \quad \omega_{\bar{m}} = -\omega_m. \tag{69}$$

This verifies conjugacy explicitly from the formula and real coefficients, consistently with J being real.

A label is self-conjugate precisely when $2m = 0 \pmod{n}$. For odd n there is the singleton 0 and the $(n-1)/2$ pairs $\{m, n-m\}$ with $1 \le m \le (n-1)/2$. For even n the singletons are $0, n/2$, with $n/2-1$ pairs $1 \le m < n/2$. Each singleton necessarily has a real eigenvalue. The converse is false: a non-singleton pair can have $\omega_m = 0$, and then its two labels give the same real eigenvalue.

Proposition 6.2 (Numerical eigenvalue multiplicities)**.** *For a numerical eigenvalue η, define its label set $M_\eta = \{m : 0 \le m < n,\ \lambda_m = \eta\}$. Its algebraic and geometric multiplicities over $\mathbb{C}$ both equal $|M_\eta|$, and*

$$\ker(J - \eta I) = \operatorname{span}_{\mathbb{C}}\{v^{(m)} : m \in M_\eta\}.$$

Every eigenvalue is semisimple. Each Fourier label contributes one one-dimensional Fourier coordinate, but this does not imply that its numerical eigenvalue is simple.

Proof. Equation (66) counts the algebraic multiplicity. Write $z = Qw$; then $(J - \eta I)z = 0$ if and only if $(\lambda_m - \eta)w_m = 0$ for every m. Exactly the coordinates in M_η are free, proving the eigenspace formula and dimension. Diagonalization also excludes nontrivial Jordan blocks. □

Thus "Fourier-mode multiplicity" means the number of labels producing a given value, if that term is used at all; it is not a second source of multiplicity. A nonreal conjugate pair consists of two different numerical eigenvalues, each of which can also coincide with eigenvalues at other labels. A tie $r_m = r_\ell$ alone says nothing about equality of λ_m, λ_ℓ. Singleton labels can likewise share their numerical value with other labels; singleton refers to label conjugacy, not simplicity.

For a concrete canonical example, take $q = 1$, $x^* = 1$ and $(a, k, c, g, h) = (2, 1, 4, 4, 1)$. P02's scalar equation vanishes, and P03 gives $d = -4, C = 2, G = H = 1$. Hence $J = -4I + P + P^{-1}$ and $\lambda_m = -4 + 2\cos\theta_m$; each non-singleton pair has a repeated real eigenvalue. At the origin, by contrast, $G = H = 0$, $C = c > 0$ and $d = a - g$, so (63) reduces to P03's $a - g + ce^{-\mathrm{i}\theta_m}$. The distinct roots of unity and $c > 0$ make all those origin eigenvalues simple, as established in Proposition 5.2.

6.4 Offset collisions and self-conjugate modes

P03's exhaustive collision list requires no replacement spectrum. Direct substitution in the master formula gives the following checks, with phases understood modulo 2π.

(i) If $q = 1$, then

$$J = dI + (C - H)P + GP^{-1}, \qquad \lambda_m = d + (C - H)e^{-\mathrm{i}\theta_m} + Ge^{\mathrm{i}\theta_m}.$$

The same expression follows by combining the C and $-H$ terms in (63). Here $r_m = d + (C + G - H)\cos\theta_m$ and $\omega_m = (-C + G + H)\sin\theta_m$.

(ii) If $q = n - 1$, then $P^{-q} = P$ and $P^q = P^{-1}$. Since $e^{\mathrm{i}(n-1)\theta_m} = e^{-\mathrm{i}\theta_m}$ and $e^{-\mathrm{i}(n-1)\theta_m} = e^{\mathrm{i}\theta_m}$,

$$J = dI + (C + G)P - HP^{-1}, \qquad \lambda_m = d + (C + G)e^{-\mathrm{i}\theta_m} - He^{\mathrm{i}\theta_m}.$$

Thus $r_m = d + (C + G - H)\cos\theta_m$ and $\omega_m = -(C + G + H)\sin\theta_m$, also obtained from (67)–(68).

(iii) If n is even and $q = n/2$, then $e^{\pm\mathrm{i}q\theta_m} = e^{\pm\mathrm{i}\pi m} = (-1)^m$ and $P^{-q} = P^q = P^{n/2}$. Therefore

$$J = dI + CP + (G - H)P^{n/2}, \qquad \lambda_m = d + Ce^{-\mathrm{i}\theta_m} + (G - H)(-1)^m.$$

Here $r_m = d + C\cos\theta_m + (G - H)(-1)^m$ and $\omega_m = -C\sin\theta_m$.

These equalities include cancellations such as $C = H$ in the first case or $G = H$ in the third. Adding contributions before or after Fourier transformation gives identical results; coincident entries must never be overwritten.

The singleton formulas, for every admissible q, are

$$\lambda_0 = d + C + G - H \in \mathbb{R}, \tag{70}$$

$$\lambda_{n/2} = d - C + (G - H)(-1)^q \in \mathbb{R} \quad (n \text{ even}). \tag{71}$$

For the second formula use $e^{-\mathrm{i}\pi} = -1$ and $e^{\pm\mathrm{i}q\pi} = (-1)^q$; all sine terms vanish. It equals $d - C + G - H$ for even q and $d - C - G + H$ for odd q. When also $q = n/2$, the factor is $(-1)^{n/2}$, agreeing with the half-offset formula at $m = n/2$. Thus no real singleton critical mode is omitted from subsequent all-mode tests.

6.5 Normality

Proposition 6.3 (Normality)**.** *Every homogeneous Jacobian* (61) *is normal over* $\mathbb{C}^n$*:* $JJ^* = J^*J$*.*

Proof. The unitary diagonalization in Theorem 6.1 makes J normal, as in [1, §3.1]. □

Normality here concerns homogeneous states, including negative ones in the domain; it is not asserted for heterogeneous or nonsynchronous Jacobians.

6.6 Synchronous mode

P02's scalar field, using a fresh state variable r, is

$$f(r) = ar - (k + h)r^2 + \frac{(c - g)r}{1 + r}, \qquad r > -1.$$

Differentiate $F(r\mathbf{1}) = f(r)\mathbf{1}$ and use $v^{(0)} = \mathbf{1}/\sqrt{n}$:

$$\boxed{\lambda_0 = d + C + G - H = f'(x^*).} \tag{72}$$

This holds at any synchronous state, without the equilibrium hypothesis.

Positive synchronous equilibrium specialization. Now assume additionally $x^* > 0$ and $F(x^*\mathbf{1}) = 0$. Then $C, G, H > 0$. Put $A = k + h$ as in the canonical notation. P02 and P03 give

$$a - Ax^* + \frac{c-g}{s} = 0, \qquad d = -kx^* - \frac{c}{s} - \frac{cx^*}{s^2} = -kx^* - \frac{c(1+2x^*)}{s^2}.$$

This is an optional equilibrium-only replacement for d in (63); the unsimplified expression (62) remains valid at all synchronous states. Writing $f(r) = rB(r)$ with $B(r) = a - Ar + (c-g)/(1+r)$ and $B(x^*) = 0$ yields

$$\boxed{\lambda_0 = -x^* \left(A + \frac{c-g}{s^2} \right) = \frac{x^*}{s}\big(a - A - 2Ax^*\big).} \tag{73}$$

The first equality uses $f'(x^*) = x^* B'(x^*)$; for the second, substitute $c - g = (Ax^* - a)s$. The equilibrium identities require a nonzero equilibrium; they also hold at a negative nonzero equilibrium in $\mathcal{U}$, but $G, H > 0$ is then false. At the origin use (62), which gives $\lambda_0 = a + c - g$. No comparison with other r_m or selection of a dominant mode is made here.

6.7 Linear stability and hyperbolicity

In this subsection $x^*\mathbf{1}$ is a synchronous *equilibrium* in $\mathcal{U}$. Adopt the project's spectral-abscissa notation

$$\operatorname{sa}(J) = \max_{0 \le m < n} r_m. \tag{74}$$

For $\dot{z} = Jz$, exponential asymptotic stability means there exist $M \ge 1$ and $\beta > 0$ such that $\|z(t)\|_2 \le Me^{-\beta t} \|z(0)\|_2$ for all $t \ge 0$.

Corollary 6.4 (Exact linear stability criterion)**.** *The linearized system is exponentially asymptotically stable if and only if*

$$\boxed{\operatorname{sa}(J) < 0 \quad \Longleftrightarrow \quad d + C\cos\theta_m + (G - H)\cos(q\theta_m) < 0 \quad \textit{for every } m.} \tag{75}$$

For all $t \ge 0$, its exact induced Euclidean propagator norm is

$$\boxed{\left\| e^{tJ} \right\|_2 = e^{t \operatorname{sa}(J)}.} \tag{76}$$

If $\operatorname{sa}(J) = 0$, the linear system is Lyapunov stable but not asymptotically stable. If $\operatorname{sa}(J) > 0$, it is unstable.

Proof. Theorem 6.1 gives $e^{tJ} = Q \operatorname{diag}(e^{t\lambda_m}) Q^*$, whose induced Euclidean norm is $\max_m |e^{t\lambda_m}| = e^{t \operatorname{sa}(J)}$ for $t \ge 0$. This proves the strict criterion. At zero abscissa a maximizing real Fourier block preserves nonzero norm; at positive abscissa it grows exponentially. Real and complex induced norms agree for a real matrix, so these conclusions apply to the real system as well. □

Proposition 6.5 (Hyperbolicity)**.** *The equilibrium is hyperbolic if and only if $r_m \neq 0$ for every Fourier label. It is nonhyperbolic whenever any $r_m = 0$, including when $\omega_m \neq 0$. Invertibility requires instead $\lambda_m \neq 0$ for every label, and is strictly weaker than hyperbolicity.*

Proof. The complete spectrum is (63); substitute it in the definitions of hyperbolicity and invertibility. A nonzero purely imaginary eigenvalue meets the invertibility test but fails the hyperbolicity test. For an instance within the canonical family, P03's origin example $n = 6$, $a = 2, c = 1, g = 5/2$ (any positive k, h, any canonical q) has $r_1 = r_5 = 0$ and nonzero imaginary parts, whereas no eigenvalue is zero: $ce^{-\mathrm{i}\theta_m} = 1/2$ is impossible by modulus. Thus the weaker criterion is strictly weaker here as well. □

Hyperbolicity alone is not stability: hyperbolic equilibria may have positive real parts. Nonhyperbolicity and instability may coexist when one mode has zero real part and another has positive real part.

Nonlinear implications and their limits. The following conclusions concern the nonlinear equilibrium in the open domain $\mathcal{U}$, using P01's C^∞ smoothness there:

(i) If $\mathrm{sa}(J) < 0$, the equilibrium is locally asymptotically stable; in fact it is locally exponentially stable.

(ii) If $\mathrm{sa}(J) > 0$, it is unstable, even if some other eigenvalues have zero real part.

(iii) If $\mathrm{sa}(J) = 0$, linearization alone is generally inconclusive about nonlinear stability or attraction. Normality only settles the linear system's boundedness in this case.

These are the standard linearized stability and instability results [2, Ch. 1]: P01 supplies smoothness on the open domain, and the complete spectrum verifies the respective sign condition. The instability theorem allows other eigenvalues on the imaginary axis. At zero abscissa, a normal linearization alone does not decide nonlinear stability (compare $\dot{y} = -y^3$ and $\dot{y} = y^3$).

All stability implications above are local. They establish neither global attraction nor a global basin, even given P01's separate boundedness result. At a positive equilibrium one may choose the local ball inside $\mathbb{R}^n_{>0}$, so the local assertions apply there as well. At the origin the conclusions above are stated in $\mathcal{U}$; instability relative to a restricted state set is a separate question unless its admissible perturbations are checked. A Hurwitz Jacobian is sufficient for nonlinear local asymptotic stability, not a necessary condition in general. Thus the rejected blanket nonlinear equivalence C040 and global assertion C153 are not reinstated.

Dominant modes and first loss of stability. At a fixed parameter point and specified synchronous equilibrium, the dominant-label set is $\arg\max_{0\le m<n} r_m$, retaining all ties. This is the fixed-point "selected-mode set" reserved in Section 2; it need not contain an unstable mode. A *first unstable mode along a path* additionally needs a specified equilibrium branch $x^*(\nu)\mathbf{1}$, a specified oriented parameter path $\boldsymbol{\pi}(\nu)$, a prior interval of stability, and a loss-of-stability crossing event. For a continuous branch/path a finite boundary event from $\mathrm{sa}(J) < 0$ must have $\mathrm{sa}(J) = 0$; calling it an instability crossing also requires positive real part on the unstable side, not merely tangency at zero. Several labels or pairs can participate, so uniqueness and transverse crossing cannot be presumed. No such path, crossing, or $q = 2$ selection analysis is supplied by this module.

6.8 Arbitrary offset and the step-q subgraph

Let $\kappa = \gcd(n, q)$, $n = \kappa n'$ and $q = \kappa q'$, where $\gcd(n', q') = 1$. Repeated addition of q returns an index to itself after the least positive L for which $n \mid Lq$. This is equivalent to $n' \mid Lq'$, hence to $n' \mid L$, so $L = n/\kappa$. The step-q permutation therefore has exactly κ disjoint cycles, each of length n/κ; they are the residue classes modulo κ. Reversing the step gives the same partition. Thus coprimality is required only to call this step-q subgraph a single cycle. The full canonical coupling also includes the step-one predecessor cycle, and the step-q partition is not an assertion of full-network disconnection or of a block decomposition of J by those residue classes.

The Fourier sampling grid and label count depend on n; parity of n determines whether the $n/2$ singleton exists. The general phases and possible eigenvalue coincidences depend on q modulo n as well as n and the coefficients. The value of λ_0 is independent of n, q at fixed coefficients and state; the

half-mode value depends on the parity of q. No proof of diagonalization, conjugacy, multiplicity, normality, or the all-mode stability and hyperbolicity criteria requires coprimality. The actual spectral abscissa generally depends on the grid and phases; P03's n, q-independent origin threshold is a special case.

7 P05: Exact step-two mode selection

Fix the canonical demand-modulated model with $q = 2$, $n \geq 5$ and $a, k, c, g, h > 0$. Let $x^*\mathbf{1}$ be a specified positive synchronous equilibrium and put $s = 1 + x^* > 1$. Import only the P04 coefficients and real dispersion relation:

$$\begin{aligned} C = \frac{c}{s} > 0, \qquad G = \frac{gx^*}{s^2} > 0, \qquad H = hx^* > 0, \\ r_m = d + C\cos\theta_m + (G - H)\cos(2\theta_m), \qquad \theta_m = \frac{2\pi m}{n}. \end{aligned} \tag{77}$$

Here $r_m = \operatorname{Re}\lambda_m$ and d is P04's diagonal coefficient. All selection conclusions below are derived from (77). The maximization is over all Fourier labels, including mode zero, and selection means the complete set maximizing the real part at this equilibrium, retaining all ties. No coprimality condition is used. P01 supplies smoothness, P02 supplies the equilibrium relation, and P03–P04 supply the homogeneous coefficients. No later proof module is used.

We follow the finite Fourier-dispersion selection method used by van Kekem–Sterk for Lorenz–96 [3, §2.1, Proposition 5]. Their model-specific thresholds do not transfer to DM. Here put

$$T = G - H, \qquad \chi = \frac{4(H - G)}{C} = -\frac{4T}{C}. \tag{78}$$

The constructions below verify realizability with positive DM coefficients.

7.1 Quadratic dispersion relation

Set $u_m = \cos\theta_m$. The identity $\cos(2\theta) = 2\cos^2\theta - 1$ gives

$$r_m = R(u_m), \qquad \boxed{R(u) = d - T + Cu + 2Tu^2.} \tag{79}$$

The additive constant is $d - T$; the part controlling selection is $Cu + 2Tu^2$. In particular, differences of growth rates and the complete maximizer set depend only on C, T, not on d. This is an algebraic selection statement, not permission to vary d independently in the model.

Put $N = \lfloor n/2 \rfloor$ and $u_j = \cos(2\pi j/n)$ for $0 \leq j \leq N$. Since cosine is strictly decreasing on $[0, \pi]$,

$$1 = u_0 > u_1 > \cdots > u_N, \qquad u_N = \begin{cases} -1, & n \text{ even}, \\ -\cos(\pi/n), & n \text{ odd}. \end{cases} \tag{80}$$

Equation (77) gives $r_{n-m} = r_m$ for $1 \leq m < n$. Define the distinct label sets

$$L_j = \begin{cases} \{0\}, & j = 0, \\ \{n/2\}, & n \text{ even and } j = N, \\ \{j, n - j\}, & \text{otherwise}. \end{cases} \tag{81}$$

For a maximizing representative set S, the full dominant-label set is $\bigcup_{j \in S} L_j$, a disjoint union.

7.2 Continuous maximizer

If $T \geq 0$, then $R(1) - R(u) = (1-u)[C + 2T(1+u)] > 0$ for $-1 \leq u < 1$. If $T < 0$, R is strictly concave with vertex

$$u_* := \frac{C}{4(H-G)} = \frac{1}{\chi} > 0. \tag{82}$$

Thus zero is uniquely selected when $u_* \geq 1$; otherwise selection is the nearest-grid problem below. The continuous boundary $\chi = 1$ need not be a discrete switch.

7.3 Exact discrete selection rule

For $T < 0$, direct completion of the square gives

$$R(u) = d + (H-G) + \frac{C^2}{8(H-G)} - 2(H-G)(u-u_*)^2. \tag{83}$$

Consequently maximizing the growth rate is exactly minimizing $|u_j - u_*|$. This holds for every $u_* > 0$, including $u_* \geq 1$. Distances are in cosine space; the grid in that space is nonuniform. No angular rounding rule is invoked.

Define the strictly decreasing midpoint sequence

$$\beta_j = \frac{u_j + u_{j+1}}{2} = \cos\frac{\pi}{n}\cos\frac{(2j+1)\pi}{n}, \qquad 0 \leq j < N. \tag{84}$$

Strict decrease follows from $\beta_j - \beta_{j+1} = (u_j - u_{j+2})/2 > 0$.

Theorem 7.1 (A: Exact selection, including endpoints and all ties)**.** *Under the stated canonical equilibrium assumptions, if $T \geq 0$ then $S = \{0\}$. If $T < 0$, the selected representatives are exactly those whose closed cells below contain u_*:*

$$\begin{aligned} V_0 &= [\beta_0, \infty), \\ V_j &= [\beta_j, \beta_{j-1}] \quad (1 \leq j < N), \\ V_N &= (-\infty, \beta_{N-1}]. \end{aligned} \tag{85}$$

A representative is the unique maximizer precisely when u_* lies in the corresponding open cell. At $u_* = \beta_j$ the selected set is exactly $\{j, j+1\}$. No nonadjacent or triple representative tie can be maximal. Only positive midpoints can be attained when $T < 0$. The full Fourier-label set is given by (81)*.*

Proof. The endpoint inequality handles $T \geq 0$. For $T < 0$, square completion (83) reduces the problem to the nearest cosine. Adjacent squared distances differ by $2(u_j - u_{j+1})(\beta_j - u_*)$, so the midpoint cells are exactly (85). In an ordered one-dimensional grid the nearest points are either one point or two adjacent points at their midpoint. Thus no other maximal tie occurs; $u_* > 0$ excludes nonpositive midpoints. □

7.4 Mode ties and selection cells

The real-part comparison also has the useful exact form

$$R(u_j) - R(u_{j+1}) = C(u_j - u_{j+1})(1 - \chi\beta_j). \tag{86}$$

For $T < 0$ an adjacent *maximal* tie therefore occurs if and only if $u_* = \beta_j > 0$, equivalently $\chi = 1/\beta_j$. The word maximal is essential: adjacent equalities outside the concave regime need not be ties at the spectral abscissa.

Let $w_j = |L_j|$. A strict cell has w_j selected Fourier labels; a representative tie has $w_j + w_{j+1}$. Thus two nonsingleton representatives give four labels, a tie involving 0 gives three, and an even-n endpoint would contribute only one label. Below we prove that the $N = n/2$ endpoint is never selected here. In the canonical reachable cells every nonzero selected representative gives exactly two labels, and every reachable nonzero/nonzero switch gives four. These counts are counts of labels, not of distinct numerical eigenvalues.

7.5 Reachable Fourier representatives

Theorem 7.2 (B: Reachable cells and parity)**.** *The full list of representatives whose nearest-cosine cell intersects* $(0,1)$*, and also the full list attainable at some canonical positive synchronous equilibrium, is*

$$\boxed{0, 1, \ldots, J, \qquad J = j_{\max}(n) = \left\lfloor \frac{n+1}{4} \right\rfloor .} \tag{87}$$

Each is attainable as the unique representative. Precisely the midpoints $\beta_0, \ldots, \beta_{J-1}$ *are positive and give attainable switches. Every* $j > J$*, in particular the even-n singleton* $n/2$*, is excluded. Canonical attainability and robustness are proved in Theorem 7.3.*

Proof. For $1 \le j \le N$, the cell intersects $(0,1)$ if and only if $\beta_{j-1} > 0$. Necessity follows from its upper endpoint. Sufficiency follows by taking a point strictly between $\max(0, \beta_j)$ and β_{j-1} when $j < N$, with the endpoint case interpreted using V_N. The upper endpoint is less than 1. The cell for 0 intersects $(0,1)$ since $0 < \beta_0 < 1$. By (84) and $\cos(\pi/n) > 0$,

$$\beta_j > 0 \iff 2j + 1 < n/2 \iff j < (n-2)/4.$$

Indeed the relevant midpoint angles lie in $(0, \pi)$, where cosine changes sign only at $\pi/2$. Hence $\beta_{j-1} > 0$ is equivalent to $j < (n+2)/4$, giving exactly (87). In particular $J < N$ for every $n \ge 5$, $\beta_{J-1} > 0$, and $\beta_J \le 0$. All retained cell intersections have nonempty interiors; there is no additional representative reachable only through an isolated tie. When $T \ge 0$ or $u_* \ge 1$ only 0 is selected, so neither regime enlarges the list. □

The parity information, with $n = 4\ell + r$ in each row, is exact:

r	J	u_J	Nearest representatives at the formal point $u_* = 0$
0	ℓ	0	ℓ
1	ℓ	$\sin(\pi/(2n)) > 0$	ℓ
2	ℓ	$\sin(\pi/n) > 0$	$\ell, \ell+1$
3	$\ell+1$	$-\sin(\pi/(2n)) < 0$	$\ell+1$

To verify the last column independently, the two cosines straddling zero in the odd cases are

$$\begin{aligned} n = 4\ell + 1: &\quad u_\ell = \sin(\pi/(2n)), \quad u_{\ell+1} = -\sin(3\pi/(2n)), \\ n = 4\ell + 3: &\quad u_\ell = \sin(3\pi/(2n)), \quad u_{\ell+1} = -\sin(\pi/(2n)). \end{aligned}$$

All their arguments are in $(0, \pi/2)$, where sine increases strictly. All remaining cosines are farther from zero by monotonicity of the grid. For $4 \mid n$ there is an exact zero cosine at $j = n/4$, and its cell contains a positive interval. This representative is a conjugate pair, not the real singleton $n/2$. For $n = 4\ell + 2$, the straddling values are $\pm\sin(\pi/n)$, and $\beta_\ell = 0$.

Large restraint at fixed finite dimension. For fixed $C > 0$, as $H - G \to +\infty$, the vertex decreases to zero and J is the unique dominant representative for all sufficiently large finite $H - G$, in every parity class. The formal zero-vertex tie for $n = 4\ell + 2$ is never attained with finite positive $C, H - G$. In fact the two competing real parts have the strictly positive, $H - G$-independent difference

$$r_\ell - r_{\ell+1} = 2C \sin(\pi/n) > 0.$$

Their squared cosines are equal, so this follows directly from (79). Only a leading-order normalization by $H - G$ loses that difference. Thus a formal limiting nearest-grid tie must not be added to the finite-parameter dominant set. No $n \to \infty$ limit is taken. For $n \equiv 3 \pmod 4$, J is the sole reachable representative with a negative cosine; for the other residue classes no negative-cosine representative is reachable.

Complete regions in one dimensionless parameter. Put $\alpha_j = 1/\beta_j$ for $0 \le j < J$. Then $1 < \alpha_0 < \cdots < \alpha_{J-1}$, and for *all* real χ allowed by the canonical coefficients the entire selection rule is

Condition	S
$\chi < \alpha_0$	$\{0\}$
$\alpha_{j-1} < \chi < \alpha_j \quad (1 \le j < J)$	$\{j\}$
$\chi > \alpha_{J-1}$	$\{J\}$
$\chi = \alpha_j \quad (0 \le j < J)$	$\{j, j+1\}$.

(88)

The middle row is empty when $J = 1$. Negative and zero χ are covered by the first row. For positive χ, the inequalities follow by inverting positive midpoint bounds, reversing their order. Thus the continuous vertex boundary $\chi = 1$ is strictly inside the uniform selection region; the first discrete switch is at $\alpha_0 > 1$.

7.6 Canonical parameter realization

The concave condition and vertex in original variables are

$$T < 0 \iff h > \frac{g}{s^2}, \qquad u_* = \frac{c/s}{4x^*(h - g/s^2)} = \frac{cs}{4x^*(hs^2 - g)}, \qquad \chi = \frac{4x^*(hs^2 - g)}{cs}. \tag{89}$$

All denominators in the vertex formula are strictly positive in this regime. For $0 \le j < J$, the switch equations are exactly

$$H - G = \frac{C}{4\beta_j}, \qquad \boxed{h = \frac{g}{(1 + x^*)^2} + \frac{c}{4(1 + x^*)x^*\beta_j}.} \tag{90}$$

They are subject to P02's equilibrium condition

$$a - (k + h)x^* + \frac{c - g}{1 + x^*} = 0. \tag{91}$$

For $\beta_j \le 0$ there is no switch in the $T < 0$ regime: division by zero is not meaningful and negative β_j would require $H - G < 0$. Such negative-midpoint equalities in the convex regime, if realizable as equal lower values, are not dominant-mode switches.

Theorem 7.3 (C: Canonical realizability, open cells and switch geometry)**.** *Every strict cell and every switch in* (88) *is nonempty in the canonical positive-equilibrium parameter space. Each strict cell has a realization persisting on a nonempty open set of positive physical parameters* (a, k, c, g, h)

with a specified smooth positive synchronous branch. Both $T > 0$ and $T < 0$, and the boundary $T = 0$, are realizable. Every positive-midpoint switch is locally a smooth codimension-one hypersurface in physical parameter space along its specified branch; all its positive-equilibrium points are regular branch points. No nonpositive-midpoint dominant switch is realizable. These are local branch statements and assert no global identification of distinct branches.

Proof of realizability. Choose any $x > 0$, any $c, g > 0$, and any desired vertex $v > 0$. Define

$$s = 1 + x, \qquad h = \frac{g}{s^2} + \frac{c}{4sxv}, \qquad k > \max\left\{0, \frac{c-g}{sx} - h\right\}, \qquad a = (k+h)x - \frac{c-g}{s}. \tag{92}$$

Then h, k, a are strictly positive: the displayed strict lower bound on k is exactly what is required for $a > 0$, together with $k > 0$. There is no upper bound on k, so this choice is always possible. Equation (91) vanishes identically at $x^* = x$; substitution into the original vector field gives, in every component,

$$ax - kx^2 + \frac{cx}{s} - \frac{gx}{s} - hx^2 = x\left[a - (k+h)x + \frac{c-g}{s}\right] = 0.$$

Also $H - G = c/(4sv) > 0$ and $u_* = v$. Hence every retained open Voronoi cell and every positive midpoint can be prescribed exactly. The construction does not treat C, G, H as independent of an equilibrium.

A particularly simple exact family, used below, is

$$x^* = 1, \qquad c = g = 4, \qquad h = 1 + \chi/2, \qquad k > 0, \qquad a = k + 1 + \chi/2, \qquad \chi > -2. \tag{93}$$

Here $C = 2$, $G = 1$, $H = 1 + \chi/2$, so the dimensionless parameter is exactly the displayed χ. Every positive χ is realized. At $k = 1$, choosing $\chi = -1, 0, 1$ realizes respectively $T > 0$, $T = 0$, and $T < 0$ with $u_* = 1$, all selecting 0. Thus the uniform region has examples in both sign regimes. The geometry and openness assertions are proved next. □

7.7 Realizable switch hypersurfaces

Use the equilibrium coordinates (x, k, c, g, h) on the open set

$$\mathcal{U}_{\mathrm{eq}} = \left\{(x, k, c, g, h) \in (0, \infty)^5 : (k+h)x - \frac{c-g}{1+x} > 0\right\}. \tag{94}$$

The map

$$\Phi(x, k, c, g, h) = \left((k+h)x - \frac{c-g}{1+x},\ k, c, g, h\right)$$

projects this parameterization of the positive equilibrium graph onto physical parameters. With $s = 1 + x$ its Jacobian determinant is

$$\det D\Phi = k + h + \frac{c-g}{s^2}. \tag{95}$$

At every point with $H - G > 0$, this is $k + (h - g/s^2) + c/s^2 > 0$. Equivalently, for the scalar equilibrium function $B(x) = a - (k+h)x + (c-g)/(1+x)$, $B_x = -\det D\Phi < 0$. The inverse function theorem makes Φ a local diffeomorphism there, and the implicit function theorem gives a specified smooth positive branch $x = x^*(a, k, c, g, h)$. This regularity uses only the scalar relation, not full-system invertibility or a Hopf result.

Strict inequalities in (88) define open subsets of $\mathcal{U}_{\mathrm{eq}}$. Every strict cell has a $\chi > 0$ point from (92), including the uniform cell, so local projection supplies a nonempty open set of physical parameters for each. In fact every concave-regime point of a strict cell has this local property. For a $T > 0$ uniform example, the family at $k = 1, \chi = -1$ has $\det D\Phi = k + h = 3/2 > 0$; strict $T > 0$ persists in a neighborhood. No global openness assertion at an arbitrary scalar fold in the uniform region is required or inferred.

For $\beta_j > 0$ define in equilibrium coordinates

$$\Psi_j(x, k, c, g, h) = h - \frac{g}{(1+x)^2} - \frac{c}{4(1+x)x\beta_j}. \tag{96}$$

Its partial derivative with respect to h, holding x, k, c, g fixed, is exactly 1. Hence its zero set is a smooth codimension-one regular level set within $\mathcal{U}_{\mathrm{eq}}$. It is nonempty by the construction, and every one of its points has $H - G > 0$, so the local diffeomorphism Φ transfers this conclusion to physical parameter space on the chosen branch. Moreover, at fixed x, c, g, k,

$$\frac{\partial \chi}{\partial h} = \frac{4(1+x)x}{c} > 0, \qquad \frac{\partial}{\partial h}(r_j - r_{j+1}) = -4x(u_j - u_{j+1})\beta_j < 0.$$

Here a varies according to (91); this is a legitimate direction in physical parameter space. Distinct midpoint thresholds and the strict gap to all other representatives at the tie show that the two sides locally select j and $j+1$, respectively. This proves genuine local changes of dominance, and finishes Theorem 7.3. No global manifold structure after forgetting branch labels is claimed.

7.8 Exact examples and small-dimension checks

In (93) take $k = 1$. The tuple $(a, k, c, g, h) = (2, 1, 4, 4, 1)$ selects 0 in every dimension. For $n = 8$, $(3, 1, 4, 4, 2)$ has $\chi = 2$, $u_* = 1/2$, and selects representative 1, since $\sqrt{2}/4 < 1/2 < (2+\sqrt{2})/4$. For $n = 6$, $(8/3, 1, 4, 4, 5/3)$ has $\chi = 4/3 = 1/\beta_0$ and selects exactly labels $\{0, 1, 5\}$. For $n = 8$, $(2+\sqrt{2}, 1, 4, 4, 1+\sqrt{2})$ has $\chi = 2\sqrt{2} = 1/\beta_1$ and selects $\{1, 2, 6, 7\}$. Every example has $x^* = 1$, all five coefficients positive, and original DM residual $a - k - h + (c - g)/2 = 0$.

For exact highest-mode examples across all four residue classes, use $k = 1$ in the same family with $\chi = 2/\beta_{J-1}$. Then $u_* = \beta_{J-1}/2 \in (0, \beta_{J-1})$, while $\beta_J \le 0$, so J is strict. In particular the following are exact specifications:

n	J	χ	u_J
8	2	$4\sqrt{2}$	0
9	2	$4/\cos(\pi/9)$	$\sin(\pi/18)$
10	2	$2/[\cos(\pi/10)\cos(3\pi/10)]$	$\sin(\pi/10)$
11	3	$2/[\cos(\pi/11)\cos(5\pi/11)]$	$-\sin(\pi/22)$

The corresponding tuples are exactly $(2 + \chi/2, 1, 4, 4, 1 + \chi/2)$. No numerical search is used.

The next table records the complete representative grid and positive midpoints for $5 \le n \le 12$. Within each row only, write $\gamma_r = \cos(r\pi/n)$. All listed midpoint values are exact. Their reciprocal values are the switch thresholds in (88); the reachable set is $\{0, \ldots, J\}$. The terminal midpoint β_J is negative except at $n = 6, 10$, where it is zero and unreachable. The five-node row is solely a check; the golden-ratio specialization is reserved for P07.

n	$(u_0, \ldots, u_N)$	J	$(\beta_0, \ldots, \beta_{J-1})$
5	$(1, \gamma_2, -\gamma_1)$	1	(γ_1^2)
6	$(1, 1/2, -1/2, -1)$	1	$(3/4)$
7	$(1, \gamma_2, -\gamma_3, -\gamma_1)$	2	$(\gamma_1^2, \gamma_1\gamma_3)$
8	$(1, \sqrt{2}/2, 0, -\sqrt{2}/2, -1)$	2	$((2+\sqrt{2})/4, \sqrt{2}/4)$
9	$(1, \gamma_2, \gamma_4, -1/2, -\gamma_1)$	2	$(\gamma_1^2, \gamma_1/2)$
10	$(1, \gamma_2, \gamma_4, -\gamma_4, -\gamma_2, -1)$	2	$(\gamma_1^2, \gamma_1\gamma_3)$
11	$(1, \gamma_2, \gamma_4, -\gamma_5, -\gamma_3, -\gamma_1)$	3	$(\gamma_1^2, \gamma_1\gamma_3, \gamma_1\gamma_5)$
12	$(1, \sqrt{3}/2, 1/2, 0, -1/2, -\sqrt{3}/2, -1)$	3	$((2+\sqrt{3})/4, (\sqrt{3}+1)/4, 1/4)$

7.9 Dominant modes versus first instability

The preceding results concern $\arg\max_m r_m$ at a fixed equilibrium. The scalar d cancels from that comparison but affects whether the maximum is negative, zero, or positive. A first loss of stability would require a specified oriented physical path, a positive equilibrium branch, a prior interval with $\max_m r_m < 0$, and a crossing to a positive maximum. Neither a selection switch nor a real-part tie establishes that.

Indeed every reachable switch can occur on a path that stays strictly stable. Fix its threshold α_j, and choose a compact interval $[b_-, b_+] \subset (0, \infty)$ with α_j in its interior and no other threshold in it. In (93) let χ increase across this interval, hold $k > b_+/2$ fixed, and keep $x^* = 1$. The equilibrium is regular, all parameters remain positive, and P04's equilibrium diagonal gives $d = -k - 3$. Thus directly from (77),

$$r_m = -k - 3 + 2\cos\theta_m - \frac{\chi}{2}\cos(2\theta_m) \le -k - 1 + \frac{b_+}{2} < -1. \tag{97}$$

The dominant representative changes from j to $j+1$, with the exact tie at α_j, while every mode remains strictly stable. This also shows that every strict cell has a stable canonical realization, since k can be enlarged without changing χ or $x^* = 1$ in this family. No first-instability subresult is asserted; that separate crossing question is deferred to P11/P12 under their branch/path gates.

Real-part ties are not eigenvalue coincidences. At every switch P05 proves $r_j = r_{j+1}$ only. P04 supplies the signed imaginary parts

$$\omega_m = -C\sin\theta_m + (G+H)\sin(2\theta_m).$$

For particular labels, equality of the complex eigenvalues would additionally require equality of these signed imaginary parts. For conjugate labels the sign changes. None of the selection, realization, or geometry proofs needs an imaginary-part condition. Numerical eigenvalue simplicity, nonzero critical frequency, simultaneous critical pairs and Hopf exclusions remain P11's tasks. Counting two or four selected labels does not certify a simple pair or a multiple-eigenvalue collision.

8 P06: Large-size limit (unfinished)

The supplied file contains an outline, not a completed limit proof.

8.1 Selected-wave-number limit

Use P05's nearest-cosine rule. The method is analogous to the Lorenz–96 limit in [3, Proposition 6], but its model-specific limit cannot be imported as a DM theorem.

8.2 Discretization and uniformity

Still required: a specified parameter regime, control of the maximizing grid point, endpoint and tie cases, and any uniform-in-n error estimate.

9 P07: Five-node golden-ratio specialization

This is a corollary package for the P03–P05 demand-modulated results. The direct dependency remains P05 → P07; P03–P04 and the P01–P02 equilibrium foundation are inherited. P08 is present in the current project but supplies no theorem used here.

Fix $a, k, c, g, h > 0$ in the canonical model

$$\dot{x}_i = ax_i - kx_i^2 + \frac{cx_{i-1}}{1+x_i} - \frac{gx_i}{1+x_{i+2}} - hx_ix_{i-2}, \qquad i \in \mathbb{Z}_5. \tag{98}$$

Let $x^*\mathbf{1}$, $x^* > 0$, be a specified synchronous equilibrium. Write

$$s = 1 + x^* > 1, \quad C = \frac{c}{s} > 0, \quad G = \frac{gx^*}{s^2} > 0, \quad H = hx^* > 0, \quad T = G - H, \quad S = G + H. \tag{99}$$

Here s denotes the denominator, not P02's scalar state variable; S is local to P07 and denotes effective DM coefficients, not the physical sum $g+h$ used locally in P08. We use $D_5 := H - G = -T$, leaving P02's discriminant D unchanged. The symbols ω_1, ω_2 below are signed imaginary parts, not a reserved positive Hopf frequency. By P02 and P03,

$$a - (k+h)x^* + \frac{c-g}{s} = 0, \tag{100}$$

$$d = a - 2kx^* - \frac{cx^*}{s^2} - \frac{g}{s} - hx^* = -kx^* - C - \frac{Cx^*}{s}. \tag{101}$$

The last equality uses equilibrium and will only be used when stated.

9.1 Exact fifth-root constants and Fourier spectrum

Lemma 9.1 (Golden-ratio trigonometry)**.** *Set*

$$\phi = \frac{1+\sqrt{5}}{2}, \qquad u_1 = \cos\frac{2\pi}{5}, \qquad u_2 = \cos\frac{4\pi}{5}, \qquad \sigma = \sin\frac{\pi}{5}. \tag{102}$$

Then

$$\phi^2 = \phi + 1, \quad \phi^{-1} = \phi - 1, \quad \phi^{-2} = 2 - \phi = \frac{3-\sqrt{5}}{2}, \quad \phi + \phi^{-1} = \sqrt{5}, \tag{103}$$

$$u_1 = \frac{\sqrt{5}-1}{4} = \frac{1}{2\phi}, \qquad u_2 = -\frac{\sqrt{5}+1}{4} = -\frac{\phi}{2}, \qquad u_1 - u_2 = \frac{\sqrt{5}}{2}, \tag{104}$$

$$\sigma = \frac{\sqrt{10-2\sqrt{5}}}{4} > 0, \qquad \sin\frac{2\pi}{5} = \phi\sigma = \frac{\sqrt{10+2\sqrt{5}}}{4}. \tag{105}$$

Proof. These are the exact fifth-root trigonometric values; substituting $\phi = (1+\sqrt{5})/2$ verifies the displayed identities. □

There are three representative classes, $\{0\}$, $\{1,4\}$ and $\{2,3\}$. Substitute the constants above into P04 with $\theta_m = 2\pi m/5$.

Theorem 9.2 (Exact five-node Fourier spectrum)**.** *The five eigenvalues, with their Fourier labels and algebraic multiplicities retained, are*

$$\begin{aligned} \lambda_0 &= d + C + T, \\ \lambda_1 &= d + \frac{C}{2\phi} - \frac{\phi T}{2} + \mathrm{i}\sigma(S - \phi C), \\ \lambda_2 &= d - \frac{\phi C}{2} + \frac{T}{2\phi} - \mathrm{i}\sigma(C + \phi S), \\ \lambda_4 &= \overline{\lambda_1}, \qquad \lambda_3 = \overline{\lambda_2}. \end{aligned} \tag{106}$$

In particular the three real-part classes are

$$r_0 = d + C + T, \qquad r_1 = d + \frac{C}{2\phi} - \frac{\phi T}{2}, \qquad r_2 = d - \frac{\phi C}{2} + \frac{T}{2\phi}. \tag{107}$$

Here $r_1 = \operatorname{Re}\lambda_1 = \operatorname{Re}\lambda_4$ *and* $r_2 = \operatorname{Re}\lambda_2 = \operatorname{Re}\lambda_3$.

Proof. Specialize (67)–(68) to $n = 5, q = 2$ using Lemma 9.1; the remaining labels follow by conjugacy. □

9.2 Complete ordering and the dominant-mode switch

Theorem 9.3 (Complete ordering of the three real-part classes)**.** *Set* $t = T/C$*, which is well defined since* $C > 0$*. The exact differences are*

$$\begin{aligned} r_1 - r_2 &= \frac{\sqrt{5}}{2}(C - T) = \frac{\sqrt{5}C}{2}(1 - t), \\ r_0 - r_1 &= \frac{\sqrt{5}\phi}{2}\left(T + \frac{C}{\phi^2}\right) = \frac{\sqrt{5}\phi C}{2}(t + \phi^{-2}), \\ r_0 - r_2 &= \frac{\sqrt{5}}{2\phi}(T + \phi^2 C) = \frac{\sqrt{5}C}{2\phi}(t + \phi^2). \end{aligned} \tag{108}$$

Consequently the following table is exhaustive, including all equalities:

Condition	*Ordering*	*Tie type*
$t > 1$	$r_0 > r_2 > r_1$	*none*
$t = 1$	$r_0 > r_1 = r_2$	*subdominant*
$-\phi^{-2} < t < 1$	$r_0 > r_1 > r_2$	*none*
$t = -\phi^{-2}$	$r_0 = r_1 > r_2$	*dominant*
$-\phi^2 < t < -\phi^{-2}$	$r_1 > r_0 > r_2$	*none*
$t = -\phi^2$	$r_1 > r_0 = r_2$	*subdominant*
$t < -\phi^2$	$r_1 > r_2 > r_0$	*none*

Ordering is independent of d. No triple tie of real-part classes occurs.

Proof. Subtract the three expressions in (107) and use $\phi^2 = \phi + 1$. The positive prefactors in (108) and the strict ordering $-\phi^2 < -\phi^{-2} < 1$ give every row. Their three distinct zeros exclude a triple tie. □

Corollary 9.4 (Exact golden-ratio selection threshold)**.** *The dominant representative set and the full selected Fourier-label set are*

Condition	*Representatives*	*Labels*
$D_5/C < \phi^{-2}$	$\{0\}$	$\{0\}$
$D_5/C = \phi^{-2}$	$\{0,1\}$	$\{0,1,4\}$
$D_5/C > \phi^{-2}$	$\{1\}$	$\{1,4\}$.

(109)

Pair $\{2,3\}$ *is never dominant at a canonical positive synchronous equilibrium. The unique dominant-class switch is*

$$\boxed{H - G = \frac{C}{\phi^2}, \qquad \frac{H-G}{C} = \frac{3-\sqrt{5}}{2}.} \tag{110}$$

Proof. Apply Theorem 7.1 with $\beta_0 = \phi^2/4$ and $\beta_1 = -1/4$, or read the maxima in Theorem 9.3. □

The five-node midpoint and control values are

$$\beta_0 = \frac{\phi^2}{4}, \quad \beta_1 = -\frac{1}{4}, \quad \alpha_0 = \frac{4}{\phi^2} = 6 - 2\sqrt{5}. \tag{111}$$

$$\chi = \frac{4(H-G)}{C}, \qquad \chi \lessgtr \frac{4}{\phi^2} \iff \text{strict selection of representative 0 or 1,} \tag{112}$$

respectively; equality selects both, as in (109).

Canonical switch surface and the equilibrium constraint. Since $D_5 = x^*(h - g/s^2)$, solving (110) gives

$$\boxed{h = \frac{g}{(1+x^*)^2} + \frac{c}{\phi^2(1+x^*)x^*}.} \tag{113}$$

The divisions are valid because x^*, s, ϕ, C are positive. The equilibrium relation (100) must hold simultaneously. Thus this equation describes a switch on the positive branch-state manifold; x^* cannot be prescribed independently of a fixed physical tuple. The corresponding strict inequalities in h at that equilibrium give uniform or pair-1 selection, respectively. P05 already supplies regularity and local realizability of this switch.

The other golden-ratio equality surfaces. At $t = 1$, equivalently $G - H = C$, one has $r_1 = r_2$ but $r_0 - r_1 = r_0 - r_2 = 5C/2 > 0$. At $t = -\phi^2$, equivalently $H - G = \phi^2 C$, one has $r_0 = r_2$ but $r_1 - r_0 = r_1 - r_2 = 5\phi C/2 > 0$. These values follow from (108) and $\phi^2 = \phi + 1$. Both are subdominant real-part ties, not dominant-mode switches. No equality of complex numerical eigenvalues is implied by either tie.

9.3 Frequencies, intersection, and numerical eigenvalue equality

Proposition 9.5 (Signed imaginary parts and frequency degeneracy)**.** *At positive synchrony,*

$$\omega_1 := \operatorname{Im}\lambda_1 = \sigma(S - \phi C), \qquad \omega_2 := \operatorname{Im}\lambda_2 = -\sigma(C + \phi S) < 0. \tag{114}$$

Pair $\{2,3\}$ *is always nonreal. Pair* $\{1,4\}$ *is nonreal if and only if* $S \neq \phi C$*. At the spectral locus*

$$\boxed{G + H = \phi C,} \tag{115}$$

one has $\lambda_1 = \lambda_4 = r_1$*, a repeated real numerical eigenvalue represented by two Fourier labels. All numerical eigenvalues are semisimple; any further coincident label is counted separately below.*

Proof. The imaginary terms were derived directly in Theorem 9.2. Since $\sigma, C, S, \phi > 0$, ω_2 is strictly negative, while $\omega_1 = 0$ is equivalent to $S = \phi C$. Conjugacy then proves all reality assertions. Semisimplicity is Proposition 6.2, which does not assume distinct numerical eigenvalues. These assertions classify spectra only. □

Proposition 9.6 (Intersection with the dominant-mode switch)**.** *The switch and the frequency-degeneracy locus intersect in the strictly positive coefficient domain exactly at*

$$\boxed{\frac{G}{C} = \frac{1}{\phi}, \qquad \frac{H}{C} = 1.} \tag{116}$$

They define a nonempty codimension-two spectral locus in $(C, G, H) > 0$*, and locally on the canonical positive equilibrium branch. At that locus*

$$\lambda_0 = \lambda_1 = \lambda_4 = d + \frac{C}{\phi} = -kx^* - \frac{Cx^*}{s} - \frac{C}{\phi^2} < 0. \tag{117}$$

This numerical eigenvalue is semisimple of multiplicity three, and $r_2 < r_0$*. In particular the full linearization is Hurwitz there. This is a spectral codimension statement, not a bifurcation classification.*

Proof. Adding and subtracting $H - G = C\phi^{-2}$ and $G + H = \phi C$ gives

$$\frac{G}{C} = \frac{\phi - \phi^{-2}}{2} = \phi - 1 = \phi^{-1}, \qquad \frac{H}{C} = \frac{\phi + \phi^{-2}}{2} = 1.$$

Both are positive. The gradients in (G, H) of $H - G - C/\phi^2$ and $G + H - \phi C$ have determinant -2, so the two linear conditions are independent. For a canonical witness take

$$x^* = 1, \qquad (a, k, c, g, h) = (\sqrt{5}, 1, 2, 4/\phi, 1). \tag{118}$$

The equilibrium bracket equals $\sqrt{5} - 2 + (2 - 4/\phi)/2 = 0$, using $1 + 2/\phi = \sqrt{5}$. The effective coefficients are $C = 1, G = \phi^{-1}, H = 1$.

For the local branch assertion use P05's verified equilibrium chart $(x, k, c, g, h) \mapsto ((k + h)x - (c-g)/(1+x), k, c, g, h)$. Its determinant is positive wherever $H - G > 0$; the intersection lies in that regime. At fixed x, k, c, the two constraint gradients in (g, h) have determinant $-2x^2/(1 + x)^2 \neq 0$. The chart therefore transfers their independent codimension-two conditions locally to physical parameters on the specified branch. There is no claim about a parameter-path crossing or an unfolding.

At the intersection, conjugacy, $\omega_1 = 0$, and $r_0 = r_1$ give the triple value. Substitute (101) and $1 - \phi^{-1} = \phi^{-2}$ for its strict negativity. The ordering table gives $r_2 < r_0$; $\omega_2 < 0$ keeps labels 2 and 3 distinct from the real value and from each other. P04 counts precisely three independent Fourier eigenvectors for the triple value. □

Proposition 9.7 (Complete numerical coincidence classification)**.** *Under the stated positive-equilibrium assumptions, the only numerical eigenvalue coincidences are as follows:*

(i) If $S \neq \phi C$, all five numerical eigenvalues are distinct.

(ii) If $S = \phi C$ and $T \neq -C/\phi^2$, only $\lambda_1 = \lambda_4$ coincide; their real value has multiplicity two.

(iii) If $S = \phi C$ and $T = -C/\phi^2$, exactly $\lambda_0 = \lambda_1 = \lambda_4$ coincide as in (117)*; their value has multiplicity three.*

All multiplicities are algebraic and geometric. Labels 2 and 3 always give distinct, simple numerical eigenvalues, sharing no value with any other label. Every nonreal pair-1 eigenvalue is simple.

Proof. Within pairs, (114) proves that labels 1 and 4 coincide exactly when $S = \phi C$, while labels 2 and 3 never coincide. The real λ_0 cannot equal either nonreal pair-2 eigenvalue. Equality between λ_0 and a pair-1 label requires both $r_0 = r_1$ and $\omega_1 = 0$, exactly the intersection just analyzed.

A cross-pair numerical equality requires $r_1 = r_2$, hence $T = C$. Then $G = H + C$ and $S = C + 2H > C$. At this real-part tie,

$$\begin{aligned} \omega_1 - \omega_2 &= \sigma[(1+\phi)S - (\phi-1)C] > 0, \\ -\omega_2 - \omega_1 &= \sigma[(1+\phi)C + (\phi-1)S] > 0. \end{aligned}$$

For the first inequality use $S > C$ and $(1+\phi) - (\phi-1) = 2$; the second has strictly positive summands. Thus $\omega_2 < \omega_1 < -\omega_2$, so ω_1 is neither ω_2 nor $-\omega_2$; conjugating excludes all four possible cross-pair equalities. This exhausts all unordered label pairs. P04 then gives the asserted multiplicities. □

The frequency-degeneracy locus itself has $-\phi < t < \phi$, since $G, H > 0$ imply $|T| < S$. It can intersect the subdominant tie $t = 1$ without a cross-pair coincidence, but cannot intersect $t = -\phi^2$. Equal real parts, repeated real values, and distinct nonreal conjugates are consequently kept separate throughout.

9.4 Exact stability test and critical-mode ledger

Corollary 9.8 (Piecewise five-node linear stability)**.** *The linearization is exponentially asymptotically stable exactly when*

$$\max\{r_0, r_1, r_2\} < 0 \quad \Longleftrightarrow \quad \begin{cases} r_0 < 0, & t > -\phi^{-2}, \\ r_0 = r_1 < 0, & t = -\phi^{-2}, \\ r_1 < 0, & t < -\phi^{-2}. \end{cases} \tag{119}$$

The middle row means that the common value must be negative. At canonical positive equilibria on that tie it is automatically negative:

$$r_0 = r_1 = -kx^* - \frac{Cx^*}{s} - \frac{C}{\phi^2} < 0. \tag{120}$$

Proof. P04 Corollary 6.4 gives the strict all-mode test; Theorem 9.3 gives its piecewise maximum. At the dominant tie, substitute $T = -C/\phi^2$ and (101) into $r_0 = d + C + T$. □

If the maximum is zero, P04 implies that the normal linear system is Lyapunov stable but not asymptotically stable. This does not settle nonlinear stability or establish a bifurcation. A strictly positive maximum means linear instability. The dominant tie cannot itself be a zero-abscissa boundary in the canonical domain, by (120).

Proposition 9.9 (Fixed-state critical-mode ledger)**.** *Only the following classes can attain the spectral abscissa; this is a fixed-state eligibility ledger, not a first-instability theorem.*

Class	*Eligibility for the maximum*	*Signed imaginary part*
0	$t \geq -\phi^{-2}$	0
$\{1,4\}$	$t \leq -\phi^{-2}$	$\pm\sigma(S-\phi C)$
$\{2,3\}$	*never*	$\mp\sigma(C+\phi S)$

At a zero-abscissa state, the inequalities in the first two rows must be strict, because the shared boundary is strictly stable. If $r_2 = 0$, some other class has strictly positive real part, so pair 2 cannot be the sole or tied controlling class of a stability boundary. No oriented parameter path, prior stability, or crossing is asserted.

Proof. The eligibility and exclusions are precisely Corollary 9.4. At every point at least one of r_0, r_1 exceeds r_2 strictly. The zero-abscissa statements follow from that strict comparison and (120). □

9.5 Realizability checks, downstream inputs, and scope

All ordering regions have canonical representatives. For any $C, G, H > 0$, the following single construction gives a canonical positive equilibrium with those exact effective coefficients:

$$x^* = 1, \quad k = C, \quad c = 2C, \quad g = 4G, \quad h = H, \quad a = H + 2G > 0, \quad d = -\frac{5C}{2}. \tag{121}$$

Indeed (100) becomes $H + 2G - (C + H) + (2C - 4G)/2 = 0$. Every original component equals x^* times that zero bracket. For any real t, take $C = 1$, $G = 1 + \max(t, 0)$ and $H = 1 + \max(-t, 0)$. Then $G - H = t$ and both coefficients are positive. Thus no row of the ordering table is merely an unattached-coefficient possibility. This elementary five-node check uses no MB comparison or new general selection theorem.

Exact inputs available to P11. The unique nonuniform dominant class, when present, is $\{1, 4\}$. Its signed imaginary part is $\sigma(S - \phi C)$, and its exact nonzero-frequency condition is $S \neq \phi C$. Whenever pair 1 is maximal, $r_2 < r_1$ strictly, including the uniform/pair-1 dominant tie. When pair 1 is uniquely dominant, $r_0 < r_1$ as well. Every nonreal pair-1 eigenvalue is simple; pair 2 never coincides numerically with another label. The only uniform/pair-1 numerical coincidence is the strictly stable triple value (117); more generally the entire dominant tie is strictly stable. These are exact spectral inputs, without a Hopf existence or first-crossing conclusion. No transversality, crossing derivative, Lyapunov coefficient, normal form, periodic orbit, or rotating wave is analyzed here; Later sections address their own hypotheses.

Remark 9.10 (Golden ratio is network geometry). The constants ϕ and σ arise solely from the fifth roots of unity and the five-node Fourier grid. They establish no physiological golden-ratio law, biological optimum, medical significance, or empirical TCM validation. P07 is a supporting specialization of the general program, not standalone golden-ratio novelty.

10 P08: Matched bilinear analysis and exact comparison

The global-interior theorem and its comparison corollary were already present as unaudited extensions in the supplied edition. Their shortened proofs below retain that status; the earlier A08 audit did not cover them. All coefficients are strictly positive, $n \geq 5$, $1 \leq q < n$, and indices are modulo n. The comparator is exactly the five-parameter model ($\mathrm{MB}_{n,q}$). The same tuple (a, k, c, g, h), n, q and time variable are used in DM and MB; each equilibrium amplitude is derived for its own equation. The foundation and selection inputs are P01–P05.

The substantive results are open same-tuple stability contrasts in both directions at $n = 7$, a universal forward contrast, and realization of every ordered pair of reachable dominant representatives with both models stable. None is a first-instability, Hopf, criticality, or literature-priority theorem. Here "strictly stable" means Hurwitz; "unstable" means a strictly positive spectral abscissa. Zero abscissa is treated separately. Selection maximizes over all labels, including zero, retaining every maximal tie.

10.1 MB local flow, synchrony, and synchronous growth

Put $S = g + h > 0$ and $M = k + S - c$; the subspace remains denoted by $\mathcal{S}$.

Theorem 10.1 (MB foundation)**.** *For arbitrary canonical (n, q) the following hold.*

(i) MB has a unique maximal classical solution through every point of $\mathbb{R}^n$. Every coordinate face is invariant. Nonnegative initial data stay nonnegative, and initially positive coordinates remain strictly positive on the maximal interval. No uniform asymptotic lower bound is meant by this finite-time positivity statement.

(ii) $\mathcal{S}$ is invariant, with exact restriction

$$\dot{s} = as - Ms^2. \tag{122}$$

The real synchronous equilibria are $0, a/M$ if $M \neq 0$, and only 0 if $M = 0$. A positive synchronous equilibrium exists if and only if $M > 0$, and is uniquely $y^* = a/M$. For $M < 0$ the nonzero root is negative.

(iii) For $s_0 > 0, M > 0$, synchronous dynamics are global, $0 < s(t) \leq \max(s_0, a/M)$ and $s(t) \to a/M$. For $M = 0$ they are $s(t) = s_0 e^{at}$, global and unbounded. For $M < 0$ they blow up at

$$t_b = \frac{1}{a} \log \left(1 + \frac{a}{|M| s_0} \right). \tag{123}$$

These classify nonnegative synchronous dynamics, not all MB trajectories.

Proof. The polynomial field is smooth and locally Lipschitz on $\mathbb{R}^n$, so local existence, uniqueness, continuous dependence and maximal continuation apply. On the coordinate face $y_i = 0$, direct substitution in the original five terms gives $F_i^{\mathrm{MB}}(y) = 0$, even when other coordinates are negative. Along any solution the integrating-factor identity is

$$y_i(t) = y_i(0) \exp \int_0^t \left[a - k y_i(r) + c y_{i-1}(r) - g y_{i+q}(r) - h y_{i-q}(r) \right] dr. \tag{124}$$

The integrand is continuous on each compact subinterval of the maximal interval. The identity holds also for zero initial coordinates and proves sign preservation and face invariance without assuming global existence.

At $y = s\mathbf{1}$, each component of the original field is $as - ks^2 + cs^2 - gs^2 - hs^2 = as - Ms^2$. The scalar solution lifts to a vector solution, uniqueness identifies the two, and continuation makes their maximal intervals coincide. Zeros of $s(a - Ms)$ give the complete synchronous equilibrium list, retaining zero before division. For $s_0 > 0$, solve $(1/s)' = -a/s + M$ to obtain

$$s(t) = \frac{s_0 e^{at}}{1 + (Ms_0/a)(e^{at} - 1)}. \tag{125}$$

For $M > 0$ the denominator stays positive and the bound and limit follow; for $M = 0$ it is one; for $M < 0$ its first zero is exactly (123), approached from above with positive numerator. The zero initial state stays zero in every case. □

10.2 Sufficient full-system bounds and their scope

Proposition 10.2 (Maximum comparison)**.** *For nonnegative MB solutions and* $Y = \max_i y_i$,

$$D^+ Y \le aY - (k - c)Y^2. \tag{126}$$

Thus $k \ge c$ *suffices for global nonnegative existence. If* $k > c$, *put* $R = a/(k - c)$. *Then* $Y(t) \le \max\{Y(0), R\}$, $[0, R]^n$ *is invariant and uniformly attracts bounded subsets, and every* $[0, R+\varepsilon]^n$, $\varepsilon > 0$, *absorbs bounded subsets in finite time. MB is therefore dissipative in this sufficient regime. The comparison alone asserts neither finite-time absorption into the exact cube* $[0, R]^n$ *nor boundedness when* $k = c$.

Proof. Apply the generic finite-maximum identity of Lemma 3.1. At any maximizing coordinate $y_i = Y$, all neighbors lie in $[0, Y]$, so $cy_i y_{i-1} \le cY^2$ and both inhibitory terms are nonpositive. This includes tied maxima and coincident offsets. For $k \ge c$, the inequality gives $Y(t) \le Y(0)e^{at}$, excluding finite-time escape; polynomial continuation yields global existence. For $k > c$, logistic comparison bounds Y and proves invariance. A logistic solution starting from the supremum of any bounded set of initial maxima converges to R and enters $[0, R + \varepsilon]$ in finite time, proving the uniform attraction and absorption assertions. □

Consequently $M > 0$ is necessary for boundedness of every nonnegative MB trajectory, and $M \ge 0$ is necessary for global existence of every such trajectory, by the synchronous counterexamples. No converse is inferred. In particular existence of positive synchrony is not used to assert full-system boundedness. The sufficient condition $k > c$ is not necessary, as an explicit example below will prove.

10.3 Homogeneous MB Jacobian and arbitrary-offset spectrum

At $y = s\mathbf{1}$, differentiate the original field before using equilibrium. None of $i - 1, i + q, i - q$ equals i in the canonical range, so the diagonal is

$$d_{\mathrm{MB}}(s) = a - 2ks + cs - gs - hs = a - (k + M)s.$$

The three off-diagonal contributions are $cs, -gs, -hs$. Therefore

$$DF^{\mathrm{MB}}(s\mathbf{1}) = [a - (k + M)s]I + s(cP - gP^{-q} - hP^q). \tag{127}$$

In full, with row i and column j, the general derivative entry is

$$[a - 2ky_i + cy_{i-1} - gy_{i+q} - hy_{i-q}]\delta^{(n)}_{ji} + cy_i\delta^{(n)}_{j,i-1} - gy_i\delta^{(n)}_{j,i+q} - hy_i\delta^{(n)}_{j,i-q}.$$

Coincident columns add; none is overwritten. At $y^* = a/M > 0$ use $a = My^*$ only now, giving $d_{\mathrm{MB}} = -ky^*$.

Theorem 10.3 (Full MB spectrum, normality, and uniform direction)**.** *For $M > 0$, positive MB synchrony has*

$$J_{\mathrm{MB}} = y^*[-kI + cP - gP^{-q} - hP^q], \tag{128}$$

$$\lambda_m^{\mathrm{MB}} = y^*[-k + ce^{-\mathrm{i}\theta_m} - ge^{\mathrm{i}q\theta_m} - he^{-\mathrm{i}q\theta_m}], \tag{129}$$

$$\operatorname{Re}\lambda_m^{\mathrm{MB}} = y^*[-k + c\cos\theta_m - S\cos(q\theta_m)], \tag{130}$$

$$\operatorname{Im}\lambda_m^{\mathrm{MB}} = y^*[-c\sin\theta_m + (h-g)\sin(q\theta_m)]. \tag{131}$$

The n Fourier labels give the full spectrum with algebraic multiplicity; J_{MB} is normal and unitarily diagonalized by the project's DFT. In particular

$$\boxed{\lambda_0^{\mathrm{MB}} = -a < 0.} \tag{132}$$

Any linear instability at positive MB synchrony involves a nonuniform mode. This does not classify a bifurcation as Hopf or steady.

Proof. The preceding differentiation gives (128). Apply the circulant framework of P04 (Gray, [1, Theorem 3.1]) to the coefficients $-k, c, -g, -h$, then take real and imaginary parts. At label zero the value is $y^*(-k + c - g - h) = -My^* = -a$. □

The complete offset-collision checks, using P03/P04 only for their index geometry, are

$$\begin{array}{r|l} q = 1 & J_{\mathrm{MB}}/y^* = -kI + (c-h)P - gP^{-1} \\ q = n-1 & J_{\mathrm{MB}}/y^* = -kI + (c-g)P - hP^{-1} \\ q = n/2 \ (n \text{ even}) & J_{\mathrm{MB}}/y^* = -kI + cP - SP^{n/2}. \end{array}$$

Their imaginary parts are respectively $y^*(-c+h-g)\sin\theta_m$, $y^*(-c+g-h)\sin\theta_m$, and $-y^*c\sin\theta_m$. The last real restraint term is $-y^*S(-1)^m$. Each also follows from the master spectrum, with cancellation retained. For even n, the singleton is $\lambda_{n/2}^{\mathrm{MB}} = y^*[-k-c-S(-1)^q]$; the uniform value is unchanged in every collision case. Label conjugacy and semisimplicity follow from the real circulant matrix and its unitary diagonalization; a pair may have zero imaginary part or share an eigenvalue with other labels. No numerical-eigenvalue simplicity is inferred from label counts.

Corollary 10.4 (Exact stability and another sufficient global bound)**.** *Define*

$$K_{n,q}(c,S) = \max_{0 \le m < n} [c\cos\theta_m - S\cos(q\theta_m)]. \tag{133}$$

At positive MB synchrony, strict stability is equivalent to $k > K_{n,q}$, and instability to $k < K_{n,q}$. At equality the normal linear system is Lyapunov stable but not asymptotically stable; nonlinear stability is undecided here. Hyperbolicity requires every real part to be nonzero, not merely a nonzero maximum.

Moreover $k > K_{n,q}$ itself implies $M > 0$, global boundedness and dissipativity for nonnegative MB data. This is another sufficient bound, not a necessary condition or an identification with $k > c$.

Proof. The abscissa is $y^*(K_{n,q} - k)$ and unitary diagonalization gives $\|e^{tJ_{\mathrm{MB}}}\|_2 = e^{t\,\mathrm{sa}(J_{\mathrm{MB}})}$, proving the linear claims. For the bound put $\delta = k - K_{n,q} > 0$, $V = \sum_i y_i$. Summing the original equations yields

$$V' = aV + y^{\mathsf{T}}[-kI + cP - gP^{-q} - hP^q]y \le aV - \delta\|y\|_2^2 \le aV - \frac{\delta}{n}V^2. \tag{134}$$

The symmetric part of the displayed matrix has eigenvalues $-k + c\cos\theta_m - S\cos(q\theta_m) \le -\delta$, and its skew part contributes zero to the real quadratic form. Logistic comparison bounds V before any maximal endpoint and proves global continuation. The simplex $V \le na/\delta$ attracts bounded subsets, and every strictly larger one absorbs them. Finally $K_{n,q} \ge c - S$ by its zero label, so $M > 0$. □

For example, for $q = 2$ the tuple $(9/2, 7, 8, 1, 1)$ has $K_{n,2} = 6$ for every $n \geq 5$: $2 + 8u - 4u^2$ uniquely maximizes at $u = 1$ on the grid. MB is dissipative although $k = 7 < c = 8$. Its own root is $y^* = 9/2$, whereas DM's unique positive root is $x^* = 1$. Direct original residuals are $9/2-8+7/2 = 0$ for DM after division by x^*, and $9/2-(9/2) = 0$ for MB after division by y^*. This refutes necessity of $k > c$. Conversely, $k > c$ need not imply spectral stability when S is large, so the sufficient bounds have different scopes.

10.4 Global asymptotic stability in the positive interior

Theorem 10.5 (Strict MB stability implies global interior stability)**.** *Let $n \geq 5$, $1 \leq q < n$, and $a, k, c, g, h > 0$, with the canonical predecessor shift $(Py)_i = y_{i-1}$. Put $S = g + h$ and assume*

$$\delta := k - K_{n,q}(c, S) > 0,$$

where $K_{n,q}$ is exactly (133)*, including label zero. Then $M = k+S-c > 0$, and $\bar{y} = y^*\mathbf{1}$, $y^* = a/M$, is the unique equilibrium in $\mathbb{R}^n_{>0}$ and is globally asymptotically stable relative to $\mathbb{R}^n_{>0}$. Every solution starting in $\mathbb{R}^n_{>0}$ exists for all forward time, remains in a compact subset of $\mathbb{R}^n_{>0}$ depending on its initial condition, and converges to $\bar{y}$. No solution starting with a zero coordinate can converge to $\bar{y}$. No assumption $k > c$ or $\gcd(n, q) = 1$ is required.*

Proof. We use Goh's generalized Lotka–Volterra Lyapunov method [4], with the following direct verification for MB. Write

$$\dot{y} = \operatorname{diag}(y)(a\mathbf{1} + By), \qquad B = -kI + cP - gP^{-q} - hP^q. \tag{135}$$

The zero label gives $M \geq \delta > 0$, hence $\bar{y} = (a/M)\mathbf{1} > 0$ and $B\bar{y} = -a\mathbf{1}$. Define

$$V(y) = \sum_i [y_i - y^* - y^* \log(y_i/y^*)]. \tag{136}$$

For $z = y - \bar{y}$, direct differentiation cancels the population factors:

$$\dot{V} = z^\mathsf{T} B z = z^\mathsf{T} B_{\mathrm{sym}} z, \quad B_{\mathrm{sym}} = -kI + \tfrac{c}{2}(P + P^{-1}) - \tfrac{S}{2}(P^q + P^{-q}). \tag{137}$$

P04's Fourier framework gives $\max \operatorname{spec}(B_{\mathrm{sym}}) = K_{n,q} - k = -\delta$, so

$$\dot{V} \leq -\delta \|y - \bar{y}\|_2^2. \tag{138}$$

Each summand of V is nonnegative, vanishes only at $y_i = y^*$, and diverges as $y_i \downarrow 0$ or $y_i \to \infty$. Thus sublevels are compact subsets of the positive interior. Positivity from (124), decreasing V, and ODE continuation give global trajectories trapped in those sublevels. The strict Lyapunov inequality then gives stability and convergence to $\bar{y}$, and excludes any other positive equilibrium. Invariant coordinate faces prevent boundary data from converging to $\bar{y}$. □

This application does not classify $k = K_{n,q}$ or assert a convergence rate uniform over initial data or sizes. The DM–MB constructions below are separate model-specific arguments.

10.5 Exact step-two MB selected-mode rule

Set $q = 2$ and $u_m = \cos\theta_m$. Since $\cos(2\theta) = 2u^2 - 1$,

$$p(u_m) := \operatorname{Re}\lambda_m^{\mathrm{MB}}/y^* = -k + S + cu_m - 2Su_m^2, \tag{139}$$

$$p(u) = -k + S + \frac{c^2}{8S} - 2S(u - v_{\mathrm{MB}})^2,$$

$$v_{\mathrm{MB}} = \frac{c}{4S} = \frac{1}{\chi_{\mathrm{MB}}}, \qquad \chi_{\mathrm{MB}} = \frac{4S}{c} > 0. \tag{140}$$

This is always strictly concave, $p'' = -4S < 0$. The unconstrained vertex is v_{MB}; on $[-1, 1]$ the continuous maximizer is $\min(1, v_{\mathrm{MB}})$. Its departure from 1 need not switch a discrete mode.

Use P05's notation

$$N = \lfloor n/2 \rfloor, \quad u_j = \cos(2\pi j/n), \quad \beta_j = (u_j + u_{j+1})/2, \quad J = \lfloor (n+1)/4 \rfloor, \quad \alpha_j = 1/\beta_j \ (0 \le j < J).$$

P05 proves $\beta_j = \cos(\pi/n)\cos((2j+1)\pi/n)$ and exactly $\beta_0, \ldots, \beta_{J-1} > 0$. Its nearest-grid-cell argument, midpoint ordering, parity and label bookkeeping are generic finite-grid facts. Its DM realization theorem is not an MB realization proof.

Theorem 10.6 (MB cells, all ties, reachability, and cell stability)**.** *On $M > 0$, the complete MB representative selection is*

$$\begin{array}{c|l} 0 < \chi_{\mathrm{MB}} < \alpha_0 & \{0\} \\ \alpha_{j-1} < \chi_{\mathrm{MB}} < \alpha_j \ (1 \le j < J) & \{j\} \\ \chi_{\mathrm{MB}} > \alpha_{J-1} & \{J\} \\ \chi_{\mathrm{MB}} = \alpha_j \ (0 \le j < J) & \{j, j+1\}. \end{array} \tag{141}$$

The middle row is empty for $J = 1$. Equivalently, closed vertex cells are $[\beta_0, \infty)$, $[\beta_j, \beta_{j-1}]$ $(1 \le j < N)$, and $(-\infty, \beta_{N-1}]$, intersected with $v_{\mathrm{MB}} > 0$. Every listed strict cell and tie is realizable with $M > 0$; the exact reachable representatives are $0, \ldots, J$.

A nonzero selected representative gives $\{j, n-j\}$, while zero is a singleton. A tie involving zero gives three labels, other reachable ties four. No nonadjacent or triple representative maximum occurs. The even Nyquist singleton is excluded. These are label counts, not numerical-eigenvalue multiplicities or frequency assertions. On the cell for j strict stability is exactly

$$\boxed{k > S + cu_j - 2Su_j^2.} \tag{142}$$

Either selected representative gives the same bound at a tie. Reverse strict inequality gives instability and equality gives zero abscissa. In the uniform cell, including a maximal tie with zero, $M > 0$ already makes the maximum $-a < 0$.

Proof. Square completion reduces selection to minimizing $|u_j - v_{\mathrm{MB}}|$. Apply the generic parts of Theorems 7.1 and 7.2. An independent sign check is

$$p(u_j) - p(u_{j+1}) = c(u_j - u_{j+1})(1 - \chi_{\mathrm{MB}}\beta_j),$$

confirming the switch orientation. For MB-specific realizability, prescribe any $v > 0$, take $g = h = 1/2$, $c = 4v$, $k > \max(0, c-1)$ and any $a > 0$. Then $M > 0$ and $v_{\mathrm{MB}} = v$. Each required strict cell and positive midpoint can be attained; strict cells are open by continuity. The grid theorem gives the labels, exclusions and ties. Taking the maximum of the derived quadratic proves the exact cell-stability test. □

P05's parity table applies unchanged to the cosine grid. At every finite positive tuple $v_{\mathrm{MB}} > 0$, so the formal zero-vertex tie for $n = 4\ell+2$ is unattainable: its two real parts differ by $2y^* c \sin(\pi/n) > 0$. Only the gap divided by y^* is independent of S. At fixed n, sufficiently large finite χ_{MB} selects J uniquely in every parity class.

10.6 Common domain, named branches, and structural comparison

For DM set $A = k + h$, $\mu = a + c - g$, $b = a - A$, and $D = b^2 + 4A\mu$. By Theorem 4.4, the full common existence set is

$$\begin{aligned}\mathcal{C} = \{(a,k,c,g,h) > 0 : M > 0, \quad [\mu > 0 \text{ or } (\mu = 0, b > 0) \\ \text{or } (\mu < 0, b > 0, D \geq 0)]\}.\end{aligned} \tag{143}$$

It is not entirely an open set. The branch-labelled comparison pairs a tuple with one of its allowed positive DM roots $x_\pm = (b \pm \sqrt{D})/(2A)$ and the sole MB root a/M. Most constructions use the open domain

$$\mathcal{C}_+ = \{(a,k,c,g,h) > 0 : M > 0,\ \mu > 0\}, \qquad x^* = x_+. \tag{144}$$

DM then has exactly one positive synchronous equilibrium; its scalar uniform derivative is $-x^*\sqrt{D}/(1 + x^*) < 0$ by P04. A separate construction below explicitly follows x_- on a two-positive-root domain.

At a specified positive DM equilibrium put $s = 1 + x^*$. Quote P03–P05:

$$\begin{gathered} d = -kx^* - c(1 + 2x^*)/s^2, \quad C = c/s, \quad G = gx^*/s^2, \quad H = hx^*, \\ r_m^{\mathrm{DM}} = d + Cu_m + (G - H)(2u_m^2 - 1), \\ \chi_{\mathrm{DM}} = \frac{4(H - G)}{C} = \frac{4x^*[h(1 + x^*)^2 - g]}{c(1 + x^*)}, \qquad \chi_{\mathrm{MB}} = \frac{4(g + h)}{c}. \end{gathered} \tag{145}$$

DM's parameter can be negative, zero or positive; its positive-vertex interpretation applies only if $h > g/s^2$. P05's full selection rule uses $\chi_{\mathrm{DM}} < \alpha_0$ for every uniform case. At fixed k, c, g, h, MB selection is independent of a, whereas DM selection can vary with a through the equilibrium branch.

Proposition 10.7 (Structural functional-form distinctions)**.** *MB real dispersion for $q = 2$ is always strictly concave in u. DM dispersion is strictly convex, linear, or strictly concave according as $h < g/(1 + x^*)^2$, $h = g/(1 + x^*)^2$, or $h > g/(1 + x^*)^2$. When $h \leq g/(1 + x^*)^2$, DM uniquely selects zero. All three curvature regimes occur in $\mathcal{C}_+$.*

MB has invariant coordinate faces. DM has $F_i^{\mathrm{DM}}|_{x_i=0} = cx_{i-1}$ and propagates positivity from any nonzero nonnegative initial state by P01. MB's uniform eigenvalue at positive synchrony is always $-a$. DM's is $-x^*\sqrt{D}/s$ on x_+, $+x^*\sqrt{D}/s$ on a distinct positive x_-, and zero at a positive double root.

DM is globally nonnegative and dissipative for the entire canonical positive class by P01. MB has the sufficient dissipative regimes above, but global unbounded synchronous growth at $M = 0$ and finite-time synchronous blow-up at $M < 0$. Thus the DM functional form is structurally dissipative over the entire canonical class, whereas MB is not. The growth/blow-up examples are outside $\mathcal{C}$.

Proof. The second derivatives are $-4y^*S < 0$ and $4(G - H)$, respectively; use P05 for uniform selection. To realize the three curvature signs, set $x^* = 1$, $c = g = 4$, $k = 1$, $a = 1 + h$, with $h = 1/2, 1, 2$. Then $\mu = a > 0$, $M = 1 + h > 0$, $y^* = 1$, $G - H = 1 - h$, and both original scalar residuals vanish. The face and global statements quote P01 and the MB proofs above. The uniform DM values result from inserting P02's roots in (73). No full-system boundedness is inferred from MB equilibrium existence. □

Remark 10.8 (Meaning and limits of matching). Matching fixes n, q, the five positive coefficients, directed offsets and coefficient roles, and the original time variable. It does not calibrate equilibrium amplitude, local derivatives, empirical behavior, or nonlinear response. The successor derivative is positive for DM and negative for MB at positive synchrony, despite the negative term in both fields. No biological or quantitative-superiority claim follows. For a fixed source $z > 0$, as receiver $r \downarrow 0$, the predecessor inputs satisfy $cz/(1+r) \to cz$ for DM and $crz \to 0$ for MB. This last assertion is only a limit under the explicitly fixed-source convention.

10.7 Every ordered dominant-mode pair with both models stable

Theorem 10.9 (Independent realization of the two selection parameters)**.** *For each fixed $n \geq 5, q = 2$, every ordered pair $(j_{\mathrm{DM}}, j_{\mathrm{MB}}) \in \{0, \dots, J\}^2$ occurs with unique selected representatives and both equilibria strictly stable on a nonempty open subset of $\mathcal{C}_+$. In particular reverse uniform/nonuniform selection is possible; distinct nonuniform/nonuniform pairs are possible when $J \geq 2$, beginning at $n = 7$. This is not a full stability partition.*

Proof. Choose positive parameters in the desired strict cells; explicit choices for representative j are

$$z_0 = \alpha_0/2, \quad z_j = (\alpha_{j-1} + \alpha_j)/2 \ (1 \leq j < J), \quad z_J = 2\alpha_{J-1}.$$

When $J = 1$ use only the first and last expressions. Put $\eta = z_{j_{\mathrm{DM}}}$, $\zeta = z_{j_{\mathrm{MB}}}$, and define

$$\begin{gathered} c = 4, \quad S = \zeta, \quad x = 1 + \eta/\zeta, \quad s = 1 + x, \\ h = \frac{\zeta + s\eta/x}{s^2 + 1}, \quad g = \zeta - h, \\ k > \max\{c + S,\ (c-g)/(xs) - h\}, \qquad a = (k+h)x - (c-g)/s. \end{gathered} \tag{146}$$

Here $xs > \eta/\zeta$, so $0 < h < S$ and $g, h > 0$. The bound on k gives $a > 0$ and $M > 0$. Further, $\mu = x[k + h + (c-g)/s] > 0$, since $k > S > g$. Substitution into the *original* vector fields gives

$$F_i^{\mathrm{DM}}(x\mathbf{1}) = x[a - (k+h)x + (c-g)/s] = 0, \qquad F_i^{\mathrm{MB}}(y^*\mathbf{1}) = y^*(a - My^*) = 0. \tag{147}$$

Algebra gives $\chi_{\mathrm{MB}} = \zeta$ and $x(hs^2 - g)/s = \eta$, hence $\chi_{\mathrm{DM}} = \eta$. For all MB labels $\operatorname{Re}\lambda_m^{\mathrm{MB}}/y^* \leq -k + c + S < 0$. For DM the formula gives

$$\frac{r_m^{\mathrm{DM}}}{x} \leq -k - \frac{c}{s^2} + \frac{g}{s^2} + h < -k + S < 0. \tag{148}$$

Indeed the two c contributions after division by x are at most $-c/s^2$, and $|g/s^2 - h| \leq g/s^2 + h$. This proves all claimed strict inequalities. Openness in all physical parameters, not just this displayed family, is proved below. □

A rational reverse-selection witness at $n = 8$ is

$$(a, k, c, g, h) = (113/4, 9, 8, 1, 1), \quad x^* = 3, \quad y^* = 113/12. \tag{149}$$

Here $M = 3$, $\mu > 0$, $\chi_{\mathrm{MB}} = 1$ selects 0, while $\chi_{\mathrm{DM}} = 45/8 > 2\sqrt{2} = \alpha_1$ selects 2. The abscissae are $-443/16$ for DM and $-113/4$ for MB. Original residuals after division by the respective amplitudes are $113/4 - 10 \cdot 3 + 7/4 = 0$ and $113/4 - 3(113/12) = 0$.

10.8 DM stable and MB unstable for every size

Theorem 10.10 (Universal same-tuple opposite stability)**.** *For each fixed $n \geq 5, q = 2$, a nonempty open subset of $\mathcal{C}_+$ has DM strictly stable with unique dominant representative 0, while MB is unstable with a unique nonzero dominant representative. A single rational tuple works for every dimension:*

$$\boxed{(a, k, c, g, h) = (1/4, 1/8, 1, 1, 1/8).} \tag{150}$$

At this tuple the independently computed amplitudes happen to be $x^* = y^* = 1$; amplitude equality is not a comparison assumption.

Proof. All coefficients are positive, $M = 1/4$, $\mu = 1/4$. At amplitude one the original DM residual is $1/4 - 1/8 + 1/2 - 1/2 - 1/8 = 0$ and the MB residual is $1/4 - 1/8 + 1 - 1 - 1/8 = 0$. P02 ensures the unique positive DM root. Its coefficients are $d = -7/8, C = 1/2, G = 1/4, H = 1/8$. Thus $G - H = 1/8 > 0$, zero is strictly dominant and $r_0^{\mathrm{DM}} = -1/4$; every DM real part is negative. For MB,

$$S = 9/8, \qquad p(u) = 1 + u - \frac{9}{4}u^2, \qquad v_{\mathrm{MB}} = 2/9, \qquad \chi_{\mathrm{MB}} = 9/2.$$

The parity formulas give $-1/4 < u_J \leq 1/2$. In the only negative case $n \equiv 3 \pmod 4$, $n \geq 7$ and $|u_J| = \sin(\pi/(2n)) < \pi/14 < 1/4$. The other cases give the upper bound, attained at $n = 6$. Hence the actual finite label J satisfies

$$p(u_J) \geq 1 - \frac{1}{4} - \frac{9}{4}\left(\frac{1}{2}\right)^2 = \frac{3}{16} > 0.$$

MB is unstable, and its negative uniform eigenvalue excludes zero from the selected set. There is no representative tie: a tie would force $\beta_j = 2/9$, but $4\beta_j = 2\cos(2\pi j/n) + 2\cos(2\pi(j+1)/n)$ is an algebraic integer, being a sum of roots of unity and their inverses. The rational number $8/9$ is not an integer, whereas every rational algebraic integer is an integer (apply the rational-root argument to a monic integer polynomial). This excludes a tie. Strict gaps and branch regularity give openness by the argument below. □

For $n = 5, 6, 7$ this witness selects MB representative 1; for $n = 8$ it selects representative 2. Other sizes follow the exact finite-grid rule and need not retain either label. The instability proof uses a finite Fourier label rather than only a positive continuous maximum.

10.9 Reverse stability: parity obstruction and constructive regions

For a positive DM equilibrium write $x = x^*$, $s = 1 + x$, $u = \cos\theta_m$, $w = -\cos(2\theta_m)$. The DM formula and the derived MB formula give

$$\frac{r_m^{\mathrm{DM}}}{x} = -k + (h - g/s^2)w - \frac{c(1 + 2x - su)}{xs^2}, \qquad \frac{r_m^{\mathrm{MB}}}{y^*} = -k + Sw + cu. \tag{151}$$

Notice $1 + 2x - su \geq x > 0$ for $u \leq 1$.

Theorem 10.11 (Reverse stability on the unique positive branch)**.** *On $\mathcal{C}_+$ with $q = 2$, a tuple with DM unstable and MB strictly stable exists if and only if $n \equiv 3 \pmod 4$. For every such fixed size there is a nonempty open region realizing this contrast, with both models uniquely selecting J. For all other sizes, MB strict stability implies DM strict stability on $\mathcal{C}_+$.*

Proof of the obstruction. On $\mathcal{C}_+$ the uniform DM eigenvalue is strictly negative. If $G - H \geq 0$, P05 makes it largest, so DM is strictly stable. Otherwise $h - g/s^2 > 0$. A nonnegative DM real part in (151) forces $w > 0$. If its cosine $u \geq 0$,

$$\frac{r_m^{\mathrm{MB}}}{y^*} - \frac{r_m^{\mathrm{DM}}}{x} = g(1 + s^{-2})w + cu + \frac{c(1 + 2x - su)}{xs^2} > 0. \tag{152}$$

That MB mode is therefore positive. Unless $n \equiv 3 \pmod 4$, all reachable DM representatives have nonnegative cosines by P05. Any nonnegative DM maximum would contradict MB strict stability. This also excludes a zero DM abscissa and proves the strict implication. □

Proof of the remaining-size realization. Let $n \equiv 3 \pmod 4$, $\tau = \sin(\pi/(2n))$, so $0 < \tau < 1/4$, $u_J = -\tau$, and $W = 1 - 2\tau^2 > 7/8$. Also $u_{J-1} = \sin(3\pi/(2n)) = 3\tau - 4\tau^3$, giving $\beta_{J-1} = \tau W$. Set

$$\begin{gathered} h = 1, \quad c = \tau, \quad g = \tau^2/4, \quad k = 1 - 5\tau^2/2, \quad x = 2/\tau, \quad s = 1 + x, \\ a = (k + 1)x - (\tau - \tau^2/4)/s. \end{gathered} \tag{153}$$

Every coefficient is positive: $k > 27/32$, $x > 8$, the subtracted term in a is less than $1/4$, and $(k + 1)x > 8$. Further $M = 2 - \tau - 9\tau^2/4 > 0$, and $\mu > 0$ since $c > g$. The original residuals vanish by (147).

For MB, $v_{\mathrm{MB}} = \tau/(4 + \tau^2) < \tau/4 < \beta_{J-1}$ and $\beta_J < 0$, so it selects J strictly. Its maximum divided by y^* is

$$-k + (1 + \tau^2/4)W - \tau^2 = -\tau^2(1/4 + \tau^2/2) < 0.$$

For DM the J-mode divided by x is

$$\frac{r_J^{\mathrm{DM}}}{x} = \frac{\tau^2}{2} - \frac{gW}{s^2} - \frac{\tau(1 + 2x + s\tau)}{xs^2}.$$

Since $1 + 2x + s\tau \leq (2 + \tau)s$, $s > x$, and $x = 2/\tau$, the two subtracted terms are less than

$$\frac{g + \tau(2 + \tau)}{x^2} < \frac{37}{256}\tau^2.$$

Here $g < 1/64$ and $\tau(2 + \tau) < 9/16$. Thus $r_J^{\mathrm{DM}}/x > 91\tau^2/256 > 0$. DM is concave, and its vertex obeys

$$v_{\mathrm{DM}} = \frac{\tau}{4x(s - g/s)} < \frac{\tau}{256} < \tau W = \beta_{J-1},$$

since $x > 8$ and $s - g/s > 8$. So DM also strictly selects J. These are strict canonical common-domain inequalities; openness is established below. □

Rational reverse-stability witness. At $n = 7$ a compact exact witness is

$$(a, k, c, g, h) = (20301/1100, 17/20, 1/2, 1/100, 1), \quad x^* = 10, \quad M = 34/25, \quad y^* = 20301/1496. \tag{154}$$

Original residuals divided by their amplitudes are $20301/1100 - (37/20)10 + (49/100)/11 = 0$ for DM and $20301/1100 - (34/25)(20301/1496) = 0$ for MB. All coefficients and μ are positive. The ordered seven-node cosines u_1, u_2, u_3 are the three roots of $8u^3 + 4u^2 - 4u - 1$: use $1 + 2(\cos\theta + \cos 2\theta + \cos 3\theta) = 0$ at nonzero seventh roots and the Chebyshev identities. Rational isolating intervals are

$$31/50 < u_1 < 63/100, \quad -23/100 < u_2 < -11/50, \quad -91/100 < u_3 < -9/10. \tag{155}$$

The endpoint signs of the cubic are respectively $(-,+),(+,-),(-,+)$. Three disjoint sign changes exhaust its degree; cosine ordering assigns the labels. Thus no decimal trigonometric value is a proof premise. MB representatives $1,2,3$ have normalized real parts $-17/20+u_1/2-101u_2/100$, $-17/20+u_2/2-101u_3/100$, and $-17/20+u_3/2-101u_1/100$. The interval upper bounds are respectively $-3027/10000$, $-409/10000$, $-19262/10000$, all negative; mode zero is $-34/25$. DM instead satisfies

$$\frac{r_2^{\mathrm{DM}}}{10} > -\frac{17}{20}-\frac{21}{2420}-\frac{23}{22000}+\frac{12099}{12100}\frac{9}{10}=\frac{9729}{242000}>\frac{1}{25}>0.$$

Both select 2 strictly: $v_{\mathrm{MB}}=25/202<39/200<\beta_1$, $v_{\mathrm{DM}}=55/48396<39/200<\beta_1$, and $\beta_2<0$.

The full domain requires explicit branch labels. The preceding obstruction cannot be extended to all of $\mathcal{C}$ without qualification. A lower-branch construction gives the exact existence answer on that larger branch-labelled domain.

Proposition 10.12 (Reverse contrast when both DM branches are allowed)**.** *If $4\mid n$ and MB is strictly stable, every positive DM synchronous equilibrium is strictly stable. If $4\nmid n$, a nonempty open common region has MB strictly stable and the specified lower positive DM branch unstable with unique dominant representative zero.*

Proof. When $4\mid n$, MB's label $n/4$ has normalized real part $-k+S$. Its strict stability forces $k>S$. Bound (148), which did not require $\mu>0$, makes every DM real part negative.

Otherwise put $\rho_n=\max_m[-\cos(2\theta_m)]$. One has $0<\rho_n<1$: equality to one requires $4m=n$ modulo $2n$, impossible when $4\nmid n$, and a label nearest $n/4$ has negative double-angle cosine. Define

$$\begin{gathered}\varepsilon=(1-\rho_n)/24,\quad k=1,\quad c=h=\varepsilon,\quad g=(1+\varepsilon)^3+2\varepsilon,\\ a=(1+\varepsilon)(1+2\varepsilon)+\frac{\varepsilon}{1+\varepsilon}.\end{gathered}\tag{156}$$

All coefficients are positive and $M=1+g>0$. With $A=1+\varepsilon$, the two DM roots are exactly

$$x_-=\varepsilon,\qquad x_+=\varepsilon+\frac{\varepsilon}{(1+\varepsilon)^2}.$$

Direct substitution of x_- in $a-Ax+(c-g)/(1+x)$ gives zero. The second is checked by the quadratic factorization $A(x-x_-)(x-x_+)$ of P02's equilibrium polynomial, or by substitution. In particular $\mu=-A\varepsilon^2-\varepsilon^2/(1+\varepsilon)<0$, $b>0$, $D>0$, and both roots are positive. On the named lower branch,

$$\lambda_0^{\mathrm{DM}}=\frac{\varepsilon^2}{(1+\varepsilon)^2}>0,\qquad G-H=\varepsilon\left(1+\frac{2\varepsilon}{(1+\varepsilon)^2}\right)>0.$$

Thus zero is uniquely dominant and unstable. For MB, $S=(1+\varepsilon)^3+3\varepsilon<1+10\varepsilon$, since $0<\varepsilon<1$, and every normalized real part obeys

$$-1+\varepsilon\cos\theta_m-S\cos(2\theta_m)\le-1+\varepsilon+S\rho_n<-\frac{13}{24}(1-\rho_n)<0.$$

The MB original residual vanishes at $y^*=a/M$. Distinct simple positive DM roots and all strict inequalities persist locally on the specified lower branch. At $n=6$, $\rho_6=1/2$, so $\varepsilon=1/48$ gives an entirely rational witness. □

10.10 Mode/stability combinations and physical-parameter openness

All five requested types of combination have realizations, without a claim that each occurs in every dimension. Both stable with identical or different modes follows from Theorem 10.9; DM stable / MB unstable is universal by Theorem 10.10; the reverse has the branch and size restrictions just proved. For both unstable with different dominant modes take $n = 8$ and

$$(a, k, c, g, h) = (16351/1100, 1/2, 3/2, 1/100, 1), \quad x^* = 10, \quad M = 1/100, \quad y^* = 16351/11. \tag{157}$$

The original residuals divided by their amplitudes are $16351/1100 - 15 + (149/100)/11 = 0$ and $16351/1100 - (1/100)(16351/11) = 0$. Coefficients and μ are positive. MB's vertex $75/202$ lies strictly between $\sqrt{2}/4$ and $(2+\sqrt{2})/4$, so MB selects 1 with normalized maximum $-1/2+3\sqrt{2}/4 > 0$. DM's vertex $55/16132 < \sqrt{2}/4$ selects 2, whose normalized maximum is

$$-\frac{1}{2} - \frac{63}{2420} + \frac{12099}{12100} = \frac{2867}{6050} > 0.$$

The small value of M is strictly positive. MB's large amplitude is its own computed equilibrium, not an amplitude imported from DM.

Proposition 10.13 (Openness in all five physical coefficients)**.** *Every strict contrast and strict mode pair constructed here persists on a nonempty open set in $(a, k, c, g, h) \in (0, \infty)^5$ for its stated fixed size, offset, and DM branch. No openness assertion rests merely on an interval inside a displayed lower-dimensional family.*

Proof. On $\mathcal{C}_+$, $D > 0$ and the unique positive root x_+ is smooth in all five coefficients. Equivalently, the scalar equilibrium function $B(x) = a-(k+h)x+(c-g)/(1+x)$ has $B_x = -\sqrt{D}/(1+x_+) < 0$, so the implicit function theorem applies. MB's branch a/M is smooth for $M > 0$. Coefficient positivity, $M > 0$, and $\mu > 0$ are strict open conditions. At fixed n there are finitely many spectral inequalities; each is strict and continuous. A unique selected representative has a strictly positive gap to every other representative, while its conjugate labels remain tied. A sufficiently small intersection of these neighborhoods preserves all branch, sign, stability and selection assertions.

For the lower-branch family use instead $\mu < 0, b > 0, D > 0$. Both positive roots are simple and the specified x_- is smooth, with $B_x = +\sqrt{D}/(1+x_-) > 0$. The same continuity argument preserves that branch and its strict conclusions. No full-system IFT, later module, uniform neighborhood radius in n, or global branch identification is used. □

10.11 Global strengthening of the strict comparison regimes

Corollary 10.14 (Reverse contrast with globally attracting MB)**.** *Every reverse witness in Theorem 10.11 and Proposition 10.12, and every strict reverse region preserved by Proposition 10.13, has the following stronger conclusion: the specified positive DM homogeneous equilibrium is unstable, while MB's positive equilibrium is globally asymptotically stable in $\mathbb{R}^n_{>0}$. Likewise the MB equilibrium in every stable mode-pair realization of Theorem 10.9 is globally asymptotically stable in $\mathbb{R}^n_{>0}$.*

Proof. Each cited construction has strictly stable MB, equivalently $\delta > 0$ by Corollary 10.4; apply Theorem 10.5. The precise margins can also be checked directly. In (153), the maximizing representative is J and $\cos\theta_J = -\tau$, so

$$K_{n,2} = (1 + \tau^2/4)(1 - 2\tau^2) - \tau^2, \qquad \delta = \tau^2/4 + \tau^4/2 > 0.$$

The DM estimate already proved gives $r_J^{\mathrm{DM}}/x > 91\tau^2/256 > 0$. For (156), $4 \nmid n$ and $K_{n,2} \le \varepsilon + S\rho_n$, whence

$$\delta > \frac{13}{24}(1-\rho_n) > 0, \qquad \lambda_0^{\mathrm{DM}} = \frac{\varepsilon^2}{(1+\varepsilon)^2} > 0.$$

Both are the original tuples, with their original DM branch labels. □

Explicit certificates and exclusions. For the rational $n = 7$ tuple (154),

$$K_{7,2} = \tfrac{1}{2}\cos(4\pi/7) - \tfrac{101}{100}\cos(8\pi/7), \qquad \delta > 409/10000, \qquad r_2^{\mathrm{DM}} > 9729/24200.$$

These strict bounds follow from the rational isolating intervals (155); decimal trigonometry is not a proof premise. At $n = 6$, the lower-branch construction gives the exact tuple

$$(a, k, c, g, h) = (61177/56448, 1, 1/48, 122257/110592, 1/48),$$

with $x_- = 1/48$, $y^* = 5872992/11409601$, $K_{6,2} = 126865/221184$, $\delta = 94319/221184$, and $\lambda_0^{\mathrm{DM}} = 1/2401 > 0$. The universal witness (150) is instead the *forward* contrast: MB is unstable, with $K_{n,2} - k \ge 3/16$. It does not satisfy the global theorem. Nor does the both-unstable tuple (157), for which $\delta = 1/2 - 3\sqrt{2}/4 < 0$.

The exact residue-class restrictions retain their original strict-MB qualification: on $\mathcal{C}_+$ the reverse contrast exists exactly when $n \equiv 3 \pmod 4$, and allowing the specified lower branch gives existence exactly when $4 \nmid n$. These statements are not promoted to an unrestricted classification allowing $k = K_{n,q}$. P12 remains an unproved transversality module and is not used here.

10.12 Interpretation, deferred gates, and internal checks

The main comparison results are a universal forward stability contrast, both stability directions at $n = 7$ on the unique-positive-root domain, every ordered pair of reachable modes with both models stable, and exact parity/branch limitations on reverse stability. Identical abstract reachable labels do not imply identical behavior at the same physical tuple. The unrestricted reverse-stability hope fails on $\mathcal{C}_+$ unless $n \equiv 3 \pmod 4$; even allowing both DM branches cannot produce it when $4 \mid n$. The rejected phrase "MB is generically unstable" is not reinstated: MB has explicit open stable regions and an exact stability criterion.

For structural growth examples, $(a, k, c, g, h) = (1, 1, 3, 1, 1)$ gives $M = 0$, and $(1, 1, 4, 1, 1)$ gives $M = -1$. At $s_0 = 1$ the latter blows up at $\log 2$. DM has its P01 bounded flow for both. Its own positive root is 1 or $(\sqrt{33} - 1)/4$, respectively, each satisfying the original residual $x[a - (k + h)x + (c - g)/(1 + x)] = 0$. MB has no positive synchronous root in either case; its origin has zero residual, and the stated scalar trajectories have the exact original MB derivative. These are not common-equilibrium stability comparisons.

The six-term model is explicitly deferred. Existing source-provenance records are retained, but no archived six-term equation is imported. Full MB Hopf, first Lyapunov coefficient, crossing transversality, and criticality comparisons are deferred as well. The spectra and open regions are inputs for later P11–P13, not proofs of their hypotheses. A selected nonuniform pair, positive real part, or candidate imaginary crossing is insufficient for Hopf.

11 P09: Transcritical bifurcation at the origin

Fix canonical $a, k, c, h > 0$, $n \ge 5$, and $1 \le q < n$, and vary g. Throughout this module put

$$A = k + h > 0, \qquad b = a - A, \qquad g_c = a + c > 0, \qquad \mu = g_c - g. \tag{158}$$

Here b is a local abbreviation, not an archived comparator coefficient. Write $F(x,\mu)$ for the original canonical DM field with $g = g_c - \mu$:

$$F_i(x,\mu) = ax_i - kx_i^2 + \frac{cx_{i-1}}{1+x_i} - \frac{(g_c-\mu)x_i}{1+x_{i+q}} - hx_ix_{i-q}. \tag{159}$$

For $|\mu| < g_c/2$ all coefficients remain positive. The field is jointly C^∞ on $\mathcal{U} \times (-g_c/2, g_c/2)$, $\mathcal{U} = (-1,\infty)^n$, by its nonvanishing denominators and P01's local foundation (Proposition 3.2). No global invariance of $\mathcal{U}$ is needed. Neighborhoods may depend on all fixed parameters and on n; no uniform-in-n assertion is made.

The direct proof inputs are P02, P03 and P04. P02's Proposition 4.1 and (25) supply

$$F(r\mathbf{1},\mu) = f(r,\mu)\mathbf{1}, \qquad f(r,\mu) = \frac{r}{1+r}(\mu + br - Ar^2), \qquad r > -1. \tag{160}$$

This is the imported exact invariant restriction, not a new global classification.

11.1 Critical spectrum and full local equilibrium reduction

Lemma 11.1 (Simple synchronous kernel and transverse gap)**.** *At* $(x,\mu) = (0,0)$,

$$J_c = D_xF(0,0) = c(P-I), \qquad \lambda_m^c = c(e^{-\mathrm{i}\theta_m} - 1), \quad \theta_m = \frac{2\pi m}{n}. \tag{161}$$

Zero is algebraically simple, with real eigenspace $\operatorname{span}\{\mathbf{1}\}$. *Every transverse eigenvalue satisfies*

$$\operatorname{Re}\lambda_m^c \le -\delta_n \quad (1 \le m < n), \qquad \delta_n = c\left(1 - \cos\frac{2\pi}{n}\right) > 0. \tag{162}$$

No condition on q *or* $\gcd(n,q)$ *is imposed beyond the canonical range.*

Proof. P03 gives $D_xF(0,\mu) = \mu I + c(P-I)$ and its simple origin spectrum. For $m \neq 0$, $\cos\theta_m \le \cos(2\pi/n) < 1$, which gives the stated gap, independently of q. □

Theorem 11.2 (Every sufficiently small equilibrium is synchronous)**.** *For every fixed canonical tuple, including* $b = 0$, *there are state and parameter neighborhoods of zero in which the entire full-system equilibrium equation is equivalent to*

$$x = r\mathbf{1} + y, \qquad y = 0, \qquad r(\mu + br - Ar^2) = 0. \tag{163}$$

In particular, no nearby nonsynchronous equilibrium branch is omitted.

Proof. Set $w^{\mathsf{T}} = n^{-1}\mathbf{1}^{\mathsf{T}}$, $E = \ker w^{\mathsf{T}}$, $\Pi_\perp = I - \mathbf{1}w^{\mathsf{T}}$, and $x = r\mathbf{1}+y$, $y \in E$. The transverse equation $T(r,y,\mu) = \Pi_\perp F(r\mathbf{1}+y,\mu) = 0$ has derivative $D_yT(0,0,0) = J_c|_E$, invertible by Lemma 11.1. The implicit-function theorem [2, Ch. 1] gives a unique nearby solution $y = \psi(r,\mu)$. Synchrony already gives $T(r,0,\mu) = 0$, so uniqueness implies

$$\psi(r,\mu) \equiv 0. \tag{164}$$

The remaining equation is $f(r,\mu) = 0$; since $1+r > 0$ it is exactly (163), including when $b = 0$. □

11.2 Local derivatives and full-vector nondegeneracy

Proposition 11.3 (Transcritical coefficients and parameter orientation)**.** *The local scalar derivatives are*

$$f(0,\mu)=0, \quad f_r(0,0)=0, \quad f_{r\mu}(0,0)=1, \quad f_{rr}(0,0)=2b. \tag{165}$$

Choose $v=\mathbf{1}$, $w^{\mathsf{T}}=n^{-1}\mathbf{1}^{\mathsf{T}}$, *so* $J_c v=0$, $w^{\mathsf{T}}J_c=0$, *and* $w^{\mathsf{T}}v=1$. *The full vector field satisfies*

$$F(0,\mu)=0, \quad w^{\mathsf{T}}F_{x\mu}(0,0)v=1, \quad w^{\mathsf{T}}F_{xx}(0,0)[v,v]=2b. \tag{166}$$

The quadratic Taylor coefficient includes the factor $1/2$ *and is* b. *With* g *as parameter the mixed coefficient is instead* $w^{\mathsf{T}}F_{xg}(0,g_c)v=-1$. *Ordinary transcritical nondegeneracy holds exactly when* $b\neq 0$.

Proof. Equation (160) gives $f_r(0,\mu)=\mu$ and $f_\mu(r,\mu)=r/(1+r)$, whose r derivative at zero is one. At $\mu=0$, $f(r,0)=br^2-(A+b)r^3+O(r^4)$, so $f_{rr}=2b$.

Independently, every term of the original field vanishes at $x=0$. Its parameter derivative is $(F_\mu)_i = x_i/(1+x_{i+q})$, whose state derivative at zero is the identity. Using g instead gives $-x_i/(1+x_{i+q})$, hence minus the identity. The second directional derivatives of the five original terms along v are respectively $0,-2k,-2c,2g_c,-2h$. Their sum is $2(g_c-c-k-h)=2b$ in each component. Averaging by w proves the full-vector identities without a missing factor of n. Also $F_\mu(0,0)=0$, as required by persistence of the origin branch. □

11.3 Generic full-system transcritical theorem

Theorem 11.4 (Nondegenerate transcritical bifurcation at $g_c=a+c$)**.** *Suppose* $b=a-k-h\neq 0$. *For sufficiently small* $|\mu|$, *all full-system equilibria sufficiently near the origin are exactly*

$$x=0 \quad \textit{and} \quad x=r_*(\mu)\mathbf{1}, \qquad r_*(0)=0, \quad r_*(\mu)\neq 0 \ (\mu\neq 0), \tag{167}$$

where r_* *is smooth and*

$$r_*'(0)=-\frac{1}{b}, \qquad r_*(\mu)=-\frac{\mu}{b}+\frac{A\mu^2}{b^3}+O(\mu^3), \qquad \frac{dr_*}{dg}(g_c)=\frac{1}{b}. \tag{168}$$

The two curves have distinct tangent directions in the synchronous state–parameter plane. The critical eigenspace is one-dimensional and synchronous; all transverse modes are strictly stable. In $\mathcal{U}$, *the full nonlinear stability exchange is*

Side	*Origin*	*Nearby nonzero branch*
$\mu<0$ $(g>g_c)$	*Locally exponentially stable*	*Unstable*
$\mu>0$ $(g<g_c)$	*Unstable*	*Locally exponentially stable*

Together with the full-vector nondegeneracy coefficients, these facts give a nondegenerate full-system local transcritical bifurcation.

Proof. Theorem 11.2 first makes the equilibrium problem complete in the full state space. Apart from the persistent origin, solve $Q(r,\mu)=Ar^2-br-\mu=0$. Since $Q_r(0,0)=-b\neq 0$, the implicit-function theorem gives a unique smooth root $r_*(\mu)$ in a product neighborhood. The identity $Q(0,\mu)=-\mu$ shows that this root is nonzero exactly for $\mu\neq 0$. This proves the full local list. For fixed $b\neq 0$, the other root b/A at $\mu=0$ is excluded by shrinking the state neighborhood.

Differentiating $Q(r_*(\mu),\mu) = 0$ gives $(2Ar_* - b)r'_* = 1$, hence $r'_*(0) = -1/b$. Differentiating again gives $2A(r'_*)^2 + (2Ar_* - b)r''_* = 0$, hence $r''_*(0) = 2A/b^3$. Smoothness proves the expansion. In (r,μ) coordinates the tangents are $(0,1)$ and $(-1/b,1)$, which are independent. This is transverse crossing in that plane, not a claim that two curves span the full $(n+1)$-dimensional ambient space. Since $d\mu/dg = -1$, the physical g slope has the opposite sign. Increasing g decreases μ.

The standard transcritical and center-manifold framework is [2, Ch. 3]. Here the center dynamics is explicit: At $\mu = 0$ the synchronous line is smooth, invariant, and tangent to the entire center eigenspace, so is a local one-dimensional center manifold. For the extended system $(\dot{x},\dot{\mu}) = (F(x,\mu),0)$, its product with the parameter interval is an exact two-dimensional center manifold: $F_\mu(0,0) = 0$, the extended linearization is $\operatorname{diag}(J_c,0)$, and all remaining eigenvalues are strictly stable. On this product use

$$u = Ar^2 - br, \qquad \frac{du}{dr}(0) = -b \neq 0.$$

This is a local smooth coordinate. Direct substitution yields

$$\dot{u} = \alpha(r)u(\mu - u), \qquad \alpha(r) = \frac{2Ar - b}{(Ar - b)(1 + r)}, \qquad \alpha(0) = 1. \tag{169}$$

Near zero the denominators do not vanish and $\alpha > 0$. Thus $d\tau/dt = \alpha(r)$ gives exactly $du/d\tau = u(\mu - u)$, the nondegenerate transcritical center dynamics. This is orbital equivalence of the exact center restriction, not exact decoupling of the full field.

Full stability still requires all Fourier modes. Differentiating first,

$$f_r(r,\mu) = \frac{\mu + br - Ar^2}{(1 + r)^2} + \frac{r}{1 + r}(b - 2Ar).$$

At the nonzero equilibrium the first numerator vanishes. By P04's (72), its uniform eigenvalue is

$$\lambda_0^{\mathrm{nz}} = \frac{r_*}{1 + r_*}(b - 2Ar_*) = -\mu\frac{b - 2Ar_*}{(1 + r_*)(b - Ar_*)}. \tag{170}$$

The last equality uses $\mu = -r_*(b - Ar_*)$. Choose $|r_*| < 1/2$ and $2A|r_*| < |b|/2$. Both $b - Ar_*$ and $b - 2Ar_*$ have the sign of b, so the final fraction is positive. Therefore, exactly,

$$\operatorname{sgn}\lambda_0^{\mathrm{nz}} = -\operatorname{sgn}\mu, \qquad \lambda_0^{\mathrm{nz}} = -\mu + O(\mu^2). \tag{171}$$

The sign argument, rather than the expansion, gives stability.

For the transverse modes use the general P04 coefficients, valid also at zero or negative $r > -1$:

$$\begin{aligned} d(r,\mu) &= a - 2kr - \frac{cr}{(1 + r)^2} - \frac{g_c - \mu}{1 + r} - hr, \\ C(r) &= \frac{c}{1 + r}, \quad G(r,\mu) = \frac{(g_c - \mu)r}{(1 + r)^2}, \quad H(r) = hr, \\ \lambda_m(r,\mu) &= d + Ce^{-\mathrm{i}\theta_m} + Ge^{\mathrm{i}q\theta_m} - He^{-\mathrm{i}q\theta_m}. \end{aligned} \tag{172}$$

Every expression is continuous at $(r,\mu) = (0,0)$, where it equals λ_m^c. Finitely many labels allow one common neighborhood with $|\lambda_m(r,\mu) - \lambda_m^c| < \delta_n/2$ for every $m \neq 0$. Thus for some $\eta > 0$,

$$|r| + |\mu| < \eta \quad \Longrightarrow \quad \operatorname{Re}\lambda_m(r,\mu) < -\delta_n/2 \quad (m \neq 0). \tag{173}$$

No mode-selection theorem is used.

At the origin $\lambda_0 = \mu$, and every other real part is $\mu - c(1 - \cos\theta_m) < -\delta_n/2$ if $|\mu| < \delta_n/2$. On the nonzero branch use (171) and (173). A Hurwitz Jacobian gives local exponential stability of the smooth nonlinear equilibrium; a positive eigenvalue gives nonlinear instability, as in P04's nonlinear implications. Every unstable equilibrium here has exactly one unstable real direction. This proves the table and the full-system theorem, without using P05, P07 or P08. □

Corollary 11.5 (Physical positive-branch orientation)**.** *For sufficiently small nonzero μ, $r_*(\mu) > 0$ exactly when $\operatorname{sgn}\mu = -\operatorname{sgn} b$. If $a < k + h$, this branch is positive and locally exponentially stable for $\mu > 0$, equivalently $g < g_c$, while the origin is unstable. If $a > k + h$, it is positive and unstable for $\mu < 0$, equivalently $g > g_c$, while the origin is locally exponentially stable. Origin instability for $\mu > 0$ is also visible relative to $\mathbb{R}^n_{\geq 0}$; for $\mu < 0$ exponential stability restricts to that state space.*

Proof. Use the exact relation $\mu = -r_*(b - Ar_*)$ and the sign of $b - Ar_*$, then the full stability table. The unstable origin eigenvector $v = \mathbf{1}$ is strictly positive. Explicitly, at any fixed small $\mu > 0$, $f(r,\mu)/r \to \mu$ as $r \downarrow 0$. Choose $\varepsilon > 0$ such that $f(r,\mu) \geq (\mu/2)r$ on $0 < r \leq \varepsilon$. Arbitrarily small positive synchronous states reach $r = \varepsilon$ in finite time. Their full-system lifts remain nonnegative, proving relative instability. For $\mu < 0$, full local exponential stability and P01's nonnegative invariance give the restriction. An unstable positive branch point is interior to $\mathbb{R}^n_{\geq 0}$; small positive synchronous perturbations also show its relative instability. P09's local branch is not any additional positive root away from zero allowed by P02 when $b > 0$. □

Proposition 11.6 (Generic critical origin in the open domain)**.** *For $\mu = 0$, $b \neq 0$, the origin is nonlinearly unstable in $\mathcal{U}$. For $b > 0$ this is visible from nonnegative synchronous states. No full relative-orthant stability conclusion is asserted at $\mu = 0$, $b < 0$.*

Proof. The exact invariant-line dynamics is

$$\dot{r} = \frac{r^2(b - Ar)}{1 + r}. \tag{174}$$

Choose $0 < \varepsilon < 1/2$ with $A\varepsilon < |b|/2$. For $0 < |r| \leq \varepsilon$ the sign is that of b, and the magnitude is at least $|b|r^2/3$. For $b > 0$ take $r(0) > 0$; for $b < 0$ take $r(0) < 0$. On that side $s = |r|$ satisfies $\dot{s} \geq (|b|/3)s^2$ until $s = \varepsilon$, which must occur in finite time. Arbitrarily small data therefore exit a fixed neighborhood. The exact synchronous lift stays in $\mathcal{U}$ up to that exit. When $b < 0$ these initial states are negative, so this proof does not establish relative-orthant instability. □

11.4 The cubic-degenerate case and its precise scope

Proposition 11.7 (Degenerate geometry and full nonlinear stability)**.** *Let $b = 0$, that is $a = A = k + h$. Ordinary transcritical quadratic nondegeneracy fails. All sufficiently small full-system equilibria are synchronous, with the following exact local geometry and stability.*

(i) For $\mu < 0$, only the origin exists locally; it is locally exponentially stable.

(ii) At $\mu = 0$, only the origin exists locally. It is a triple zero of the scalar field and is locally asymptotically stable in the full open domain $\mathcal{U}$, but not locally exponentially stable.

(iii) For $\mu > 0$, there are precisely the origin and

$$x = r_\pm \mathbf{1}, \qquad r_\pm = \pm\sqrt{\mu/A}. \tag{175}$$

For sufficiently small $\mu > 0$, both nonzero equilibria belong to $\mathcal{U}$ and are locally exponentially stable there. The origin is unstable, also relative to $\mathbb{R}^n_{\geq 0}$. Only $r_+\mathbf{1}$ is physically nonnegative.

The two nonzero local roots occur for $g < g_c$; for $g > g_c$ only the stable origin remains locally. This is a cubic-degenerate origin with square-root nonzero branches. No pitchfork classification or two-parameter unfolding is asserted.

Proof. Theorem 11.2 does not require $b \neq 0$. Now $f(r, \mu) = r(\mu - Ar^2)/(1 + r)$, and the nonzero equation $Ar^2 = \mu$ gives the complete list. For $0 < \mu < A$, the negative root exceeds -1; shrink the interval further to put both roots within the reduction and spectral-gap neighborhoods. At $\mu = 0$,

$$f(r, 0) = -\frac{Ar^3}{1 + r}, \qquad f_{rrr}(0, 0) = -6A \neq 0. \tag{176}$$

The scalar zero is therefore triple. At either nonzero root,

$$\lambda_0 = f_r(r_\pm, \mu) = -\frac{2Ar_\pm^2}{1 + r_\pm} < 0. \tag{177}$$

The proof of (173) never used $b \neq 0$; it supplies strict stability of all other Fourier modes. Thus both full equilibria are locally exponentially stable. At the origin $\lambda_0 = \mu$ gives the off-critical stability; the positive synchronous escape argument above proves relative instability for $\mu > 0$.

At the fixed value $\mu = 0$, the smooth invariant synchronous line passes through zero and is tangent to the whole one-dimensional center subspace. All complementary eigenvalues have real part at most $-\delta_n$, and there is no unstable subspace. On this exact center manifold, $r^2/2$ has derivative $-Ar^4/(1 + r) < 0$ for small $r \neq 0$, so its origin is locally asymptotically stable. These verify the hypotheses of the finite-dimensional center-manifold stability reduction principle: a smooth local system with center and strictly stable spectra inherits local asymptotic stability from its center restriction. No parameter direction is included in this fixed-parameter stability statement.

This applies the center-manifold stability reduction principle of [2, Ch. 3]; the displayed gap and exact invariant restriction verify its hypotheses.

Convergence is not exponential. Each sufficiently small nonzero synchronous solution keeps its sign, tends to zero, and obeys

$$\frac{d}{dt}(r^{-2}) = \frac{2A}{1 + r(t)} \longrightarrow 2A.$$

It follows that $|r(t)| \sim (2At)^{-1/2}$ as $t \to \infty$. These exact full-system solutions exclude a local exponential stability estimate. No common asymptotic rate for every nonsynchronous trajectory is asserted. □

Proposition 11.8 (Codimension and the meeting of the algebraic loci)**.** *For fixed admissible (n, q), the degenerate origin locus is a smooth codimension-two submanifold of $(0, \infty)^5$, given by*

$$a - k - h = 0, \qquad g - a - c = 0. \tag{178}$$

The branch-intersection and P02 repeated-root loci meet there. This is an algebraic/spectral codimension statement only.

Proof. In coordinates (a, k, c, g, h) the gradients are

$$(1, -1, 0, 0, -1), \qquad (-1, 0, -1, 1, 0).$$

They are independent: the g coordinate forces the coefficient of the second vector to be zero, and the k coordinate then forces the first to be zero. The regular level-set theorem proves codimension

two. The locus is nonempty: any $k, c, h > 0$ give $a = k + h$, $g = a + c > 0$. The center eigenvalue remains simple, not a double-zero spectral degeneracy.

P02's Proposition 4.5 gives

$$g_f = a + c + \frac{(a-A)^2}{4A}, \qquad r_f = \frac{a-A}{2A}.$$

Hence $g_f = g_c$ exactly at $a = A$, where $r_f = 0$. The repeated root is positive only for $a > A$; the meeting point is the origin, not a positive nonzero equilibrium. Only this algebraic overlap is recorded. It is not a saddle-node proof; that remains P10. □

12 P10: Saddle-node bifurcation and true bistable region

Fix $a, k, c, h > 0$, $n \geq 5$, $1 \leq q < n$, and vary $g > 0$ in the canonical DM field ($\mathrm{DM}_{n,q}$). Put

$$A = k + h, \quad b = a - A > 0, \quad g_c = a + c, \quad g_f = g_c + \frac{b^2}{4A}, \quad r_f = \frac{b}{2A}, \quad s_f = 1 + r_f. \tag{179}$$

Here r is the synchronous amplitude and b is a local abbreviation. *Notation reconciliation:* P02's roots s_-, s_+ are the amplitudes called r_-, r_+ here. P02's repeated-root amplitude s_f in (35) is our r_f; P10's s_f denotes its denominator $1 + r_f$, following P03/P04's denominator convention. No earlier notation or file is changed. All neighborhoods below may depend on the fixed parameters and on (n, q); no uniform spectral gap as $n \to \infty$ is claimed.

The direct inputs are P02's invariant restriction and root formulas, P03's original-field Jacobian and origin stability, P04's complete Fourier spectrum, and P09's origin branch organization. P01 supplies smoothness, uniqueness and global nonnegative existence. We also use the real implicit-function theorem, the usual smooth local stable/unstable manifold theorem, and linearized stability/instability. These standard local results introduce no new project-module dependency or literature work.

12.1 Exact fold geometry

P02 gives, for $r > -1$,

$$F(r\mathbf{1}, g) = f(r, g)\mathbf{1}, \qquad f(r, g) = ar - Ar^2 + \frac{(c-g)r}{1+r} = \frac{r}{1+r}(\mu + br - Ar^2), \quad \mu = g_c - g. \tag{180}$$

The nonzero equilibrium equation is $Ar^2 - br - \mu = 0$. Completing the square, or using its discriminant, verifies independently

$$g_f = c + \frac{(a+A)^2}{4A}, \qquad s_f = 1 + \frac{a-A}{2A} = \frac{a+A}{2A}, \qquad g_f - g_c = \frac{b^2}{4A} > 0. \tag{181}$$

Define the state and parameter coordinates

$$u = r - r_f, \qquad \nu = g_f - g. \tag{182}$$

Then $\mu = \nu - b^2/(4A)$, $br_f - Ar_f^2 = b^2/(4A)$, and $b - 2Ar_f = 0$. Expanding in u therefore proves, with no approximation,

$$\boxed{\mu + br - Ar^2 = \nu - Au^2, \qquad \dot{u} = \frac{r_f + u}{s_f + u}(\nu - Au^2).} \tag{183}$$

Choose $|u| < r_f/2$. Both amplitude and denominator are positive. For $\nu < 0$ there is no nonzero synchronous equilibrium anywhere on $r > -1$; in particular none near r_f. For $\nu = 0$ there is one double positive scalar equilibrium r_f. For small $\nu > 0$ there are two distinct nearby positive roots

$$\boxed{r_\pm = r_f \pm \sqrt{\nu/A} = s_\pm \text{ of P02.}} \tag{184}$$

Both are positive exactly when $0 < \nu < b^2/(4A)$ within this real root regime. The origin always exists, but is excluded from the chosen fold state neighborhood.

12.2 Complete fold spectrum and the real transverse reduction

For a synchronous state let $s = 1 + r$ and import P03/P04:

$$d = a - 2kr - \frac{cr}{s^2} - \frac{g}{s} - hr, \qquad C = \frac{c}{s}, \quad G = \frac{gr}{s^2}, \quad H = hr.$$

$$\begin{aligned} J(r, g) &= dI + CP + GP^{-q} - HP^q, \\ \lambda_m &= d + Ce^{-\mathrm{i}\theta_m} + Ge^{\mathrm{i}q\theta_m} - He^{-\mathrm{i}q\theta_m}, \end{aligned} \tag{185}$$

where $\theta_m = 2\pi m/n$ and $0 \le m < n$. Colliding offsets are added, as in P03; none is removed. At a positive equilibrium P04 gives

$$\lambda_0 = f_r(r, g) = \frac{r}{1 + r}(b - 2Ar). \tag{186}$$

Thus $\lambda_0^f = 0$, also obtained by differentiating (183) at $(u, \nu) = (0, 0)$.

Lemma 12.1 (Automatic transverse stability at every positive fold)**.** *At the fold, with subscript f denoting evaluation there,*

$$\boxed{G_f - H_f = r_f \left(k + \frac{4A^2 c}{(a + A)^2} \right) > 0.} \tag{187}$$

The exact complex fold spectrum is (185)*, with $d_f = -C_f - G_f + H_f$. In particular,*

$$\operatorname{Re} \lambda_m^f = C_f(\cos \theta_m - 1) + (G_f - H_f)(\cos(q\theta_m) - 1), \tag{188}$$

$$\operatorname{Im} \lambda_m^f = -C_f \sin \theta_m + (G_f + H_f) \sin(q\theta_m). \tag{189}$$

For every $m \neq 0$,

$$\boxed{\operatorname{Re} \lambda_m^f \le -\delta_f < 0, \qquad \delta_f = \frac{c}{s_f} \left(1 - \cos \frac{2\pi}{n} \right) > 0.} \tag{190}$$

The only eigenvalue on the imaginary axis is zero. It has algebraic and geometric multiplicity one and real eigenspace $\operatorname{span}\{\mathbf{1}\}$.

Proof. Using (181), directly

$$\frac{g_f}{s_f^2} - h = \frac{c + As_f^2}{s_f^2} - h = k + \frac{c}{s_f^2} = k + \frac{4A^2 c}{(a + A)^2} > 0. \tag{191}$$

Multiplication by $r_f > 0$ proves (187). Since $0 = \lambda_0^f = d_f + C_f + G_f - H_f$, taking real and imaginary parts of the complete P04 formula proves the two identities. For $1 \le m < n$, set

$j = \min(m, n-m)$. The angle $2\pi j/n$ belongs to $[2\pi/n, \pi]$, so $\cos\theta_m \le \cos(2\pi/n) < 1$. The first cosine difference is strictly negative, while the second is nonpositive. This proves the margin. For even n the real singleton $m = n/2$ is included; for odd n the same argument ends before π. Collisions $q = 1, n-1, n/2$ and $\gcd(n,q) > 1$ change neither argument. Even if $q\theta_m$ is a multiple of 2π, the first term is strictly negative. P04's unitary diagonalization lists exactly n eigenvalues with algebraic multiplicity. Only label zero can vanish, so its multiplicities are one and its eigenvector is $\mathbf{1}$. No simplicity of the other numerical eigenvalues is asserted. □

Proposition 12.2 (Full-system local equilibrium completeness)**.** *In a state neighborhood of $r_f\mathbf{1}$ and parameter interval about g_f, every full-system equilibrium is synchronous. The entire local equilibrium equation reduces exactly to $\nu - Au^2 = 0$.*

Proof. Use $x = r\mathbf{1} + y$ with $E = \{y : \mathbf{1}^\mathsf{T} y = 0\}$ and $\Pi_\perp = I - \mathbf{1}\mathbf{1}^\mathsf{T}/n$. By Lemma 12.1, the transverse derivative of $\Pi_\perp F(r\mathbf{1} + y, g)$ at the fold is the invertible map $J_f|_E$. Apply the implicit-function and synchrony-uniqueness argument of Theorem 11.2 [2, Ch. 1]. It gives $y = 0$ for every nearby equilibrium. Since $r > 0$, the remaining equation is $\nu - Au^2 = 0$ by (183). □

12.3 Full-system nondegeneracy and exact center dynamics

Proposition 12.3 (Original-vector-field saddle-node coefficients)**.** *With the above v, w, one has $J_f v = 0$, $w^\mathsf{T} J_f = 0$, $w^\mathsf{T} v = 1$, and*

$$\alpha_{\mathrm{SN}} := w^\mathsf{T} F_g(r_f\mathbf{1}, g_f) = -\frac{r_f}{s_f} < 0, \tag{192}$$

$$w^\mathsf{T} F_{xx}(r_f\mathbf{1}, g_f)[v, v] = -\frac{2Ar_f}{s_f} \neq 0, \tag{193}$$

$$\beta_{\mathrm{SN}} := \tfrac{1}{2} w^\mathsf{T} F_{xx}[v, v] = -\frac{Ar_f}{s_f} < 0. \tag{194}$$

Derivatives hold all other physical coefficients fixed. With ν as parameter the parameter coefficient has the opposite sign, $+r_f/s_f$.

Proof. The row and column sums of J_f are both $\lambda_0^f = 0$, giving the right and left nullvectors and their normalization. In the original vector field, $(F_g)_i = -x_i/(1 + x_{i+q})$, hence its average at the fold is $-r_f/s_f$.

Independently differentiate each original summand twice along $x = (r_f + t)\mathbf{1}$ at fixed $g = g_f$. The five componentwise second derivatives, retaining all terms even at coincident offsets, are

$$0, \qquad -2k, \qquad -\frac{2c}{s_f^3}, \qquad \frac{2g_f}{s_f^3}, \qquad -2h.$$

Their sum is $-2A + 2(g_f - c)/s_f^3$. Since $g_f - c = As_f^2$, this equals $-2A + 2A/s_f = -2Ar_f/s_f$ in every component. Averaging produces no extra factor of n. The Taylor coefficient of u^2 is half this full contraction; the derivatives themselves contain no Taylor factorial. □

Theorem 12.4 (Nondegenerate full-system positive saddle-node)**.** *For every fixed canonical $a > k + h$, $c, k, h > 0$ and admissible (n, q), the equilibrium $r_f\mathbf{1}$ at*

$$\boxed{g = g_f = a + c + \frac{(a - k - h)^2}{4(k + h)}}$$

is a nondegenerate full-system saddle-node. It has a simple uniform zero eigenvalue and strictly stable transverse spectrum. Locally there are no equilibria for $g > g_f$, one double equilibrium for $g = g_f$, and precisely the two positive equilibria (184) for $g < g_f$ sufficiently close. The exact center restriction is orbitally equivalent, by a positive change of time, to

$$\boxed{\frac{du}{d\tau} = \nu - Au^2, \qquad \frac{d\nu}{d\tau} = 0.} \tag{195}$$

Proof. The field is smooth near the positive fold. Lemma 12.1 and Proposition 12.3 verify the simple-zero and nonzero parameter/quadratic coefficients of the saddle-node theorem [5, 2]; Proposition 12.2 gives local equilibrium completeness. For the parameter extension, the linearization in (u, y, ν) is

$$\dot{u} = (r_f/s_f)\nu, \qquad \dot{y} = (J_f|_E)y, \qquad \dot{\nu} = 0. \tag{196}$$

Thus the invariant surface $y = 0$ is tangent to the whole generalized center subspace and is an exact center manifold. On it, (183) and the positive factor $(r_f + u)/(s_f + u)$ for $|u| < r_f/2$ give (195) by $d\tau/dt = (r_f + u)/(s_f + u)$. The roots and their sides follow immediately. At $\nu = 0$, negative u escapes, so the fold is not locally asymptotically stable. □

12.4 Both branches and all Fourier modes

Proposition 12.5 (Exact uniform signs throughout the two-root interval)**.** *For every $g_c < g < g_f$, define D as in P02 and $\Delta = \sqrt{D}$. Then*

$$0 < \Delta = \sqrt{b^2 + 4A(a + c - g)} = 2\sqrt{A(g_f - g)} < b,$$

$$r_\pm = \frac{b \pm \Delta}{2A}, \qquad b - 2Ar_\pm = \mp\Delta, \tag{197}$$

$$\boxed{\lambda_0(r_\pm) = \mp\frac{r_\pm}{1 + r_\pm}\Delta.} \tag{198}$$

Thus the lower branch is always unstable in the uniform direction, whereas the upper uniform eigenvalue is always strictly negative. Any upper-branch loss of stability within this open interval must involve a nonuniform mode.

Proof. P02's discriminant factorization is also obtained directly by inserting $g_f = g_c + b^2/(4A)$. The strict parameter inequalities give $0 < \Delta < b$, so both roots are positive. Substitution into $b - 2Ar_\pm$ cancels b and leaves $\mp\Delta$. Equation (186) gives the exact signs, and the positive lower eigenvalue implies nonlinear instability. This last implication needs no hyperbolicity of the other lower-branch eigenvalues. □

Corollary 12.6 (Near-fold full branch stability)**.** *There exists $\varepsilon_{\mathrm{sp}} > 0$ such that both branches for $0 \le \nu < \varepsilon_{\mathrm{sp}}$ satisfy*

$$\operatorname{Re}\lambda_m(r_\pm, g_f - \nu) < -\delta_f/2 \quad (m \neq 0).$$

For $0 < \nu < \min(\varepsilon_{\mathrm{sp}}, b^2/(4A))$, the upper branch is Hurwitz and locally exponentially stable. The lower branch is a hyperbolic saddle with exactly one simple positive real eigenvalue and $n - 1$ stable real dimensions. Its local stable manifold has dimension $n-1$ and codimension one, and its unstable manifold has dimension one.

Proof. Every labeled eigenvalue in (185) is continuous in (r, g) near (r_f, g_f). There are finitely many labels. A common neighborhood makes each transverse real part differ from its fold value by less than $\delta_f/2$. Both continuous roots $r_\pm(\nu)$ tend to r_f as $\nu \downarrow 0$; smoothness in ν at zero is not needed. Shrink $\varepsilon_{\mathrm{sp}}$ so both branches lie in that neighborhood and remain positive. Combine the margin with (198). Only label zero has positive real part on the lower branch, so its eigenvalue is simple. Real conjugate-pair blocks and real singleton modes together give $n-1$ stable dimensions, even if stable numerical eigenvalues repeat. P04's nonlinear stability implications and the smooth local stable/unstable manifold theorem apply. □

12.5 The nonempty open bistability set

Theorem 12.7 (Stable coexistence and openness in physical parameters)**.** *For each fixed $a, k, c, h > 0$ with $a > k + h$ and each admissible (n, q), there is $\varepsilon > 0$ with*

$$(g_f - \varepsilon, g_f) \subset (g_c, g_f) \tag{199}$$

such that throughout this interval the full system has the locally exponentially stable origin, a positive index-one saddle $r_-\mathbf{1}$, and the locally exponentially stable positive equilibrium $r_+\mathbf{1}$. At every strict point of a sufficiently small such interval this configuration persists on an open neighborhood in $(0, \infty)^5$ with coordinates (a, k, c, g, h) and fixed (n, q). Here "bistability" means coexistence of at least two locally stable equilibria; it does not mean that these are the only attractors.

Proof. At the origin P03 gives

$$\operatorname{Re}\lambda_m(0) = g_c - g - c(1 - \cos\theta_m) \le g_c - g < 0 \quad (g_c < g < g_f).$$

Thus the origin is Hurwitz and locally exponentially stable in $\mathcal{U}$, and also relative to $\mathbb{R}^n_{\ge 0}$ using P01 invariance. Take $0 < \varepsilon < \min(\varepsilon_{\mathrm{sp}}, b^2/(4A))$ and apply Corollary 12.6. The interval is nonempty and lies strictly above g_c.

At a strict witness all three Jacobians are invertible and hyperbolic. The implicit-function theorem continues each equilibrium smoothly under changes of all five physical coefficients. The origin stays identically zero. Alternatively the positive roots are explicitly smooth because $D > 0$; uniqueness identifies them with the IFT continuations. Their separation and strict positivity persist after shrinking the parameter neighborhood. Spectral continuity preserves both Hurwitz properties and the lower saddle's unstable dimension. The inequalities $a > A$, $g_c < g < g_f$ and positivity of every coefficient also persist. Such neighborhoods can depend on (n, q); no common all-dimension persistence neighborhood for the index-one configuration is asserted. □

Proposition 12.8 (Exact coexistence set and a simple sufficient criterion)**.** *For fixed (n, q) define the open two-positive-root domain*

$$\mathcal{D} = \{\boldsymbol{\pi} \in (0, \infty)^5 : a > k + h,\ g_c < g < g_f\}.$$

For $\boldsymbol{\pi} \in \mathcal{D}$, evaluate (185) *at its upper root and define*

$$\boxed{\mathcal{B}_{n,q} = \left\{\boldsymbol{\pi} \in \mathcal{D} : \max_{0 \le m < n} \operatorname{Re}\lambda_m(r_+, g) < 0\right\}.} \tag{200}$$

This set is open and nonempty. It is exactly the set in $\mathcal{D}$ where the origin and upper synchronous equilibrium are both locally exponentially stable. Every point has an unstable lower positive branch,

although its unstable dimension is asserted to be one only in the near-fold neighborhood above. A sufficient condition for membership in $\mathcal{B}_{n,q}$ is

$$\boxed{G \geq H \quad \Longleftrightarrow \quad \frac{g}{(1+r_+)^2} \geq h.} \tag{201}$$

This condition allows equality.

Proof. The positive upper root and all its labeled eigenvalues are continuous (and smooth before taking the finite maximum) on $\mathcal{D}$. The strict finite set of inequalities defines an open subset of $(0,\infty)^5$, hence of $\mathbb{R}^5$. Nonemptiness follows from Theorem 12.7. The origin is automatically Hurwitz on $\mathcal{D}$. A Hurwitz Jacobian gives nonlinear local exponential stability. Conversely, a smooth equilibrium satisfying a local exponential estimate has a Jacobian semigroup satisfying that estimate by differentiation of the flow at the equilibrium, so all eigenvalues have strictly negative real part. Thus the definition is exact for exponential stable coexistence, not a classification of possible nonhyperbolic asymptotically stable boundary points or other attractors.

Subtracting the uniform row sum in (185) gives, at either branch and in fact at any synchronous state,

$$\operatorname{Re}\lambda_m = \lambda_0 + C(\cos\theta_m - 1) + (G-H)(\cos(q\theta_m) - 1). \tag{202}$$

On the upper branch $\lambda_0 < 0$ and $C > 0$. If $G - H \geq 0$, every mode has negative real part, including $m = 0$; for $m \neq 0$ the C contribution is strictly negative. Since $r_+ > 0$, dividing $G - H$ by it gives the physical-coefficient condition. The lower uniform sign follows from Proposition 12.5. □

One can give an explicit, dimension-independent sufficient interval for the two stable equilibria (not necessarily for the saddle's index). Set

$$M = g_f - hs_f^2 = c + ks_f^2 > 0, \qquad t_* := \min\left\{\frac{r_f}{2}, \frac{M}{4hs_f}, \sqrt{\frac{M}{2(A+h)}}\right\} > 0. \tag{203}$$

For $0 < \nu < At_*^2$, put $t = \sqrt{\nu/A} < t_*$. Then $g = g_f - At^2$, $r_+ = r_f + t$, and directly

$$g - h(1+r_+)^2 = M - 2hs_f t - (A+h)t^2 > 0. \tag{204}$$

Indeed each of the last two terms is strictly less than $M/2$. Also $At_*^2 \leq b^2/(16A) < g_f - g_c$, so this whole interval lies in $\mathcal{D}$ and in $\mathcal{B}_{n,q}$ for every admissible (n,q). This supplies a quantitative alternative to the continuity argument for stable coexistence. Equation (191) also proves directly that $G > H$ persists on the upper branch near the fold. Intersecting this interval with that of Corollary 12.6 retains the index-one conclusion.

Remark 12.9 (An exact obstruction to whole-interval stability). Take

$$n = 6, \quad q = 1, \qquad (a,k,c,g,h) = \left(6, \frac{1}{10}, \frac{1}{10}, \frac{101}{10}, \frac{9}{10}\right). \tag{205}$$

Here $A = 1$, $b = 5$, $g_c = 61/10$, $g_f = 247/20$, $\mu = -4$, and the equilibrium polynomial is $(r-1)(r-4)$. At $r_+ = 4$ the original-field equilibrium residual is exactly zero, and

$$d = -\frac{109}{250}, \quad C = \frac{1}{50}, \quad G = \frac{202}{125}, \quad H = \frac{18}{5}, \quad G - H = -\frac{248}{125}.$$

Consequently $\lambda_0 = -12/5 < 0$, but the real mode $m = 3$ has

$$\lambda_3 = d - C - G + H = \frac{191}{125} > 0. \tag{206}$$

The origin is stable and the upper equilibrium unstable at this strict interior point of the two-root interval. This refutes universal upper-branch stability, hence universal origin/upper stable coexistence on that interval. At the lower root $r_- = 1$, one has $\lambda_0 = 3/2$ and $\operatorname{Re}\lambda_1 = \operatorname{Re}\lambda_5 = 53/80 > 0$; the other three real parts are negative. Thus its unstable dimension is three here, also precluding a whole-interval index-one claim. This makes no assertion about other attractors there. It is a static full-spectrum counterexample, not a theorem about a crossing, a Hopf bifurcation or a periodic branch.

12.6 Synchronous threshold and the precise local basin statement

Proposition 12.10 (Exact basin threshold on the nonnegative synchronous ray)**.** *For every $g_c < g < g_f$, the scalar flow on the ray satisfies*

r	$(0, r_-)$	(r_-, r_+)	(r_+, ∞)
$f(r, g)$	$-$	$+$	$-$

If $0 < r(0) < r_-$, then $r(t) \to 0$. If $r(0) > r_-$, then $r(t) \to r_+$; this includes the constant solution $r(0) = r_+$. The data $r(0) = r_-$ and $r(0) = 0$ give their respective constant solutions. These are exact full-system solutions by synchrony, even when the upper equilibrium is transversely unstable.

Proof. Factor (180) as

$$f(r, g) = -\frac{Ar}{1 + r}(r - r_-)(r - r_+). \tag{207}$$

The prefactor is negative for $r > 0$, giving the three signs. Uniqueness prevents crossing an equilibrium. P01 and P02 give global nonnegative existence and boundedness. More specifically solutions below r_- decrease and are bounded below by zero; solutions between the roots increase and are bounded above by r_+; those above r_+ decrease and are bounded below by it. Each therefore has a limit. If that limit were not a zero of f, continuity would bound $|f|$ away from zero near the limit, contradicting convergence. The monotone interval and the initial strict inequality select the claimed zero. No nonsynchronous basin conclusion follows from this argument alone. □

Proposition 12.11 (Unstable synchrony and a local separating stable disk)**.** *At a near-fold parameter tuple covered by Corollary 12.6 and Theorem 12.7, let $p = r_-\mathbf{1}$. The local stable manifold is a smooth embedded $(n-1)$-disk tangent to E. Its germ locally separates a sufficiently small neighborhood into two sides. The local unstable manifold is exactly a segment of the synchronous line through p. Its two forward branches tend, respectively, to the origin and to $r_+\mathbf{1}$.*

In fact one may choose a neighborhood $V \subset \mathbb{R}^n_{>0}$ of p such that its local stable disk S has

$$V \setminus S = V_- \dot\cup V_+, \qquad V_- \subset \mathcal{A}(0), \quad V_+ \subset \mathcal{A}(r_+\mathbf{1}), \tag{208}$$

where the two sets are connected and $\mathcal{A}(z)$ denotes the basin of the equilibrium z relative to $\mathbb{R}^n_{\geq 0}$. Every point of S tends to p. Thus S is the common basin boundary relative to this neighborhood. *No assertion identifies the full global basin boundary or globally continued stable manifold.*

Proof. The stable/unstable manifold theorem [2, Ch. 1] applies to the hyperbolic index-one saddle. On a small synchronous interval, every point converges exponentially to p in negative time; uniqueness of the local unstable manifold therefore identifies it with that interval. Proposition 12.10 supplies its two forward limits.

To identify the basin sides, write the stable disk as $s = \phi(y)$ in $x = p + s\mathbf{1} + y$, $y \in E$, and set $\xi = s - \phi(y)$, $z = y$. Invariance and smoothness give

$$\dot{\xi} = \xi(\lambda + \alpha(\xi, z)), \qquad \dot{z} = Bz + R(\xi, z), \quad \lambda = \lambda_0(p) > 0, \quad B = J(p)|_E, \tag{209}$$

with $\alpha(0,0) = 0$, $R(\xi, 0) = 0$, and $D_z R(0,0) = 0$. Normality and the stable gap give $z^\mathsf{T} B z \leq -\gamma \|z\|^2$. Shrink the neighborhood until $|\alpha| < \lambda/2$ and $\|R\| \leq (\gamma/2)\|z\|$. Then $\|z\|$ decreases at least exponentially and every nonzero $|\xi|$ grows at rate at least $\lambda/2$.

Choose small exit points $(\pm\rho, 0)$ on the two unstable branches. Each belongs to its respective sink's open basin. Hence for sufficiently small $\sigma > 0$ both caps $\{(\pm\rho, z) : \|z\| \leq \sigma\}$ remain in those basins. In the box $|\xi| < \rho$, $\|z\| < \sigma$, a point with $\xi \neq 0$ exits through the cap of its sign and converges to that sink; points with $\xi = 0$ converge to p. The two connected half-boxes and the stable disk give exactly (208). This argument establishes the boundary only relative to that neighborhood. □

Global scope limit. The ray threshold is global only within the invariant synchronous ray. The preceding full-space statement concerns one sufficiently small neighborhood of the lower saddle at a fixed near-fold tuple. It neither identifies the complete global boundary between the sinks nor gives a partition of the whole orthant. Other equilibria, periodic attractors, or other invariant attracting sets elsewhere are not excluded. In particular "bistability" here does not assert exactly two global attractors.

12.7 Connection to P09, degenerate limit, and the P11 boundary

For $b > 0$, as $g \downarrow g_c$ from above,

$$r_- \longrightarrow 0, \qquad r_+ \longrightarrow b/A, \qquad \frac{dr_-}{dg} = \frac{1}{\Delta} \longrightarrow \frac{1}{b}. \tag{210}$$

Thus P09's small positive branch on $g > g_c$ is precisely P02/P10's lower branch, unstable in the uniform direction throughout (g_c, g_f). It meets the upper branch at g_f. No upper stability all the way to g_c follows; Remark 12.9 shows why this would be false in general.

For fixed $k, c, h > 0$, as $b = a - A \downarrow 0$,

$$r_f = \frac{b}{2A} \to 0, \qquad g_f - g_c = \frac{b^2}{4A} \to 0, \qquad \alpha_{\mathrm{SN}}, \beta_{\mathrm{SN}} \to 0. \tag{211}$$

The limiting locus is P09's $a = A$, $g = a + c$, where the center field is $-Ar^3/(1+r)$, its cubic derivative is $-6A$, and the center zero remains simple. P09 supplies the codimension-two algebraic/spectral statement. The P10 nondegeneracy theorem excludes this limit. There is no two-parameter unfolding or cusp classification here.

The exact facts available for later P11 work are the uniform signs on the entire open two-root interval, the complete fold spectrum, and automatic all-mode upper stability sufficiently near the fold. A subsequent loss of upper stability inside this interval must be nonuniform. P10 establishes no Hopf existence or frequency, critical pair simplicity, crossing transversality, Lyapunov coefficient, or periodic branch. The static counterexample does not begin those tasks. The frozen direct edge into P10 remains P02/P03/P04/P09, with P01 inherited; recording potential later inputs does not alter the frozen DAG.

13 P11: Hopf spectral conditions and spectral simplicity

We prove spectral assertions for the canonical demand-modulated model at positive synchrony. Direct inputs are P02, P04 and P05; P07 is used only for the five-node specialization. P03's coefficients are inherited through P04. P08–P10 and P12 or later results supply no mathematical premise. In particular, an imaginary pair here is only a *spectral Hopf candidate*. No parameter crossing, nonlinear nondegeneracy, periodic orbit or propagation theorem is asserted. The physical path is frozen for P12 without differentiating along it. The optional realizability result below concerns spectral boundaries, not first instability.

13.1 Setting, representative classes and frequency

Fix $q = 2$, $n \geq 5$, $a, k, c, g, h > 0$, and a specified positive synchronous equilibrium $x^*\mathbf{1}$. Set

$$s = 1 + x^*, \qquad C = \frac{c}{s} > 0, \qquad G = \frac{gx^*}{s^2} > 0, \qquad H = hx^* > 0. \tag{212}$$

Use P04's verified spectrum and unitary diagonalization:

$$\begin{aligned} \lambda_m &= d + Ce^{-\mathrm{i}\theta_m} + Ge^{2\mathrm{i}\theta_m} - He^{-2\mathrm{i}\theta_m}, \quad \theta_m = 2\pi m/n, \\ r_m &= d + C\cos\theta_m + (G - H)\cos 2\theta_m, \\ \omega_m &= -C\sin\theta_m + (G + H)\sin 2\theta_m, \\ J &= \mathsf{Q}\,\mathrm{diag}(\lambda_m)\mathsf{Q}^*, \qquad \lambda_{n-m} = \overline{\lambda_m} \quad (1 \leq m < n). \end{aligned} \tag{213}$$

Here $J = DF^{\mathrm{DM}}(x^*\mathbf{1})$; ω_m is a signed imaginary part. The reserved positive frequency is chosen separately when needed.

Let $N = \lfloor n/2 \rfloor$, $u_j = \cos(2\pi j/n)$ and L_j be P05's representative label classes: $L_0 = \{0\}$, $L_N = \{n/2\}$ if n is even, and otherwise $L_j = \{j, n - j\}$. Write

$$\mathcal{S} = \operatorname*{arg\,max}_{0 \leq j \leq N} r_j$$

for the *complete* selected representative set, including every tie. The selected Fourier labels are $\bigcup_{j \in \mathcal{S}} L_j$. This notation distinguishes $\mathcal{S}$ from the scalar $G + H$.

Lemma 13.1 (Reachable nonuniform representatives are genuine pairs)**.** *Every reachable nonuniform representative satisfies* $1 \leq j \leq j_{\max} := \lfloor (n + 1)/4 \rfloor < n/2$. *Consequently* $0 < \theta_j < \pi$, $\sin\theta_j > 0$, *and* $L_j = \{j, n - j\}$ *has two different labels. No even-n Nyquist singleton is included.*

Proof. P05 Theorem 7.2 supplies the reachable list. Since $(n + 1)/4 < n/2$ for $n > 1$, its nonzero members satisfy the claimed strict bounds. The angle and sine statements follow. This counts Fourier labels only, not numerical eigenvalue multiplicities. □

Proposition 13.2 (Exact signed frequency and its zero locus)**.** *For any genuine nonuniform pair representative* $0 < j < n/2$,

$$\boxed{\omega_j = \sin\theta_j\,[2(G + H)u_j - C].} \tag{214}$$

Thus $\omega_j = 0$ *if and only if* $2(G + H)u_j = C$. *If* $u_j \leq 0$, *then* $\omega_j < 0$ *automatically. If* $u_j > 0$, *the zero-frequency locus is exactly*

$$G + H = \frac{C}{2u_j}. \tag{215}$$

For positive u_j *the sign of* ω_j *is the sign of* $G + H - C/(2u_j)$. *These are spectral statements, not bifurcation classifications.*

Proof. Insert $\sin 2\theta_j = 2\sin\theta_j\cos\theta_j$ in (213). The factor $\sin\theta_j$ is strictly positive. If $u_j \le 0$, the bracket is at most $-C < 0$; otherwise divide by $2u_j > 0$. The Nyquist singleton is outside the proposition's scope. □

13.2 Strict candidates, simplicity and the stable complement

Definition 13.3 (Strict nonuniform spectral Hopf candidate)**.** A positive synchronous equilibrium of the canonical $q = 2$ model is a *strict nonuniform spectral Hopf candidate in representative j* if $j \neq 0$, $\mathcal{S} = \{j\}$, $r_j = 0$, and $\omega_j \neq 0$. Uniqueness refers to representatives and includes the exclusion of all maximal ties. This definition does not assert a Hopf bifurcation.

Theorem 13.4 (General strict spectral criterion)**.** *At a strict candidate, the entire imaginary-axis spectrum is exactly*

$$\lambda_j = \mathrm{i}\omega_j, \qquad \lambda_{n-j} = -\mathrm{i}\omega_j, \qquad \omega_j \neq 0. \tag{216}$$

Each value is algebraically and geometrically simple over $\mathbb{C}$ and the two values are distinct. Every other Fourier label has $r_\ell < 0$. The number

$$\eta = -\max_{\ell \notin L_j} r_\ell \tag{217}$$

is strictly positive, and all noncritical eigenvalues satisfy $\operatorname{Re}\lambda_\ell \le -\eta$. The real critical space has dimension two and its real invariant complement has dimension $n-2$. Moreover

$$0,\ 2\mathrm{i}\omega_j \notin \operatorname{spec}(J), \qquad J^{-1} \text{ and } (2\mathrm{i}\omega_j I - J)^{-1} \text{ exist.} \tag{218}$$

Proof. Strict representative selection leaves precisely labels $j, n-j$ at real part zero; their values are distinct because $\omega_j \neq 0$. No other label can share either value. P04's multiplicity formula therefore proves simplicity. The finite complement has maximum $-\eta < 0$ and real dimension $n-2$. Neither 0 nor $2\mathrm{i}\omega_j$ equals $\pm\mathrm{i}\omega_j$ or a stable eigenvalue, proving the inverse gates. These are the spectral hypotheses used in the usual Hopf theorem [6]; crossing and nonlinear nondegeneracy remain separate. □

Proposition 13.5 (Exact real and adjoint normalization)**.** *At the strict candidate, choose*

$$v_H = v^{(j)}, \qquad p_H = v_H, \qquad e_c = \sqrt{2}\,\operatorname{Re} v_H, \quad e_s = \sqrt{2}\,\operatorname{Im} v_H. \tag{219}$$

Under the project's Hermitian pairing $\langle p, v\rangle = p^* v$,

$$Jv_H = \mathrm{i}\omega_j v_H, \qquad J^* p_H = -\mathrm{i}\omega_j p_H, \qquad \langle p_H, v_H\rangle = 1. \tag{220}$$

The real vectors e_c, e_s are orthonormal and span the real critical plane. In the ordered basis (e_c, e_s) the matrix is

$$J|_{\mathrm{crit}} = \begin{pmatrix} 0 & \omega_j \\ -\omega_j & 0 \end{pmatrix}. \tag{221}$$

In its coefficient plane the signed angular velocity is $-\omega_j$.

Proof. Unitary diagonalization gives the adjoint equation with conjugate eigenvalue and $p_H = v_H$. Orthogonality of v_H and $\bar{v}_H$ gives $\|\operatorname{Re} v_H\|^2 = \|\operatorname{Im} v_H\|^2 = 1/2$ and their orthogonality. Taking real and imaginary parts of $Jv_H = \mathrm{i}\omega_j v_H$ gives $Je_c = -\omega_j e_s$ and $Je_s = \omega_j e_c$, hence the stated block and orientation. If a positive frequency is required when $\omega_j < 0$, use the conjugate label. □

Remark 13.6 (Ties and zero frequency are excluded separately). If two different representative classes have $r_j = r_t = 0$, their labels give additional imaginary-axis values or larger multiplicities. Either possibility fails the single-simple-pair-with-stable-complement hypothesis. A tie of real parts alone is not equality of numerical eigenvalues. In particular choosing one member of a maximal tie does not satisfy Definition 13.3.

If a genuine pair has $r_j = \omega_j = 0$, both labels produce the same real zero eigenvalue. P04 makes it semisimple, with complex algebraic and geometric multiplicity exactly $|\{m : \lambda_m = 0\}| \geq 2$. Under strict selection of that representative its multiplicity is exactly two and its real kernel has dimension two. Without strict selection other zero labels must also be counted. This is ineligible for the simple Hopf package. No named multiple-criticality or steady-state bifurcation is classified. The two-dimensional real plane in Theorem 13.4, by contrast, corresponds to two distinct simple complex eigenvalues.

13.3 Regular spectral surfaces and physical admissibility

Fix a reachable nonuniform representative $1 \leq j \leq j_{\max}$ throughout this subsection, so $0 < \theta_j < \pi$. Use P02's nonzero equilibrium equation in branch coordinates:

$$a = (k+h)x - \frac{c-g}{1+x} > 0, \qquad x, k, c, g, h > 0. \tag{222}$$

This is exactly P05's open chart domain $\mathcal{U}_{\mathrm{eq}}$. Combining P04's diagonal coefficient with the equilibrium equation gives

$$d = a - 2kx - \frac{cx}{(1+x)^2} - \frac{g}{1+x} - hx = -kx - \frac{c(1+2x)}{(1+x)^2}.$$

For fixed representative j, put $v_j = \cos 2\theta_j = 2u_j^2 - 1$; this scalar v_j is not a Fourier vector $v^{(j)}$. Define

$$\begin{aligned} R_j(x,k,c,g,h) = & -kx - \frac{c(1+2x)}{(1+x)^2} + \frac{c}{1+x}u_j \\ & + x\left(\frac{g}{(1+x)^2} - h\right)v_j. \end{aligned} \tag{223}$$

Substitution in P04's real dispersion shows exactly $R_j = r_j$.

Proposition 13.7 (Regular spectral boundary and excluded frequency locus)**.** *At fixed x, c, g, h,*

$$\frac{\partial R_j}{\partial k} = -x < 0, \qquad k_H = \frac{1}{x}\left[-\frac{c(1+2x)}{(1+x)^2} + \frac{c}{1+x}u_j + x\left(\frac{g}{(1+x)^2} - h\right)v_j\right]. \tag{224}$$

Thus $R_j = 0$ is a regular codimension-one level set wherever it meets $\mathcal{U}_{\mathrm{eq}}$, explicitly the graph $k = k_H$. A physical point requires $k_H > 0$ and $a_H = (k_H + h)x - (c-g)/(1+x) > 0$. Within a strict nonuniform P05 cell, the portions with $\omega_j \neq 0$ are strict candidates, and this regular surface transfers locally to physical parameter space along the specified positive branch.

For $u_j > 0$, any nonempty intersection $R_j = \omega_j = 0$ is locally a regular codimension-two level set in branch coordinates. It is codimension one inside $R_j = 0$. The same statement transfers to physical parameters wherever P05's concave-chart hypotheses hold. For $u_j \leq 0$ this intersection is empty.

Proof. In (223) only $-kx$ depends on k, proving the partial derivative and every sign in the solved formula. Strict selection and $\omega_j \neq 0$ are open conditions. P05 Theorem 7.1 implies $H - G > 0$ at any nonuniform selected representative. On precisely this concave part of the positive chart, P05 supplies

$$\Phi(x,k,c,g,h) = (a,k,c,g,h), \qquad \det D\Phi = k + h + \frac{c-g}{(1+x)^2} = k + \left(h - \frac{g}{(1+x)^2}\right) + \frac{c}{(1+x)^2} > 0.$$

Its local diffeomorphism therefore transfers the regular level set. No global injectivity or topology after forgetting the branch is claimed. The derivative $\partial_k R_j$ holds x, c, g, h fixed and lets a follow (222); it is not a derivative along the later fixed-a continuation path.

For the additional frequency equation, at fixed branch coordinates

$$\partial_k \omega_j = 0, \qquad \partial_g \omega_j = \frac{2x u_j \sin\theta_j}{(1+x)^2} > 0 \quad (u_j > 0).$$

The Jacobian of (R_j, ω_j) in columns (k, g) has determinant $-2x^2 u_j \sin\theta_j/(1+x)^2 \neq 0$. The regular level set theorem proves the local codimension assertion, conditional on existence and physical positivity; the local diffeomorphism applies under the same concavity hypotheses just stated. Proposition 13.2 excludes all $u_j \leq 0$ intersections. No unfolding is inferred. □

Proposition 13.8 (Exact obstruction and cell realizability criterion)**.** *Fix a reachable nonuniform representative $1 \leq j \leq j_{\max}$. Let $\beta_j = (u_j + u_{j+1})/2$ be P05's midpoint, and put*

$$w_j = 1 - 2u_j^2, \quad L_j^{\mathrm{vert}} = \max(0, \beta_j), \qquad B_j = \frac{w_j}{4(1-u_j)}. \tag{225}$$

Its strict P05 cell contains a physical strict spectral Hopf candidate if and only if

$$\boxed{w_j > 0 \quad \textit{and} \quad L_j^{\mathrm{vert}} < B_j.} \tag{226}$$

More precisely, for a prescribed positive vertex $z = C/[4(H-G)]$ in that strict cell and prescribed $x > 0$, such a candidate can be realized by some positive physical coefficients if and only if

$$\frac{w_j}{4z} > 1 - u_j + \frac{x}{1+x}. \tag{227}$$

Consequently not every reachable nonuniform cell supports a candidate. For example $n = 8, j = 1$ has $w_j = 0$ and never reaches $r_j = 0$. For $n = 12, j = 2$, $u_j = 1/2$, $L_j^{\mathrm{vert}} = B_j = 1/4$; this cell also has no physical candidate despite $w_j > 0$.

Proof. Write $D_c = H - G > 0$ and $C = c/(1+x)$ within the selected cell. At a candidate, (223) reads

$$kx = D_c w_j - C\left(1 - u_j + \frac{x}{1+x}\right), \qquad \frac{kx}{C} = \frac{w_j}{4z} - \left(1 - u_j + \frac{x}{1+x}\right). \tag{228}$$

Because $k, x, C > 0$ and $u_j < 1$, this forces $w_j > 0$ and (227). It further forces $z < B_j$. P05's strict cell is $L_j^{\mathrm{vert}} < z < \beta_{j-1}$, where $\beta_{j-1} > 0$; β_j exists since $j_{\max} < N$. Thus necessity of (226) follows.

Conversely, if (226) holds, choose

$$L_j^{\mathrm{vert}} < z < \min(\beta_{j-1}, B_j).$$

Both upper bounds exceed the lower bound. Choose $x > 0$ sufficiently small that (227) holds. For fixed x, z satisfying that inequality, the following construction proves the more precise sufficiency as well. Choose any $C > 0$, set $D_c = C/(4z)$, and define $k > 0$ by (228). For a free $G > 0$, set

$$c = C(1+x), \qquad g = \frac{G(1+x)^2}{x}, \qquad h = \frac{G + D_c}{x}, \qquad a = kx + D_c - C + G\left(1 + \frac{1+x}{x}\right). \tag{229}$$

These equations give $H - G = D_c$ and exactly $a = (k+h)x - (c-g)/(1+x)$. Increasing G makes $a > 0$ without changing z, k or $r_j = 0$. The signed frequency bracket is $2(D_c + 2G)u_j - C$. For $u_j \le 0$ it is automatically negative; for $u_j > 0$ it vanishes for at most one G, which can be avoided while retaining positivity. Hence a strict candidate is obtained. The two examples follow by exact substitution of their cosines and midpoints. They also show why P05's realization of dominance alone cannot imply realization of a spectral boundary. □

13.4 An exact five-node physical candidate

Use the P07 constants and spectra, without a new P07 derivation:

$$\phi = \frac{1+\sqrt{5}}{2}, \quad \sigma = \frac{\sqrt{10 - 2\sqrt{5}}}{4} > 0, \quad u_1 = \frac{1}{2\phi}, \quad u_2 = -\frac{\phi}{2}.$$

P07 shows that $\{1, 4\}$ is the only potentially dominant nonuniform pair and that $\omega_1 = \sigma[(G+H) - \phi C]$.

Theorem 13.9 (Exact five-node spectral Hopf candidate)**.** *For $n = 5, q = 2$ set*

$$\begin{gathered} x_H = 1, \quad c = 2, \quad g_H = 16, \quad h = 6, \\ k_H = \frac{3\phi - 4}{2} = \frac{3\sqrt{5} - 5}{4}, \qquad a_H = \frac{3\phi + 22}{2} = \frac{47 + 3\sqrt{5}}{4}. \end{gathered} \tag{230}$$

This is a positive physical tuple with a positive synchronous equilibrium $x_H\mathbf{1}$. It is a strict spectral candidate in representative 1*. Its full labeled spectrum is*

$$\begin{aligned} &\lambda_0 = -\frac{3\phi + 1}{2}, && && \\ &\lambda_1 = \mathrm{i}\omega_H, \qquad \lambda_4 = -\mathrm{i}\omega_H, && && \omega_H = \sigma(10 - \phi) > 0, \\ &\lambda_2 = -\frac{3\sqrt{5}}{2} - \mathrm{i}\sigma(1 + 10\phi), && && \lambda_3 = \overline{\lambda_2}. \end{aligned} \tag{231}$$

The critical values are each algebraically and geometrically simple. The ordering is $r_1 = 0 > r_0 > r_2$, with exact noncritical gap

$$\eta_H = \frac{3\phi + 1}{2} > 0. \tag{232}$$

The equilibrium is the upper of two distinct positive synchronous roots. The other root is

$$x_- = \frac{3(2 - \phi)}{3\phi + 8} \in (0, 1). \tag{233}$$

Proof. Since $5/3 < \sqrt{5} < 3$, $k_H > 0$ and $a_H > 0$; the remaining three coefficients are positive. In the original canonical field, each component at $\mathbf{1}$ is exactly

$$F_i^{\mathrm{DM}}(\mathbf{1}) = a_H - k_H + \frac{2}{2} - \frac{16}{2} - 6 = a_H - k_H - 13 = 0.$$

This verifies the original model residual before using any spectral data. Direct substitution gives

$$\begin{gathered} s=2, \quad C=1, \quad G=4, \quad H=6, \quad T=G-H=-2, \quad S=G+H=10, \\ d=-k_H-\frac{3}{2}=\frac{1-3\phi}{2}. \end{gathered} \tag{234}$$

The raw P04 expression independently gives $d=a_H-2k_H-1/2-8-6=-k_H-3/2$. P07's exact real-part formulas now yield

$$\begin{aligned} r_1 &= \frac{1-3\phi}{2}+\frac{1}{2\phi}+\phi=0, \\ r_0 &= \frac{1-3\phi}{2}+1-2=-\frac{3\phi+1}{2}, \\ r_2 &= \frac{1-3\phi}{2}-\frac{\phi}{2}-\frac{1}{\phi}=\frac{3-6\phi}{2}=-\frac{3\sqrt{5}}{2}. \end{aligned}$$

Here $\phi^{-1}=\phi-1$ and $2\phi-1=\sqrt{5}$. Their noncritical difference is $r_0-r_2=(3\phi-4)/2=k_H>0$. Equivalently, $-\phi^2<-2<-\phi^{-2}$, so P07's full ordering table gives the same strict ordering. In particular pair $\{2,3\}$ is excluded.

For the imaginary parts, P07 gives $\omega_1=\sigma(10-\phi)$ and $\omega_2=-\sigma(1+10\phi)$. The exact bounds $1<\phi<2$ imply $10-\phi>8>0$, so the critical frequency is nonzero without decimal evidence. P04 conjugacy fixes the signs at labels $4,3$. This proves every entry of (231). The general theorem proves simplicity and the inverse gates, and the strict noncritical ordering gives (232).

For the branch identification, use only P02's abbreviations and polynomial:

$$A=k_H+h=\frac{3\phi+8}{2}, \quad b=a_H-A=7, \quad \mu=a_H+2-16=\frac{3\phi-6}{2}<0. \tag{235}$$

Because 1 is a root and $A-b=\mu$, that polynomial factors as

$$Q(X)=AX^2-bX-\mu=A(X-1)(X-x_-), \qquad x_-=-\mu/A.$$

The displayed x_- is positive because $\phi<2$, and is less than one since $3(2-\phi)<3\phi+8$. Thus the discriminant is $A^2(1-x_-)^2>0$. Together with $b>0$ and $\mu<0$ this checks P02's two-positive-root regime fully; those two signs alone would not suffice without the positive discriminant. The larger root is exactly $x_H=1$. No P10 theorem is used. □

13.5 Frozen continuation path and exact spectral data

Proposition 13.10 (Local branch, path, vectors and resolvents for P12/P13)**.** *Hold $a=a_H, k=k_H, c=2, h=6$ fixed and freeze the physical path*

$$\boxed{\pi(g)=(a_H,k_H,2,g,6), \qquad g_H=16.} \tag{236}$$

There is a neighborhood of 16 with a unique smooth positive synchronous root near 1, denoted $x_+(g)$, with $x_+(16)=1$. It continues the upper root. With the project's node ordering $0,\dots,4$,

$$\begin{gathered} v_H=\frac{1}{\sqrt{5}}(1,e^{2\pi \mathrm{i}/5},e^{4\pi \mathrm{i}/5},e^{6\pi \mathrm{i}/5},e^{8\pi \mathrm{i}/5})^{\mathsf{T}}, \\ p_H=v_H, \qquad \overline{v_H}=v^{(4)}, \qquad J_Hv_H=\mathrm{i}\omega_Hv_H, \quad J_H^*p_H=-\mathrm{i}\omega_Hp_H, \quad p_H^*v_H=1. \end{gathered} \tag{237}$$

The real orthonormal basis $e_c = \sqrt{2}\,\mathrm{Re}\,v_H$, $e_s = \sqrt{2}\,\mathrm{Im}\,v_H$ *has block* $\left(\begin{smallmatrix} 0 & \omega_H \\ -\omega_H & 0 \end{smallmatrix}\right)$. *Both required inverse operators exist and are exactly*

$$\begin{aligned} J_H^{-1} &= \sum_{m=0}^{4} \lambda_m^{-1} v^{(m)} v^{(m)*}, \\ (2\mathrm{i}\omega_H I - J_H)^{-1} &= \sum_{m=0}^{4} (2\mathrm{i}\omega_H - \lambda_m)^{-1} v^{(m)} v^{(m)*}. \end{aligned} \tag{238}$$

In particular the mode-0 and mode-2 denominators useful for later work are

$$\lambda_0 = -\eta_H \neq 0, \qquad 2\mathrm{i}\omega_H - \lambda_2 = \frac{3\sqrt{5}}{2} + \mathrm{i}\sigma(21 + 8\phi) \neq 0. \tag{239}$$

Proof. Let $B(X, g) = a_H - (k_H + 6)X + (2 - g)/(1 + X)$. P02's scalar field is $f(X, g) = XB(X, g)$. P04's uniform identity $\lambda_0 = \partial_X f$ at synchrony, together with $B(1, 16) = 0$, gives $B_X(1, 16) = \lambda_0 = -\eta_H < 0$. The scalar implicit-function theorem supplies the unique smooth root near 1. Positivity persists in a sufficiently small neighborhood, and P02's distinct roots and their strict ordering persist there, so this is the locally continued upper root, not uniqueness of all positive roots. The parameter tuple stays positive for g near 16. No derivative of $x_+(g)$ is calculated.

Proposition 13.5 gives all vector identities and the real block with its stated sign. The spectral theorem and Theorem 13.4 give the two sums, all of whose denominators are nonzero. Finally, $2(10 - \phi) + (1 + 10\phi) = 21 + 8\phi$ verifies the second denominator in (239); its real part is positive. No tensor, nonlinear Fourier selection rule or Taylor-factor convention is required for these spectral identities. □

Proposition 13.11 (Spectral hypothesis ledger at the five-node point)**.** *The following records the established spectral inputs and the remaining P12/P13 work.*

- *H1. All five physical coefficients in* (230) *are positive.*
- *H2.* $x_H = 1$ *is an exact positive synchronous equilibrium of the original DM field.*
- *H3. The path varying* g *in* (236) *has the smooth local upper branch* $x_+(g)$ *through* $x_+(16) = 1$.
- *H4. Labels* $1, 4$ *give* $\pm\mathrm{i}\omega_H$ *with* $\omega_H = \sigma(10 - \phi) > 0$.
- *H5. Each critical complex eigenvalue is algebraically and geometrically simple.*
- *H6. Every other eigenvalue has real part at most* $-\eta_H < 0$, *where* $\eta_H = (3\phi + 1)/2$.
- *H7.* $0 \notin \mathrm{spec}(J_H)$.
- *H8.* $2\mathrm{i}\omega_H \notin \mathrm{spec}(J_H)$.
- *H9. Exact normalized right and adjoint vectors are* (237)*; the pairing and real-plane orientation are fixed.*
- *H10. Spectral transversality is* ***not established****; no state or eigenvalue derivative along* (236) *has been computed.*
- *H11. The first Lyapunov coefficient and nonlinear nondegeneracy are* ***not established****; no tensor or normal-form coefficient is computed.*

H12. No periodic orbit, periodic-branch stability or rotating wave is established.

Proof. Items H1–H2 and H4–H6 are Theorem 13.9; H3 and H9 are Proposition 13.10; H7–H8 follow from Theorem 13.4. H10–H12 explicitly delimit unproved later conclusions, rather than adding assumptions already certified here. □

14 P12: Hopf transversality (unfinished application)

The source has no derivative calculation. The formulas below specify the standard reduction; the crossing on P11's path remains to be verified.

14.1 Equilibrium derivative

For $B(x, g) = a - (k + h)x + (c - g)/(1 + x) = 0$ with a, k, c, h fixed, implicit differentiation gives

$$B_x = -(k + h) - \frac{c - g}{(1 + x)^2}, \qquad x_g = \frac{1}{(1 + x)B_x} \quad (B_x \neq 0).$$

The implicit-function theorem applies locally at P11's simple upper root; it fails in this parameterization at a scalar fold [2, Ch. 1].

14.2 Total crossing derivative

Differentiate P04's coefficients along the moving equilibrium:

$$\frac{d\lambda_m}{dg} = d_g + C_g e^{-\mathrm{i}\theta_m} + G_g e^{2\mathrm{i}\theta_m} - H_g e^{-2\mathrm{i}\theta_m}.$$

Here every subscript denotes a total derivative, including x_g.

14.3 Outstanding nondegeneracy

Evaluate and certify $\mathrm{Re}(d\lambda_j/dg) \neq 0$ at P11's exact tuple, with the orientation of increasing g explicit. This is the crossing hypothesis in [6]; the spectral conditions alone do not prove it.

15 P13: First Lyapunov coefficient (unfinished application)

No model-specific derivative tensors or sign calculation were supplied.

15.1 Derivative tensors

At the candidate, use the Taylor convention $F(x_H + z) = Jz + \frac{1}{2}\mathcal{B}(z, z) + \frac{1}{6}\mathcal{C}(z, z, z) + O(\|z\|^4)$, where $\mathcal{B} = D^2F$ and $\mathcal{C} = D^3F$. Both tensors must be differentiated from the DM field, including its rational terms and coincident offsets; DM is not a quadratic model.

15.2 Fourier and resolvent inputs

Use P11's $Jv = \mathrm{i}\omega v$, $J^*p = -\mathrm{i}\omega p$, $p^*v = 1$, $\omega > 0$, and its verified inverse gates. Cyclic equivariance restricts products of modes to their sum modulo n; retain normalization factors and possible aliasing when evaluating the actual tensors.

15.3 Standard coefficient

In this convention the standard formula is [6]

$$l_1 = \frac{1}{2\omega}\,\mathrm{Re}\big\{p^*\big[\mathcal{C}(v,v,\bar v) - 2\mathcal{B}(v, J^{-1}\mathcal{B}(v,\bar v)) \\ + \mathcal{B}(\bar v, (2\mathrm{i}\omega I - J)^{-1}\mathcal{B}(v,v))\big]\big\}.$$

The formula replaces a derivation of general normal-form theory; the DM evaluation and rigorous sign or nonvanishing certificate remain required.

15.4 Hopf conclusion

Only after P12's nonzero crossing and $l_1 \neq 0$ have been verified can the nondegenerate Hopf theorem be applied. Branch stability additionally uses P11's stable complement. No periodic branch is established here.

16 P14: Rotating waves (unfinished application)

16.1 Exact symmetry

Once P12–P13 verify a nondegenerate Hopf bifurcation, use equivariant Hopf theory [7, Theorem 3.7]. The homogeneous DM field commutes with P; the critical Fourier column obeys $Pv^{(m)} = e^{-2\pi\mathrm{i}m/n}v^{(m)}$. With m chosen to carry $+\mathrm{i}\omega$, the expected branch symmetry is $Px(t) = x(t - mT/n)$, with the branch's actual period T. This is a conditional application, not an existence proof from a cosine ansatz.

16.2 Clusters and orientation

The character kernel has size $\gcd(n,m)$, giving $n/\gcd(n,m)$ phase classes. Reconcile the exact time-shift sign with the chosen eigenvalue and predecessor convention before stating propagation direction.

16.3 Outstanding verification

Verify the relevant axial subgroup and branch hypotheses and distinguish exact nonlinear symmetry from its leading Fourier approximation. Generic ring results do not establish those DM hypotheses or exact rotating waves after heterogeneous perturbation.

17 P15: Heterogeneous persistence (unfinished application)

The heterogeneous coefficient assignment has not been specified in the source.

17.1 Local equilibrium

For a specified smooth family $F(x,\varepsilon)$ with $F(x_0,0) = 0$ and invertible $J = D_xF(x_0,0)$, the implicit-function theorem gives a unique nearby equilibrium $x^*(\varepsilon)$ with the family's regularity [2, Ch. 1]. For an analytic family, use the analytic version stated in [8, §3.2].

17.2 Positivity and stability

If $x_0 > 0$, positivity persists locally by continuity. If J is Hurwitz, local exponential stability persists by spectral continuity. Invertibility alone does not imply stability; a nonzero-frequency Hopf equilibrium can have invertible J.

17.3 Remaining scope

Specify where every perturbed coefficient enters, its regularity, and an admissible parameter neighborhood. Any quantitative radius remains to be estimated. Optional periodic-orbit persistence requires hyperbolicity modulo phase, via a transverse Poincare map [2, Ch. 1]; equilibrium persistence is insufficient.

18 P16: Static sensitivity (unfinished application)

18.1 Physical forcing

Under P15's hypotheses, differentiation gives $\xi = x^{*\prime}(0) = -J^{-1}b$, where $b = F_\varepsilon(x_0, 0)$. The vector b must be computed from the chosen heterogeneous assignment.

18.2 Fourier inverse

P04 and circulant inversion [1, Theorem 3.1] give

$$\widehat{\xi}_m = -\widehat{b}_m/\lambda_m, \qquad \|\xi\|_2 \le \frac{\|b\|_2}{\min_m |\lambda_m|}, \qquad (J^{-1})_{j\ell} = \frac{1}{n}\sum_m \frac{e^{\mathrm{i}\theta_m(j-\ell)}}{\lambda_m}.$$

18.3 Localization

Graph-distance decay may be obtained from [9, Theorem 3.4] only after verifying its spectral continuum and analytic-neighborhood hypotheses for $f(z) = 1/z$, including exclusion of zero. Pointwise invertibility does not supply uniform decay constants. Use the graph-distance convention of that theorem; any directed or wraparound refinement needs its own argument.

18.4 Remainder

The physical forcing, neighborhood, remainder constants and any uniformity in n remain uncomputed. A nonzero imaginary Hopf pair need not make the static inverse singular.

19 P17: Linear transient and harmonic response

The source is an outline. The following standard linear corollaries require no further model assumptions; quantitative nonlinear response remains open.

19.1 Transient

P04's single Fourier framework [1, Theorem 3.1] gives

$$e^{tJ} = Q\,\mathrm{diag}(e^{t\lambda_m})Q^*, \qquad \|e^{tJ}\|_2 = e^{t\,\mathrm{sa}(J)} \quad (t \ge 0).$$

19.2 Leading modes

Late-time dominance requires nonzero initial projection onto the leading eigenspace; retain every tie. Euclidean norm decay is not coordinatewise monotonicity or a propagation theorem.

19.3 Explicit harmonic extension

For the added linear forcing $\dot{z} = Jz + be^{\mathrm{i}\Omega t}$, substitution gives $z_p = e^{\mathrm{i}\Omega t}(\mathrm{i}\Omega I - J)^{-1}b$ whenever the inverse exists. If J is Hurwitz, every solution approaches this harmonic response. Real forcing and response are obtained by taking real parts.

19.4 Unfinished nonlinear scope

Forcing conventions, observables, projected resonance amplitudes and nonlinear approximation bounds remain to be specified and proved.

20 P18: Stability radius and Hopf shift (unfinished application)

20.1 Moving Jacobian

For P15's specified family set $E(\varepsilon) = D_x F(x^*(\varepsilon), \varepsilon) - J_0$. Bounds must include both direct coefficient variation and equilibrium motion; these estimates are absent from the supplied outline.

20.2 Sufficient stability radius

If J_0 is normal and Hurwitz, put $\gamma = -\operatorname{sa}(J_0) > 0$. The normal-matrix perturbation inclusion [10, §1 and Theorem 4$_\varepsilon$] guarantees stability for $\|E(\varepsilon)\|_2 < \gamma$, even when the perturbed matrix is nonnormal. Turning this into a physical parameter radius requires an explicit bound on E; it is not a sharp structured-radius assertion.

20.3 Simple eigenvalue and threshold shift

Separately, at a simple transverse Hopf point of an analytic family, [8, Theorem 1] and the implicit-function theorem give

$$g_H'(0) = -\frac{\operatorname{Re}[p^* J_\varepsilon v]}{\operatorname{Re}[p^* J_g v]}, \qquad p^* v = 1, \quad \operatorname{Re}[p^* J_g v] \neq 0.$$

Here J_ε, J_g differentiate the Jacobian evaluated at the moving equilibrium. These are not derivatives with the equilibrium frozen.

20.4 Remaining Hopf conditions

Verify the family's regularity, remaining spectral gap, crossing and first-Lyapunov-coefficient nondegeneracy. A spectral threshold shift alone does not establish a nonlinear Hopf bifurcation.

References

[1] Robert M. Gray. Toeplitz and circulant matrices: A review. *Foundations and Trends in Communications and Information Theory*, 2(3):155–239, 2006. https://ee.stanford.edu/~gray/toeplitz.pdf.

[2] John Guckenheimer and Philip Holmes. *Nonlinear Oscillations, Dynamical Systems, and Bifurcations of Vector Fields*, volume 42 of *Applied Mathematical Sciences*. Springer, New York, 1983. https://doi.org/10.1007/978-1-4612-1140-2.

[3] Dirk L. van Kekem and Alef E. Sterk. Travelling waves and their bifurcations in the Lorenz-96 model. *Physica D: Nonlinear Phenomena*, 367:38–60, 2018. https://arxiv.org/abs/1704.05442v3.

[4] B. S. Goh. Global stability in many-species systems. *The American Naturalist*, 111(977):135–143, 1977. https://doi.org/10.1086/283144.

[5] Yuri A. Kuznetsov. Saddle-node bifurcation. *Scholarpedia*, 1(10):1859, 2006. https://doi.org/10.4249/scholarpedia.1859.

[6] Yuri A. Kuznetsov. Andronov–hopf bifurcation. *Scholarpedia*, 1(10):1858, 2006. https://doi.org/10.4249/scholarpedia.1858.

[7] Ian Stewart. Hopf bifurcation and phase patterns in symmetric ring networks, 2024. https://arxiv.org/html/2403.15814v1.

[8] Anne Greenbaum, Ren-Cang Li, and Michael L. Overton. First-order perturbation theory for eigenvalues and eigenvectors. *SIAM Review*, 62(2):463–482, 2020. https://arxiv.org/abs/1903.00785.

[9] Michele Benzi and Nader Razouk. Decay bounds and $O(n)$ algorithms for approximating functions of sparse matrices. *Electronic Transactions on Numerical Analysis*, 28:16–39, 2007. https://www.cs.emory.edu/~benzi/Web_papers/benzi_razouk.pdf.

[10] Mark Embree and Lloyd N. Trefethen. Generalizing eigenvalue theorems to pseudospectra theorems. *SIAM Journal on Scientific Computing*, 23(2):583–590, 2001. https://people.maths.ox.ac.uk/~trefethen/publication/PDF/2001_94.pdf.